ALMA MATER STUDIORUM
UNIVERSITÀ DI BOLOGNA

Department of Physics and Astronomy "A. Righi"

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Interactions and Correlations in Condensed Matter

Lectures held by
Prof. Domenico Di Sante

Notes by
Gioele Mancino

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Contents

Introduction

1 Second Quantization

In the first quantization formalism, we describe a system of N particles by a wavefunction $\Psi(\mathbf{r}_1, \mathbf{r}_2, \ldots, \mathbf{r}_N)$, which is a function of the coordinates of all the particles. This wavefunction must satisfy the appropriate symmetry properties, depending on whether we are dealing with bosons or fermions. For bosons, the wavefunction must be symmetric under the exchange of any two particles, while for fermions, the wavefunction must be antisymmetric under the exchange of any two particles:

$$\Psi(\mathbf{r}_1, \mathbf{r}_2, \ldots, \mathbf{r}_i, \ldots, \mathbf{r}_j, \ldots, \mathbf{r}_N) = \begin{cases} +\Psi(\mathbf{r}_1, \mathbf{r}_2, \ldots, \mathbf{r}_j, \ldots, \mathbf{r}_i, \ldots, \mathbf{r}_N) & \text{for bosons,} \\ -\Psi(\mathbf{r}_1, \mathbf{r}_2, \ldots, \mathbf{r}_j, \ldots, \mathbf{r}_i, \ldots, \mathbf{r}_N) & \text{for fermions.} \end{cases}$$

But an important limitation of this formalism is that it cannot describe systems with a variable number of particles, since the wavefunction is defined for a fixed number of particles

$$\Psi(\mathbf{r}_1, \mathbf{r}_2, \ldots, \mathbf{r}_N) = \sum_{\nu_1 \ldots \nu_N} B_{\nu_1 \ldots \nu_N} \, S_\pm \phi_{\nu_1}(\mathbf{r}_1)\phi_{\nu_2}(\mathbf{r}_2) \cdots \phi_{\nu_N}(\mathbf{r}_N),$$

where $S_\pm$ is the symmetrization or antisymmetrization operator, depending on whether we are dealing with bosons or fermions, and $\phi_\nu(\mathbf{r})$ are the single-particle wavefunctions. The coefficients $B_{\nu_1 \ldots \nu_N}$ are the expansion coefficients of the wavefunction in terms of the single-particle states. Thus, the wavefunction is defined for a fixed number of particles N, and it cannot describe processes that involve the creation or annihilation of particles. Furthermore we can write

$$S_\pm \phi_{\nu_1}(\mathbf{r}_1)\phi_{\nu_2}(\mathbf{r}_2) \cdots \phi_{\nu_N}(\mathbf{r}_N) = \begin{pmatrix} \psi_{\nu_1}(\mathbf{r}_1) & \cdots & \psi_{\nu_1}(\mathbf{r}_N) \\ \vdots & \ddots & \vdots \\ \psi_{\nu_N}(\mathbf{r}_1) & \cdots & \psi_{\nu_N}(\mathbf{r}_N) \end{pmatrix}_\pm ,$$

since the symmetrization or antisymmetrization operator can be expressed as a sum over all permutations of the particles, with a plus sign for bosons and a minus sign for fermions, which is equivalent to the determinant or permanent of the matrix of single-particle wavefunctions, depending on whether we are dealing with fermions or bosons.[1] Thus, the wavefunction is defined for a fixed number of particles N, and it cannot describe processes that involve the creation or annihilation of particles.

This is a problem when we want to describe systems that can exchange particles with their environment, such as in the case of open systems or in the case of systems that can undergo particle creation and annihilation, such as in quantum field theory. To overcome this limitation, we need to introduce a new formalism, which is called **second quantization**, which allows us to describe systems with a variable number of particles by introducing creation and annihilation operators.

[1] The permanent of a matrix is the sum over all permutations of the matrix elements, with a plus sign for each permutation (while the determinant has alternating signs).

1.1 | Occupation Number Representation

In the context of second quantization, we can describe the state of a system of particles by specifying the **occupation numbers of the single-particle states** (seen as a set of quantum numbers). The occupation number n_ν of a single-particle state ν is defined as the number of particles that occupy that state.

Thus, we can represent the state of a N-particle system by a vector of occupation numbers:

$$\left|n_{\nu_1}, n_{\nu_2}, \dots, n_{\nu_m}\right\rangle, \quad \sum_\nu n_\nu = N,$$

where n_{ν_i} is the occupation number of the single-particle state ν_i, and the sum of all occupation numbers must be equal to the total number of particles N. This representation is called the **occupation number representation**, and it is a convenient way to describe many-body systems, since it allows us to keep track of the number of particles in each state without having to specify the coordinates of each particle. From this it follows straightforwardly the definition of the number operator $\hat{n}_\nu$, which counts the number of particles in the single-particle state ν:

$$\hat{n}_\nu \left|n_{\nu_1}, n_{\nu_2}, \dots, n_{\nu_m}\right\rangle = n_\nu \left|n_{\nu_1}, n_{\nu_2}, \dots, n_{\nu_m}\right\rangle.$$

For bosons, the occupation number can take any non-negative integer value, while for fermions, the occupation number can only be 0 or 1, due to the Pauli exclusion principle:

$$n_\nu = \begin{cases} 0, 1, 2, \dots & \text{for bosons,} \\ 0, 1 & \text{for fermions.} \end{cases}$$

In order to use this formalism, we need to introduce the analogue of the Hilbert space for the variable number of particles, which is called the **Fock space**. The Fock space is the direct sum of the Hilbert spaces for all possible numbers of particles:

$$\mathcal{F} = \bigoplus_{N=0}^{\infty} \mathcal{H}_N,$$

where $\mathcal{H}_N$ is the Hilbert space for N particles. The Fock space allows us to describe states with a variable number of particles, and it is the natural setting for the second quantization formalism. In the Fock space, we can define the vacuum state $|0\rangle$, which is the state with no particles, and we can build all other states by applying creation operators to the vacuum state. But before let us make an example.

Example (Fock space basis). Let us consider a system of bosons with varying number of particles, and let us choose a single-particle basis with two states, which we can label as $|1\rangle$ and $|2\rangle$. The Fock space for this system is the direct sum of the Hilbert spaces for all possible numbers of particles:

$$\mathcal{F} = \bigoplus_{N=0}^{\infty} \mathcal{H}_N.$$

We can show some examples of basis for the Fock space:

$$\begin{aligned} \mathcal{H}_0 &= \left|0, 0, \dots, 0\right\rangle, \\ \mathcal{H}_1 &= \left|1, 0, \dots, 0\right\rangle, \left|0, 1, \dots, 0\right\rangle, \dots, \left|0, 0, \dots, 1\right\rangle, \\ \mathcal{H}_2 &= \left|1, 1, 0, \dots, 0\right\rangle, \dots, \left|1, 0, \dots, 0, 1, 0, \dots, 0\right\rangle, \left|2, 0, \dots, 0\right\rangle, \dots \end{aligned}$$

where the first state is the vacuum state, the second states are the single-particle states, and the third states are the two-particle states. We can see that in the occupation number representation, we can easily keep track of the number of particles in each state, and we can build all possible states by permuting the occupation numbers.

Note that for a system of fermions, the basis states would be different, since the occupation numbers can only be 0 or 1, so we would not have states with more than one particle in the same state: the last state in the two-particle Hilbert space would not be allowed, since it would violate the Pauli exclusion principle.

Now it is time for us to introduce the creation and annihilation operators, which are the fundamental building blocks of the second quantization formalism.

1.2 | Ladder Operators

Ladder operators are operators that can raise or lower the occupation number of a single-particle state. We will separate the definition of the ladder operators for bosons and fermions, since they satisfy different commutation or anticommutation relations.

1.2.1 | Bosons

For bosons, we define the **creation operator** $\hat{b}_\nu^\dagger$ and the **annihilation operator** $\hat{b}_\nu$ for a single-particle state ν as follows:

$$\begin{cases} \hat{b}_\nu \left|n_{\nu_1}, n_{\nu_2}, \ldots, n_\nu, \ldots\right\rangle = B_-(n_\nu) \left|n_{\nu_1}, n_{\nu_2}, \ldots, n_\nu - 1, \ldots\right\rangle, \\ \hat{b}_\nu^\dagger \left|n_{\nu_1}, n_{\nu_2}, \ldots, n_\nu, \ldots\right\rangle = B_+(n_\nu) \left|n_{\nu_1}, n_{\nu_2}, \ldots, n_\nu + 1, \ldots\right\rangle, \end{cases}$$

where $B_-(n_\nu)$ and $B_+(n_\nu)$ are coefficients that depend on the occupation number n_ν. The creation operator $\hat{b}_\nu^\dagger$ increases the occupation number of the state ν by one, while the annihilation operator $\hat{b}_\nu$ decreases the occupation number of the state ν by one. For bosons, these operators satisfy the following commutation relations:

$$\begin{cases} [\hat{b}_\nu, \hat{b}_{\nu'}^\dagger] = \delta_{\nu\nu'}, \\ [\hat{b}_\nu, \hat{b}_{\nu'}] = [\hat{b}_\nu^\dagger, \hat{b}_{\nu'}^\dagger] = 0, \end{cases}$$

where $[A, B] = AB - BA$ is the commutator of the operators A and B. The commutation relations reflect the fact that bosons can occupy the same state, and that the order of the operators does not matter.

Let us apply these operators to the occupation number states to find the coefficients $B_-(n_\nu)$ and $B_+(n_\nu)$, starting from the vacuum state $\left|0, 0, \ldots, 0, \ldots\right\rangle$:

$$\begin{aligned} &\hat{b}_\nu \left|0, 0, \ldots, 0, \ldots\right\rangle = 0, && B_-(0) = 0, \\ &\hat{b}_\nu^\dagger \left|0, 0, \ldots, 0, \ldots\right\rangle = \left|0, 0, \ldots, 0, 1, 0, \ldots\right\rangle, && B_+(0) = 1, \end{aligned}$$

where we understood how the annihilation operator gives zero when applied to the vacuum state, while the creation operator creates a particle in the state ν.

Now let us apply the operators to a generic normalized occupation number state $\left|\phi\right\rangle$:

$$\hat{b}_\nu \left|\phi\right\rangle, \quad \left\langle\phi\right| \hat{b}_\nu^\dagger \hat{b}_\nu \left|\phi\right\rangle = |\hat{b}_\nu \left|\phi\right\rangle| = \lambda > 0,$$

such that if we apply the creation operator to the state $\hat{b}_\nu \left|\phi\right\rangle$ we get:

$$\hat{b}_\nu^\dagger \hat{b}_\nu \left|\phi\right\rangle = \lambda \left|\phi\right\rangle = n_\nu \left|\phi\right\rangle,$$

since the number operator $\hat{n}_\nu = \hat{b}_\nu^\dagger \hat{b}_\nu$ counts the number of particles in the state ν . Now if we apply the number operator $\hat{n}_\nu = \hat{b}_\nu^\dagger \hat{b}_\nu$ to the state $\hat{b}_\nu \left|\phi\right\rangle$, using the commutation relations, we should gbe able to find the coefficient $B_-(n_\nu)$:

$$\hat{n}_\nu \hat{b}_\nu \left|\phi\right\rangle = (\hat{b}_\nu^\dagger \hat{b}_\nu) \hat{b}_\nu \left|\phi\right\rangle = (\hat{b}_\nu \hat{b}_\nu^\dagger - 1) \hat{b}_\nu \left|\phi\right\rangle = \hat{b}_\nu (\hat{b}_\nu^\dagger \hat{b}_\nu - 1) \left|\phi\right\rangle = (\lambda - 1) \hat{b}_\nu \left|\phi\right\rangle,$$

so that we can identify the coefficient $B_-(n_\nu)$ as:

$$B_-(n_\nu) = \sqrt{n_\nu},$$

and by applying the number operator to the state $\hat{b}_\nu^\dagger |\phi\rangle$, we get:

$$\hat{n}_\nu \hat{b}_\nu^\dagger |\phi\rangle = (\hat{b}_\nu^\dagger \hat{b}_\nu)\hat{b}_\nu^\dagger |\phi\rangle = \hat{b}_\nu^\dagger(\hat{b}_\nu \hat{b}_\nu^\dagger) |\phi\rangle = \hat{b}_\nu^\dagger(\hat{b}_\nu^\dagger \hat{b}_\nu + 1) |\phi\rangle = (\lambda + 1)\hat{b}_\nu^\dagger |\phi\rangle ,$$

so that we can identify the coefficient $B_+(n_\nu)$ as:

$$B_+(n_\nu) = \sqrt{n_\nu + 1}.$$

Thus, the action of the creation and annihilation operators on the occupation number states can be expressed as:

$$\begin{cases} \hat{b}_\nu |n_{\nu_1}, n_{\nu_2}, \ldots, n_\nu, \ldots\rangle = \sqrt{n_\nu} |n_{\nu_1}, n_{\nu_2}, \ldots, n_\nu - 1, \ldots\rangle , \\ \hat{b}_\nu^\dagger |n_{\nu_1}, n_{\nu_2}, \ldots, n_\nu, \ldots\rangle = \sqrt{n_\nu + 1} |n_{\nu_1}, n_{\nu_2}, \ldots, n_\nu + 1, \ldots\rangle . \end{cases}$$

1.2.2 | Fermions

For fermions, we define the **creation operator** $\hat{c}_\nu^\dagger$ and the **annihilation operator** $\hat{c}_\nu$ for a single-particle state ν with the same action on the occupation number states as for bosons, but with the coefficients $B_-(n_\nu)$ and $B_+(n_\nu)$ that depend on the occupation number n_ν in a different way, since for fermions the occupation number can only be 0 or 1:

$$\begin{cases} \hat{c}_\nu |n_{\nu_1}, n_{\nu_2}, \ldots, n_\nu, \ldots\rangle = C_-(n_\nu) |n_{\nu_1}, n_{\nu_2}, \ldots, n_\nu - 1, \ldots\rangle , \\ \hat{c}_\nu^\dagger |n_{\nu_1}, n_{\nu_2}, \ldots, n_\nu, \ldots\rangle = C_+(n_\nu) |n_{\nu_1}, n_{\nu_2}, \ldots, n_\nu + 1, \ldots\rangle , \end{cases}$$

with

$$C_-(n_\nu) = \begin{cases} 0 & \text{if } n_\nu = 0, \\ 1 & \text{if } n_\nu = 1, \end{cases} \quad C_+(n_\nu) = \begin{cases} 1 & \text{if } n_\nu = 0, \\ 0 & \text{if } n_\nu = 1, \end{cases}$$

so that the creation operator $\hat{c}_\nu^\dagger$ creates a particle in the state ν if it is not already occupied, while the annihilation operator $\hat{c}_\nu$ annihilates a particle in the state ν if it is occupied. For fermions, these operators satisfy the following anticommutation relations:

$$\begin{cases} \{\hat{c}_\nu, \hat{c}_\nu^\dagger\} = 1, \\ \{\hat{c}_\nu, \hat{c}_\nu\} = \{\hat{c}_\nu^\dagger, \hat{c}_\nu^\dagger\} = 0, \end{cases}$$

since the multiparticle wavefunction must be antisymmetric under the exchange of any two particles, and thus the order of the operators matters. The anticommutation relations reflect the fact that fermions cannot occupy the same state:

$$\hat{c}_\nu^\dagger \hat{c}_\nu^\dagger = \frac{1}{2}\{\hat{c}_\nu^\dagger, \hat{c}_\nu^\dagger\} = 0,$$

and similarly for the annihilation operator.

1.3 | Hamiltonian of an Interacting System

In a many-body system, the Hamiltonian has to contain both a kinetic term, which describes the motion of the particles, and an interaction term, which describes the interactions between the particles. Let us show how to write the Hamiltonian of an interacting system in the second quantization formalism, starting from the first quantization formalism. We will see how many results from first quantization come in handy when it will be time to build the second quantization version.

First Quantization

In first quantization, the Hamiltonian of a system of N interacting particles can be written as:

$$H = \sum_{i=1}^{N} \hat{T}(\mathbf{r}_i, \nabla_{\mathbf{r}_i}) + \frac{1}{2}\sum_{i\neq j} \hat{V}(\mathbf{r}_i - \mathbf{r}_j),$$

where $\hat{T}(\mathbf{r}_i, \nabla_{\mathbf{r}_i})$ is the **one-body operator** for the i-th particle, and $\hat{V}(\mathbf{r}_i - \mathbf{r}_j)$ is the interaction potential, or **two body operator**, between the i-th and j-th particles. The factor of $\frac{1}{2}$ is included to avoid double counting the interactions between pairs of particles.

The easiest case in which we can write the Hamiltonian of a many-body system is the Helium atom.

> **Example (Helium atom).** Let us consider the helium atom, which consists of two electrons interacting with each other and with the nucleus. The Hamiltonian of the helium atom can be written as:
>
> $$H = \left[-\frac{\hbar^2}{2m}\nabla^2_{\mathbf{r}_1} - \frac{Ze^2}{4\pi\epsilon_0 r_1}\right] + \left[-\frac{\hbar^2}{2m}\nabla^2_{\mathbf{r}_2} - \frac{Ze^2}{4\pi\epsilon_0 r_2}\right] + \frac{e^2}{4\pi\epsilon_0|\mathbf{r}_1 - \mathbf{r}_2|},$$
>
> where m is the mass of the electron, Z is the atomic number of the nucleus (which is 2 for helium), e is the elementary charge, ϵ_0 is the vacuum permittivity and r_i is the distance of the i-th electron from the nucleus. The first term in square brackets represent the contribution of the kinetic energy and the potential energy of the first electron, while the second term in square brackets for the second electron; the last term represent the interaction between the two electrons, which is the **electron-electron interaction**. This Hamiltonian is a good starting point for understanding the electronic structure of the helium atom, and it can be solved using various approximation methods, such as perturbation theory or variational methods.

Let us study more in detail the structure of the one-body and two-body operators.

One-body operator. The one-body operator $\hat{T}(\mathbf{r}_j, \nabla_{\mathbf{r}_j})$ is an operator that acts on the coordinates of a single particle, and it typically includes the kinetic energy term and the potential energy term due to external fields. It can be expressed in terms of the single-particle states as:

$$T_{\nu_a\nu_b} = \int \mathrm{d}^3\mathbf{r}_j\, \Psi^*_{\nu_a}(\mathbf{r}_j)\hat{T}(\mathbf{r}_j, \nabla_{\mathbf{r}_j})\Psi_{\nu_b}(\mathbf{r}_j),$$

where $\Psi_\nu(\mathbf{r}_j)$ are the single-particle wavefunctions, and the integral is taken over all space. The internal operator $\hat{T}(\mathbf{r}_j, \nabla_{\mathbf{r}_j})$ can be expressed as the sum of the kinetic energy operator and the potential energy operator:

$$\hat{T}(\mathbf{r}_j, \nabla_{\mathbf{r}_j}) = -\frac{\hbar^2}{2m}\nabla^2_{\mathbf{r}_j} + V_{\mathrm{ext}}(\mathbf{r}_j),$$

and since $T_{\nu_a\nu_b}$ is the matrix element of the one-body operator between the single-particle states ν_a and ν_b, we can write:

$$\hat{T}(\mathbf{r}_j, \nabla_{\mathbf{r}_j}) = \sum_{\nu_a\nu_b} T_{\nu_a\nu_b} \left|\Psi_{\nu_b}(\mathbf{r}_j)\right\rangle \left\langle\Psi_{\nu_a}(\mathbf{r}_j)\right| ,$$

where $|\Psi_{\nu_a}(\mathbf{r}_j)\rangle$ and $|\Psi_{\nu_b}(\mathbf{r}_j)\rangle$ are the single-particle states corresponding to the quantum numbers ν_a and ν_b, respectively.

Two-body operator. The two-body operator $\hat{V}(\mathbf{r}_j - \mathbf{r}_k)$ is an operator that acts on the coordinates of two particles, and it typically includes the interaction potential between the particles. It can be expressed in terms of the single-particle states as:

$$V_{\nu_c\nu_d,\nu_a\nu_b} = \int \mathrm{d}^3\mathbf{r}_j\, \mathrm{d}^3\mathbf{r}_k\, \Psi^*_{\nu_c}(\mathbf{r}_j)\Psi^*_{\nu_d}(\mathbf{r}_k)\hat{V}(\mathbf{r}_j - \mathbf{r}_k)\Psi_{\nu_a}(\mathbf{r}_j)\Psi_{\nu_b}(\mathbf{r}_k),$$

where again $\Psi_\nu(\mathbf{r}_j)$ are the single-particle wavefunctions, and the integral is taken over all space. The internal operator $\hat{V}(\mathbf{r}_j - \mathbf{r}_k)$ can then be expressed as a sum over the single-particle states as:

$$\hat{V}(\mathbf{r}_j - \mathbf{r}_k) = \sum_{\nu_a\nu_b\nu_c\nu_d} V_{\nu_c\nu_d,\nu_a\nu_b} \left|\Psi_{\nu_c}(\mathbf{r}_j)\right\rangle \left|\Psi_{\nu_d}(\mathbf{r}_k)\right\rangle \left\langle\Psi_{\nu_b}(\mathbf{r}_k)\right| \left\langle\Psi_{\nu_a}(\mathbf{r}_j)\right| ,$$

where $V_{\nu_c\nu_d,\nu_a\nu_b}$ are the matrix elements of the two-body operator between the single-particle states.

Now that we have expressed the one-body and two-body operators in terms of the single-particle states, we can rewrite the Hamiltonian of the system in terms of these operators as:

$$H = \sum_{i=1}^{N}\sum_{\nu_a\nu_b} T_{\nu_a\nu_b} \left|\Psi_{\nu_b}(\mathbf{r}_i)\right\rangle \left\langle\Psi_{\nu_a}(\mathbf{r}_i)\right| + \frac{1}{2}\sum_{i\neq j}\sum_{\nu_a\nu_b\nu_c\nu_d} V_{\nu_c\nu_d,\nu_a\nu_b} \left|\Psi_{\nu_c}(\mathbf{r}_i)\right\rangle \left|\Psi_{\nu_d}(\mathbf{r}_j)\right\rangle \left\langle\Psi_{\nu_b}(\mathbf{r}_j)\right| \left\langle\Psi_{\nu_a}(\mathbf{r}_i)\right| ,$$

where we have expressed the Hamiltonian in terms of the matrix elements of the one-body and two-body operators between the single-particle states. This Hamiltonian is still in a first quantization form, since it is expressed in terms of the coordinates of the particles, but it is a convenient starting point for expressing the Hamiltonian in terms of the creation and annihilation operators in the second quantization formalism.

Second Quantization

In order to express the Hamiltonian in terms of the creation and annihilation operators, we need to understand how to express the operators in second quantization. The key idea is to express the operators in terms of the ladder operators, which can be done by using the completeness relation of the single-particle states.

It can be shown that an operator in first quantization can be expressed in second quantization as:

$$\hat{A} = \sum_{\nu_a\nu_b} A_{\nu_a\nu_b}\hat{a}^\dagger_{\nu_a}\hat{a}_{\nu_b},$$

where $A_{\nu_a\nu_b}$ are the matrix elements of the operator in the single-particle basis, and $\hat{a}^\dagger_\nu$ and $\hat{a}_\nu$ are the creation and annihilation operators for the single-particle state ν, respectively.[2] Thus there exists a systematic way to express any operator in first quantization in terms of the creation

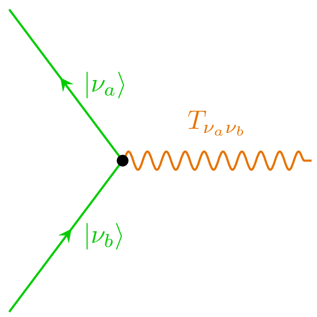
[2]To be substitued by $\hat{b}$ and $\hat{c}$ according to the statistics of the particles.

and annihilation operators in second quantization, by expressing the operator in terms of its matrix elements in the single-particle basis, and then replacing the single-particle states with the corresponding creation and annihilation operators.

We can then write the Hamiltonian of the system in second quantization as:

$$H = \sum_{\nu_a \nu_b} T_{\nu_a \nu_b} \hat{a}^\dagger_{\nu_a} \hat{a}_{\nu_b} + \frac{1}{2} \sum_{\nu_a \nu_b \nu_c \nu_d} V_{\nu_c \nu_d, \nu_a \nu_b} \hat{a}^\dagger_{\nu_c} \hat{a}^\dagger_{\nu_d} \hat{a}_{\nu_a} \hat{a}_{\nu_b},$$

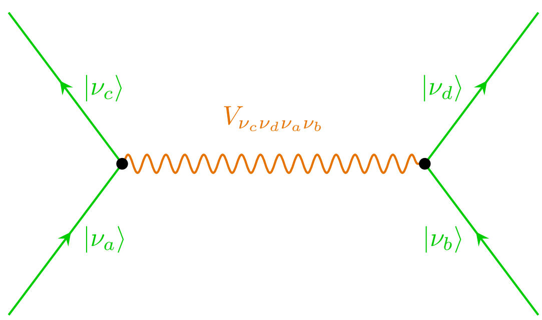
where the first term represents the one-body interactions, and the second term represents the two-body interactions between the particles. This Hamiltonian is now expressed in terms of the creation and annihilation operators, and it can be used to describe systems with a variable number of particles, since the creation and annihilation operators can create or annihilate particles in the system.



In the diagram on the left, we can see the one-body interaction between the single-particle states $|\nu_a\rangle$ and $|\nu_b\rangle$, which is represented by the matrix element $T_{\nu_a \nu_b}$. The particle represented by the line entering the vertex is in the state $|\nu_b\rangle$, while the particle represented by the line exiting the vertex is in the state $|\nu_a\rangle$. The interaction is represented by a wavy line, which indicates that it is a one-body interaction, since it involves only one particle.

The vertex represents the point where the interaction takes place, and it is associated with the matrix element $T_{\nu_a \nu_b}$, which quantifies the strength of the interaction between the two states, or else the probability amplitude for the transition from state $|\nu_b\rangle$ to state $|\nu_a\rangle$ due to the one-body operator. Basically we are annihilating a particle in the state $|\nu_b\rangle$ and creating a particle in the state $|\nu_a\rangle$ due to the one-body interaction.

For the two body interaction, represented in the diagram on the right, we have two vertices representing the interaction between the single-particle states $|\nu_a\rangle$, $|\nu_b\rangle$, $|\nu_c\rangle$ and $|\nu_d\rangle$. The particles represented by the lines entering the vertices are in the states $|\nu_a\rangle$ and $|\nu_b\rangle$, while the particles represented by the lines exiting the vertices are in the states $|\nu_c\rangle$ and $|\nu_d\rangle$.



The interaction is represented by a wavy line connecting the two vertices, which indicates that it is a two-body interaction, since it involves two particles. The matrix element $V_{\nu_c \nu_d \nu_a \nu_b}$ quantifies the strength of the interaction between the four states, or else the probability amplitude for the transition from states $|\nu_a\rangle$ and $|\nu_b\rangle$ to states $|\nu_c\rangle$ and $|\nu_d\rangle$ due to the two-body operator. We are describing a process in which we are annihilating two particles in the states $|\nu_a\rangle$ and $|\nu_b\rangle$ and creating two particles in the states $|\nu_c\rangle$ and $|\nu_d\rangle$ due to the two-body interaction.

1.4 | Field Operators

Field operators are a powerful tool in quantum mechanics, particularly in the study of many-body systems. They allow us to describe the creation and annihilation of particles in a given state. The field operator $\hat{\Psi}(\mathbf{r})$ can be expressed as a sum over a complete set of single-particle states:

$$\hat{\Psi}(\mathbf{r}) = \sum_{\nu} \phi_{\nu}(\mathbf{r})\hat{a}_{\nu},$$
$$\hat{\Psi}^{\dagger}(\mathbf{r}) = \sum_{\nu} \phi^{*}_{\nu}(\mathbf{r})\hat{a}^{\dagger}_{\nu},$$

where $\phi_{\nu}(\mathbf{r})$ are the single-particle wavefunctions on the phase space, and $\hat{a}_{\nu}$ and $\hat{a}^{\dagger}_{\nu}$ are the annihilation and creation operators for the single-particle state ν, respectively.[3] The field operator $\hat{\Psi}(\mathbf{r})$ annihilates a particle at position $\mathbf{r}$, while the field operator $\hat{\Psi}^{\dagger}(\mathbf{r})$ creates a particle at position $\mathbf{r}$; they are defined in such a way that they satisfy the appropriate commutation or anticommutation relations, depending on whether we are dealing with bosons or fermions.

If we have an **homogeneous system**, we can choose plane waves as our single-particle states to sum only over momentum states, moving to the **momentum space** representation. In this case, the field operators can be expressed as:

$$\hat{\Psi}(\mathbf{r}) = \frac{1}{\sqrt{\Omega}} \sum_{\mathbf{k}} e^{-i\mathbf{k}\cdot\mathbf{r}}\hat{a}_{\mathbf{k}},$$
$$\hat{\Psi}^{\dagger}(\mathbf{r}) = \frac{1}{\sqrt{\Omega}} \sum_{\mathbf{k}} e^{i\mathbf{k}\cdot\mathbf{r}}\hat{a}^{\dagger}_{\mathbf{k}},$$

we are summing over all quantum numbers ν, which in this case are the momentum states $\mathbf{k}$, then we are creating or annihilating particles in all of those momentum states for a precise position $\mathbf{r}$. Since this operators are based on the same annihilation and creation operators, they satisfy the same commutation or anticommutation relations, depending on whether we are dealing with bosons or fermions. For bosons, the field operators satisfy the commutation relations:

$$\begin{cases} [\hat{\Psi}(\mathbf{r}), \hat{\Psi}^{\dagger}(\mathbf{r}')] = \delta(\mathbf{r}-\mathbf{r}'), \\ [\hat{\Psi}(\mathbf{r}), \hat{\Psi}(\mathbf{r}')] = [\hat{\Psi}^{\dagger}(\mathbf{r}), \hat{\Psi}^{\dagger}(\mathbf{r}')] = 0, \end{cases}$$

while for fermions, they satisfy the anticommutation relations:

$$\begin{cases} \{\hat{\Psi}(\mathbf{r}), \hat{\Psi}^{\dagger}(\mathbf{r}')\} = \delta(\mathbf{r}-\mathbf{r}'), \\ \{\hat{\Psi}(\mathbf{r}), \hat{\Psi}(\mathbf{r}')\} = \{\hat{\Psi}^{\dagger}(\mathbf{r}), \hat{\Psi}^{\dagger}(\mathbf{r}')\} = 0, \end{cases}$$

derived from the commutation or anticommutation relations of the ladder operators. Instead of wavefunctions, we can also express the field operators in terms of the position basis.

Interacting Hamiltonian in Terms of Field Operators

Let us now express the Hamiltonian of an interacting system in terms of the field operators. Starting from the Hamiltonian in second quantization, derived in the previous section, it can be expressed as:

$$\hat{H} = \sum_{\nu_a, \nu_b} \hat{T}_{\nu_a\nu_b}\hat{a}^{\dagger}_{\nu_a}\hat{a}_{\nu_b} + \frac{1}{2} \sum_{\nu_a, \nu_b, \nu_c, \nu_d} \hat{V}_{\nu_c\nu_d\nu_a\nu_b}\hat{a}^{\dagger}_{\nu_c}\hat{a}^{\dagger}_{\nu_d}\hat{a}_{\nu_a}\hat{a}_{\nu_b},$$

[3] Here we use generic ladder operators $\hat{a}$, but according to the context, we might use $\hat{c}$ for electrons or $\hat{b}$ for bosons, along with the respective commutation relations.

where we are summing over all quantum numbers ν. Now rewriting it in terms of the field operators is a straightforward task, since we can express the one-body and two-body operators in a more explicit way as:

$$\begin{aligned}\hat{H} &= \sum_{\nu_a,\nu_b} \int \mathrm{d}^3\mathbf{r}\, \left(\Psi^*_{\nu_a}(\mathbf{r})T(\mathbf{r},\nabla_\mathbf{r})\Psi_{\nu_b}(\mathbf{r})\right) \hat{a}^\dagger_{\nu_a}\hat{a}_{\nu_b} \\ &+ \frac{1}{2}\sum_{\nu_a,\nu_b,\nu_c,\nu_d} \left(\int \mathrm{d}^3\mathbf{r}\,\mathrm{d}^3\mathbf{r}'\,\Psi^*_{\nu_c}(\mathbf{r})\Psi^*_{\nu_d}(\mathbf{r}')V(\mathbf{r}-\mathbf{r}')\Psi_{\nu_a}(\mathbf{r})\Psi_{\nu_b}(\mathbf{r}')\right) \hat{a}^\dagger_{\nu_c}\hat{a}^\dagger_{\nu_d}\hat{a}_{\nu_a}\hat{a}_{\nu_b},\end{aligned}$$

from which is now straightforward to recognize the expressions of the field operators (with $\Psi_\nu(\mathbf{r})$ the single-particle wavefunctions), so that we can rewrite the Hamiltonian as:

$$\hat{H} = \int \mathrm{d}^3\mathbf{r}\,\hat{\Psi}^\dagger(\mathbf{r})T(\mathbf{r},\nabla_\mathbf{r})\hat{\Psi}(\mathbf{r}) + \frac{1}{2}\int \mathrm{d}^3\mathbf{r}\,\mathrm{d}^3\mathbf{r}'\,\hat{\Psi}^\dagger(\mathbf{r})\hat{\Psi}^\dagger(\mathbf{r}')V(\mathbf{r}-\mathbf{r}')\hat{\Psi}(\mathbf{r}')\hat{\Psi}(\mathbf{r}),$$

where the first term represents the kinetic energy and the potential energy due to external fields, while the second term represents the interaction energy between the particles. This Hamiltonian is now expressed in terms of the field operators, and it can be used to describe systems with a variable number of particles, since the field operators can create or annihilate particles in a specific state in the system.

Let us now analyze the case in which the one-body operator is given by the kinetic term, and the two-body operator is given by the Coulomb potential, which is a common case in many-body physics.

Free kinetic term. The kinetic term of the Hamiltonian can be expressed as:

- in **first quantization**, the kinetic term is given by:

  $$\hat{T}(\mathbf{r},\nabla_\mathbf{r}) = -\frac{\hbar^2}{2m}\nabla^2_\mathbf{r}\delta_{\sigma\sigma'},$$

  where we are using the plane wave basis $|\mathbf{k},\sigma\rangle$, such that

  $$\langle \mathbf{r}|\mathbf{k},\sigma\rangle = \frac{1}{\sqrt{\Omega}}e^{i\mathbf{k}\cdot\mathbf{r}},$$

  and if we compute the matrix element of the kinetic term, we get:

  $$\langle \mathbf{k}_a,\sigma_a|\,\hat{T}(\mathbf{r},\nabla_\mathbf{r})\,|\mathbf{k}_b,\sigma_b\rangle = \frac{\hbar^2k_b^2}{2m}\delta_{\mathbf{k}_a\mathbf{k}_b}\delta_{\sigma_a\sigma_b},$$

  since the plane waves are eigenstates of the kinetic energy operator (the laplacian is diagonal), and the eigenvalue is given by $\frac{\hbar^2k^2}{2m}$. The delta functions ensure that the kinetic term only contributes when the momentum and spin states are the same.

- in **second quantization**, the kinetic term can be expressed as:

  $$\hat{T} = \sum_{\nu_a,\nu_b} T_{\nu_a\nu_b}\hat{a}^\dagger_{\nu_a}\hat{a}_{\nu_b} = \sum_{\mathbf{k},\mathbf{k}',\sigma,\sigma'} \langle \mathbf{k},\sigma|\,\hat{T}(\mathbf{r},\nabla_\mathbf{r})\,|\mathbf{k}',\sigma'\rangle\,\hat{a}^\dagger_{\mathbf{k},\sigma}\hat{a}_{\mathbf{k}',\sigma'} = \sum_{\mathbf{k},\sigma}\frac{\hbar^2k^2}{2m}\hat{a}^\dagger_{\mathbf{k},\sigma}\hat{a}_{\mathbf{k},\sigma},$$

  where we have used the fact that the kinetic term is diagonal in the plane wave basis, so that the only non-zero contributions come from terms where $\mathbf{k} = \mathbf{k}'$ and $\sigma = \sigma'$. Indeed, we can see that in the last term the kinetic term is expressed as a sum over the momentum states, with the eigenvalue of the kinetic energy operator as the coefficient, and the number operator for the corresponding momentum states:

  $$\hat{T} = \sum_{\mathbf{k},\sigma}\frac{\hbar^2k^2}{2m}\hat{n}_{\mathbf{k},\sigma}.$$

Coulomb interaction term. Starting directly from the second quantized expression of the Coulomb term of the Hamiltonian, we can express it as:

$$V = \frac{1}{2}\sum_{\sigma,\sigma'}\sum_{\mathbf{k}_1,\mathbf{k}_2,\mathbf{k}_3,\mathbf{k}_4} \langle \mathbf{k}_3,\sigma;\mathbf{k}_4,\sigma'|\, V(\mathbf{r}-\mathbf{r}')\, |\mathbf{k}_2,\sigma';\mathbf{k}_1,\sigma\rangle\, \hat{a}^\dagger_{\mathbf{k}_3,\sigma}\hat{a}^\dagger_{\mathbf{k}_4,\sigma'}\hat{a}_{\mathbf{k}_2,\sigma'}\hat{a}_{\mathbf{k}_1,\sigma},$$

where we are summing over all momentum and spin states, and the matrix element of the Coulomb potential can be expressed using the normalized plane wave basis as:

$$V = \frac{1}{2}\sum_{\sigma,\sigma'}\sum_{\mathbf{k}_1,\mathbf{k}_2,\mathbf{k}_3,\mathbf{k}_4}\left[\int\!\!\int \mathrm{d}^3\mathbf{r}\,\mathrm{d}^3\mathbf{r}'\,\frac{e^{-i\mathbf{k}_3\cdot\mathbf{r}}}{\sqrt{\Omega}}\frac{e^{-i\mathbf{k}_4\cdot\mathbf{r}'}}{\sqrt{\Omega}}V(\mathbf{r}-\mathbf{r}')\frac{e^{i\mathbf{k}_2\cdot\mathbf{r}'}}{\sqrt{\Omega}}\frac{e^{i\mathbf{k}_1\cdot\mathbf{r}}}{\sqrt{\Omega}}\right]\hat{a}^\dagger_{\mathbf{k}_3,\sigma}\hat{a}^\dagger_{\mathbf{k}_4,\sigma'}\hat{a}_{\mathbf{k}_2,\sigma'}\hat{a}_{\mathbf{k}_1,\sigma},$$

and now, exploiting the form of the Coulomb potential, we can express the matrix element as:

$$V = \frac{1}{2}\sum_{\sigma,\sigma'}\sum_{\mathbf{k}_1,\mathbf{k}_2,\mathbf{k}_3,\mathbf{k}_4}\frac{e^2}{\Omega^2}\left[\int\!\!\int \mathrm{d}^3\mathbf{r}\,\mathrm{d}^3\mathbf{r}'\,\frac{e^{i(\mathbf{k}_1-\mathbf{k}_3)\cdot\mathbf{r}}e^{i(\mathbf{k}_2-\mathbf{k}_4)\cdot\mathbf{r}'}}{|\mathbf{r}-\mathbf{r}'|}\right]\hat{a}^\dagger_{\mathbf{k}_3,\sigma}\hat{a}^\dagger_{\mathbf{k}_4,\sigma'}\hat{a}_{\mathbf{k}_2,\sigma'}\hat{a}_{\mathbf{k}_1,\sigma}.$$

We can then perform the change of variables $\boldsymbol{\xi} = \mathbf{r}-\mathbf{r}'$, so that the integral can be expressed as:

$$V = \frac{1}{2}\sum_{\sigma,\sigma'}\sum_{\mathbf{k}_1,\mathbf{k}_2,\mathbf{k}_3,\mathbf{k}_4}\frac{e^2}{\Omega^2}\left[\int \mathrm{d}^3\mathbf{r}\,e^{i(\mathbf{k}_1-\mathbf{k}_3+\mathbf{k}_2-\mathbf{k}_4)\cdot\mathbf{r}}\int \mathrm{d}^3\boldsymbol{\xi}\,e^{i(\mathbf{k}_2-\mathbf{k}_4)\cdot(\mathbf{r}-\boldsymbol{\xi})}\frac{1}{|\boldsymbol{\xi}|}\right]\hat{a}^\dagger_{\mathbf{k}_3,\sigma}\hat{a}^\dagger_{\mathbf{k}_4,\sigma'}\hat{a}_{\mathbf{k}_2,\sigma'}\hat{a}_{\mathbf{k}_1,\sigma},$$

where we have separated the integral over $\mathbf{r}$ and the integral over $\boldsymbol{\xi}$. Now the equation is basically telling us to rename $\mathbf{k}_2-\mathbf{k}_4$ as $\mathbf{q}$:

$$V = \frac{1}{2}\sum_{\sigma,\sigma'}\sum_{\mathbf{k}_1,\mathbf{k}_2,\mathbf{k}_3,\mathbf{q}}\frac{e^2}{\Omega^2}\left[\int \mathrm{d}^3\mathbf{r}\,e^{i(\mathbf{k}_1-\mathbf{k}_3+\mathbf{q})\cdot\mathbf{r}}\int \mathrm{d}^3\boldsymbol{\xi}\,e^{i\mathbf{q}\cdot(\mathbf{r}-\boldsymbol{\xi})}\frac{1}{|\boldsymbol{\xi}|}\right]\hat{a}^\dagger_{\mathbf{k}_3,\sigma}\hat{a}^\dagger_{\mathbf{k}_2-\mathbf{q},\sigma'}\hat{a}_{\mathbf{k}_2,\sigma'}\hat{a}_{\mathbf{k}_1,\sigma},$$

where we have also relabeled the indices of the creation and annihilation operators according to the change of variables. In the end we found the *Fourier transform of the Coulomb potential*:

$$\tilde{V} = \int \mathrm{d}^3\boldsymbol{\xi}\,\frac{e^{i\mathbf{q}\cdot\boldsymbol{\xi}}}{|\boldsymbol{\xi}|} = \frac{4\pi}{|\mathbf{q}|^2}.$$

As we can see, the Coulomb potential is a very **long range potential**, since its transform as $\mathbf{q}\to 0$ diverges:

$$\lim_{\mathbf{q}\to 0}\tilde{V} = \frac{4\pi}{|\mathbf{q}|^2},$$

since it is not screened by the presence of other particles in the system, and it can have a significant effect on its properties, even at large distances. In the end, integrating over $\mathbf{r}$ we get

$$V = \frac{1}{2}\sum_{\sigma,\sigma'}\sum_{\mathbf{k}_1,\mathbf{k}_3,\mathbf{q}}\frac{e^2}{\Omega^2}\Omega\,\delta_{\mathbf{k}_3,\mathbf{k}_1+\mathbf{q}}\,\hat{a}^\dagger_{\mathbf{k}_3,\sigma}\hat{a}^\dagger_{\mathbf{k}_2-\mathbf{q},\sigma'}\hat{a}_{\mathbf{k}_2,\sigma'}\hat{a}_{\mathbf{k}_1,\sigma},$$

where the delta function enforces the **conservation of momentum** during the interaction, so that the total momentum before and after the interaction is the same. Finally, we can sum over the delta function, so that we can express the Coulomb term as:

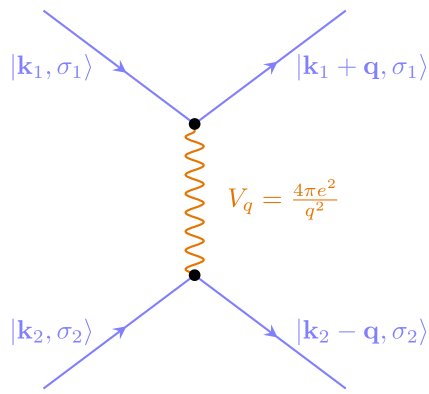
$$V = \frac{1}{2\Omega}\sum_{\sigma,\sigma'}\sum_{\mathbf{k}_1,\mathbf{k}_2,\mathbf{q}}\frac{4\pi e^2}{|\mathbf{q}|^2}\hat{a}^\dagger_{\mathbf{k}_1+\mathbf{q},\sigma}\hat{a}^\dagger_{\mathbf{k}_2-\mathbf{q},\sigma'}\hat{a}_{\mathbf{k}_2,\sigma'}\hat{a}_{\mathbf{k}_1,\sigma}.$$

Thus during the interaction, two particles with momenta $\mathbf{k}_1$ and $\mathbf{k}_2$ exchange a momentum $\mathbf{q}$ and end up with momenta $\mathbf{k}_1+\mathbf{q}$ and $\mathbf{k}_2-\mathbf{q}$, so that the total momentum is conserved during the interaction.[4] This is a consequence of the translational invariance of the system, which implies that the total momentum is a good quantum number:

$$\mathbf{k}_3 + \mathbf{k}_4 = (\mathbf{k}_1 + \mathbf{q}) + (\mathbf{k}_2 - \mathbf{q}) = \mathbf{k}_1 + \mathbf{k}_2.$$

For the spin part, we know that the *Coulomb interaction does not depend on the spin of the particles*,[5] so that the spin is also conserved during the interaction, which is a consequence of the rotational invariance of the system.

Thus we have modeled the interaction between the particles as a two-body interaction, which then scatters two particles with momenta $\mathbf{k}_1$ and $\mathbf{k}_2$ into two particles with momenta $\mathbf{k}_1 + \mathbf{q}$ and $\mathbf{k}_2 - \mathbf{q}$, where the momentum $\mathbf{q}$ is the momentum exchanged during the interaction. This is a very general form of the interaction, which can be applied to any system with translational invariance, and it is the starting point for many-body perturbation theory and other techniques used to study interacting systems.



This Hamiltonian cannot be solved exactly (for more than two particles), so we need to use some approximation methods to study the properties of the system. We are so going to develop the many body perturbation theory in the next chapters, which is a powerful tool to study interacting systems, and it is based on the idea of treating the interaction as a perturbation to the non-interacting system.

[4] The momentum conservation is enforced at the end by the delta fiction $\delta_{\mathbf{k}_1, \mathbf{k}_3 + \mathbf{q}}$.

[5] In some kind of interactions, we have other terms including all the Pauli matrices, so spin can switch.

2 | The Electron Gas

The electron gas is a fundamental model in condensed matter physics that describes a system of free electrons: it is a widely used model to study different kinds of materials and systems. The electron gas is indeed a useful tool to understand the behavior of electrons in a solid, and it captures many of the essential features of real materials, such as the Fermi surface, the density of states, and the response functions.

2.1 | Jellium Model

The electron gas is a model of a system of electrons that are free to move in a uniform background of positive charge, which is often used to study the properties of metals and other materials. This model is also known as the **jellium model**, and it is a useful tool to understand the behavior of electrons in a solid. Its main features are:

- The electrons are treated as *free particles*, which means that they do not interact with each other, but they do interact with the positive background charge. FALSE
- The positive background charge is treated as a *uniform distribution of charge*, which means that it does not have any structure or dynamics.
- The system is assumed to be *infinite and homogeneous*, which means that it has no boundaries and that the properties of the system are the same everywhere.

The Hamiltonian of the electron gas can be expressed as:

$$H = (T_{\text{ion}} + V_{\text{ion-ion}}) + (T_{\text{el}} + V_{\text{el-el}}) + V_{\text{el-ion}},$$

where T_{ion} is the kinetic energy of the ions, $V_{\text{ion-ion}}$ is the potential energy of the ion-ion interaction, T_{el} is the kinetic energy of the electrons, $V_{\text{el-el}}$ is the potential energy of the electron-electron interaction, and $V_{\text{el-ion}}$ is the potential energy of the electron-ion interaction.

2.1.1 | Jellium Hamiltonian

Let us review term by term, starting from the ionic part:[1]

$$\begin{aligned} T_{\text{ion}} + V_{\text{ion-ion}} &= \int \mathrm{d}^3\mathbf{R}\, \hat{\Psi}^\dagger_{\text{ion}}(\mathbf{R}) \left(-\frac{\hbar^2}{2M}\nabla^2_{\mathbf{R}}\right) \hat{\Psi}_{\text{ion}}(\mathbf{R}) \\ +\frac{1}{2}\int\int \mathrm{d}^3\mathbf{R}_1\,\mathrm{d}^3\mathbf{R}_2\, &\hat{\Psi}^\dagger_{\text{ion}}(\mathbf{R}_1)\hat{\Psi}^\dagger_{\text{ion}}(\mathbf{R}_2)\frac{Z^2e^2}{|\mathbf{R}_1-\mathbf{R}_2|}\hat{\Psi}_{\text{ion}}(\mathbf{R}_2)\hat{\Psi}_{\text{ion}}(\mathbf{R}_1), \end{aligned}$$

while for the electrons we have:

$$\begin{aligned} T_{\text{el}} + V_{\text{el-el}} &= \int \mathrm{d}^3\mathbf{r}\, \hat{\Psi}^\dagger_{\text{el}}(\mathbf{r}) \left(-\frac{\hbar^2}{2m}\nabla^2_{\mathbf{r}}\right) \hat{\Psi}_{\text{el}}(\mathbf{r}) \\ +\frac{1}{2}\int\int \mathrm{d}^3\mathbf{r}_1\,\mathrm{d}^3\mathbf{r}_2\, &\hat{\Psi}^\dagger_{\text{el}}(\mathbf{r}_1)\hat{\Psi}^\dagger_{\text{el}}(\mathbf{r}_2)\frac{e^2}{|\mathbf{r}_1-\mathbf{r}_2|}\hat{\Psi}_{\text{el}}(\mathbf{r}_2)\hat{\Psi}_{\text{el}}(\mathbf{r}_1). \end{aligned}$$

The remaining term is the electron-ion interaction, which can be expressed as:

$$V_{\text{el-ion}} = \sum_\sigma \int\int \mathrm{d}^3\mathbf{r}\,\mathrm{d}^3\mathbf{R}\, \hat{\Psi}^\dagger_{\text{el},\sigma}(\mathbf{r})\hat{\Psi}^\dagger_{\text{ion}}(\mathbf{R}) \left(\frac{-Ze^2}{|\mathbf{r}-\mathbf{R}|}\right) \hat{\Psi}_{\text{ion}}(\mathbf{R})\hat{\Psi}_{\text{el},\sigma}(\mathbf{r}).$$

In the jellium model, we assume that the ions are fixed in space and that they form a **uniform background of positive charge**, while the electrons are free to move in this background, such that the density can be expressed as:

$$\rho_{jel}(\mathbf{R}) = \frac{N}{\Omega},$$

where N is the number of nuclei and Ω is the volume of the system, such is constant. In the end the idea is to get rid of the ionic part (the nuclei, which have constant density) and to *focus only on*

[1] Capital letters for nuclei coordinates and small letters for electrons coordinates.

the electronic part, which is the one that determines the properties of the system. Indeed, making this assumption, we can see that the complicated *periodic potential* of the ions is replaced by a simple *uniform background*.

Ion-ion interaction. We can use the jellium assumption to simplify the ion-ion interaction, which can be expressed as

$$V_{\text{ion-ion}} = \frac{1}{2} \int \int \mathrm{d}^3\mathbf{R}_1 \, \mathrm{d}^3\mathbf{R}_2 \, \rho_{jel}(\mathbf{R}_1)\rho_{jel}(\mathbf{R}_2) \frac{1}{|\mathbf{R}_1 - \mathbf{R}_2|},$$

where we are basically multiplicating the two infinitesimal charges $\rho_{jel}(\mathbf{R}_1)\mathrm{d}^3\mathbf{R}_1$ and $\rho_{jel}(\mathbf{R}_2)\mathrm{d}^3\mathbf{R}_2$ and then integrating over all space after dividing by the distance between the two charges. Thus we can write

$$\begin{aligned} V_{\text{ion-ion}} &= \frac{1}{2} \int \int \mathrm{d}^3\mathbf{R}_1 \, \mathrm{d}^3\mathbf{R}_2 \, \frac{N^2}{\Omega^2} \frac{1}{|\mathbf{R}_1 - \mathbf{R}_2|} \\ &= \frac{N^2}{2\Omega^2} \int \mathrm{d}^3\mathbf{R} \int \mathrm{d}^3\boldsymbol{\xi} \, \frac{1}{|\boldsymbol{\xi}|} = \frac{N^2}{2\Omega^2} \, \Omega \, \lim_{q \to 0} V_q, \end{aligned}$$

where in the end we have performed the change of variables $\boldsymbol{\xi} = \mathbf{R}_1 - \mathbf{R}_2$ and we have expressed the integral in terms of the Fourier transform of the Coulomb potential, which diverges as $q \to 0$. This means that the ion-ion interaction diverges in the thermodynamic limit, which is a problem for the theory, but we will see that this divergence is cancelled by the electron-ion interaction, so that the final Hamiltonian of the electron gas is well defined and does not diverge.

Electron-ion interaction. Moving on to the electron-ion interaction (among electrons and the jellium background), we can express it as:

$$\begin{aligned} V_{\text{el-ion}} &= -\sum_{\sigma} \int \int \mathrm{d}^3\mathbf{r} \, \mathrm{d}^3\mathbf{R} \, \rho_{\text{el}}(\mathbf{r})\rho_{\text{ion}}(\mathbf{R}) \frac{1}{|\mathbf{r} - \mathbf{R}|} \\ &= -\frac{N}{\Omega} \sum_{\sigma} \int \int \mathrm{d}^3\mathbf{r} \, \mathrm{d}^3\mathbf{R} \, \rho_{\text{el}}(\mathbf{r}) \frac{1}{|\mathbf{r} - \mathbf{R}|} \\ &= -\frac{N}{\Omega} \left[\sum_{\sigma} \int \mathrm{d}^3\mathbf{r} \, \rho_{\text{el}}(\mathbf{r}) \right] \int \mathrm{d}^3\boldsymbol{\xi} \, \frac{1}{|\boldsymbol{\xi}|} = -\frac{N}{\Omega} \, N \, \lim_{q \to 0} V_q, \end{aligned}$$

where, in the end, we used the fact that the total charge of the electrons is equal to the total charge of the ions, so that $\sum_{\sigma} \int \mathrm{d}^3\mathbf{r} \, \rho_{\text{el}}(\mathbf{r}) = N$, which is a consequence of the charge neutrality of the system. This means that the electron-ion interaction also diverges in the thermodynamic limit, but it has the opposite sign of the ion-ion interaction; but they do not cancel each other.

Electron-electron interaction for $q = 0$. Moving onto the next interesting term, encoding the electron-electron interaction, we can consider the contribution for $q = 0$, which is the one that diverges, with the idea to show that this divergence is cancelled by the ion-ion and electron-ion interactions. Thus we can express the electron-electron interaction for $q = 0$ as:

$$V_{\text{el-el}}(q = 0) = \frac{1}{2} \sum_{\sigma,\sigma'} \sum_{\mathbf{k}_1,\mathbf{k}_2} \left[\lim_{q \to 0} V_q \right] \hat{c}^{\dagger}_{\mathbf{k}_1,\sigma} \hat{c}^{\dagger}_{\mathbf{k}_2,\sigma'} \hat{c}_{\mathbf{k}_2,\sigma'} \hat{c}_{\mathbf{k}_1,\sigma},$$

where now we are interested to change the order of the ladder operators, since we are dealing with fermions, so that the last term becomes:

$$\begin{aligned}
\hat{c}^\dagger_{\mathbf{k}_1,\sigma}\hat{c}^\dagger_{\mathbf{k}_2,\sigma'}\hat{c}_{\mathbf{k}_2,\sigma'}\hat{c}_{\mathbf{k}_1,\sigma} &= -\hat{c}^\dagger_{\mathbf{k}_1,\sigma}\hat{c}^\dagger_{\mathbf{k}_2,\sigma'}\hat{c}_{\mathbf{k}_1,\sigma}\hat{c}_{\mathbf{k}_2,\sigma'} \\
&= -\hat{c}^\dagger_{\mathbf{k}_1,\sigma}\left(\{\hat{c}^\dagger_{\mathbf{k}_2,\sigma'},\, \hat{c}_{\mathbf{k}_1,\sigma}\} - \hat{c}_{\mathbf{k}_1,\sigma}\hat{c}^\dagger_{\mathbf{k}_2,\sigma'}\right)\hat{c}_{\mathbf{k}_2,\sigma'} \\
&= -\hat{c}^\dagger_{\mathbf{k}_1,\sigma}\left(\delta_{\mathbf{k}_1,\mathbf{k}_2}\delta_{\sigma,\sigma'} - \hat{c}_{\mathbf{k}_1,\sigma}\hat{c}^\dagger_{\mathbf{k}_2,\sigma'}\right)\hat{c}_{\mathbf{k}_2,\sigma'},
\end{aligned}$$

where we used the anticommutation relations of the fermionic ladder operators, which state that $\{\hat{c}^\dagger_{\mathbf{k},\sigma}, \hat{c}_{\mathbf{k}',\sigma'}\} = \delta_{\mathbf{k},\mathbf{k}'}\delta_{\sigma,\sigma'}$ and that all the other anticommutators are zero. Thus we can express the electron-electron interaction for $q = 0$ as:

$$\begin{aligned}
V_{\text{el-el}}(q=0) &= \frac{1}{2\Omega}\sum_{\sigma,\sigma'}\sum_{\mathbf{k}_1,\mathbf{k}_2}\left[\lim_{q\to 0} V_q\right]\left(\hat{n}_{\mathbf{k}_1,\sigma}\hat{n}_{\mathbf{k}_2,\sigma'} - \delta_{\mathbf{k}_1,\mathbf{k}_2}\delta_{\sigma,\sigma'}\hat{n}_{\mathbf{k}_1,\sigma}\right) \\
&= \frac{1}{2\Omega}\left[\lim_{q\to 0} V_q\right]\sum_{\sigma}\sum_{\mathbf{k}_1}\left[\left(\sum_{\sigma'}\sum_{\mathbf{k}_2}\hat{n}_{\mathbf{k}_1,\sigma}\hat{n}_{\mathbf{k}_2,\sigma'}\right) - \hat{n}_{\mathbf{k}_1,\sigma}\right],
\end{aligned}$$

and remembering that the sum of the occupation numbers is equal to the total number of particles $\sum_{\mathbf{k}} \hat{n}_{\mathbf{k}} = N$ (independently from $\mathbf{k}_i$, since the sum is over all momenta), we can simplify the expression further

$$V_{\text{el-el}}(q=0) = \frac{1}{2\Omega}(N^2 - N)\lim_{q\to 0} V_q \sim \frac{N^2}{2\Omega}\lim_{q\to 0} V_q,$$

after considering the TD limit, which shows how the electron-electron interaction diverges as $q \to 0$ (a consequence of the long-range nature of the Coulomb interaction).

Cancellation of the divergences. If we sum the ion-ion and ion-electron interactions, we get the following expression:

$$V_{\text{ion-ion}} + V_{\text{ion-el}} = \frac{N^2}{2\Omega}\lim_{q\to 0} V_q - \frac{N^2}{\Omega}\lim_{q\to 0} V_q = -\frac{N^2}{2\Omega}\lim_{q\to 0} V_q,$$

which is easily recognized as the opposite of the electron-electron interaction for $q = 0$ (at the TD limit), so that we have:

$$V_{\text{ion-ion}} + V_{\text{ion-el}} + V_{\text{el-el}}(q=0) = 0,$$

effectively cancelling the divergences of the theory, so that we can get rid of the $q = 0$ contribution to the electron-electron interaction and all the terms related to the interactions with the jellium background; now we can focus on the remaining contributions, which are well defined and do not diverge.

After the removal of the divergences, we can express the **Hamiltonian of the homogeneous electron gas** as:

$$H_{\text{el}} = \sum_{\mathbf{k}\sigma}\frac{\hbar^2 k^2}{2m}\hat{c}^\dagger_{\mathbf{k},\sigma}\hat{c}_{\mathbf{k},\sigma} + \frac{1}{2\Omega}\sum_{\sigma,\sigma'}\sum_{\mathbf{k}_1,\mathbf{k}_2,\mathbf{q}\neq 0}\left(\frac{4\pi e^2}{q^2}\right)\hat{c}^\dagger_{\mathbf{k}_1+\mathbf{q},\sigma}\hat{c}^\dagger_{\mathbf{k}_2-\mathbf{q},\sigma'}\hat{c}_{\mathbf{k}_2,\sigma'}\hat{c}_{\mathbf{k}_1,\sigma},$$

where we have neglected the constant term that diverges as $q \to 0$, since it does not affect the properties of the system, along with the terms related to the interactions with the jellium background. This Hamiltonian is the starting point for many-body perturbation theory and other techniques used to study interacting systems, and it is a very important model in condensed matter physics, since it captures many of the essential features of real materials, such as metals and semiconductors.

The strength of this model is that it is simple enough to be solved exactly in some cases, but it is also rich enough to capture many of the essential features of real materials, such as the Fermi surface, the density of states, and the response functions. It is also a very useful model to study the effects of interactions on the properties of the system, such as the formation of collective excitations, the screening of the Coulomb interaction, and the emergence of new phases of matter.

2.2 | Ground State Energy and Fermi Sphere

Thus the hamiltonian of the electron gas, after having removed all the divergences, can be expressed as:

$$H_{\text{el}} = \sum_{\mathbf{k}\sigma} \frac{\hbar^2 k^2}{2m} \hat{c}^\dagger_{\mathbf{k},\sigma} \hat{c}_{\mathbf{k},\sigma} + \frac{1}{2\Omega} \sum_{\sigma,\sigma'} \sum_{\mathbf{k}_1,\mathbf{k}_2,\mathbf{q}\neq 0} \left(\frac{4\pi e^2}{q^2}\right) \hat{c}^\dagger_{\mathbf{k}_1+\mathbf{q},\sigma} \hat{c}^\dagger_{\mathbf{k}_2-\mathbf{q},\sigma'} \hat{c}_{\mathbf{k}_2,\sigma'} \hat{c}_{\mathbf{k}_1,\sigma},$$

where the idea is to use the eigenvalues and eigenvectors of the non-interacting part of the Hamiltonian, which is the kinetic energy term, as a basis on which we will apply perturbation theory to study the effects of the interactions.

The non-interacting part of the Hamiltonian is diagonal in the momentum space, and its eigenvalues are given by:

$$H^{(0)}_{\text{jel}} \Psi_{\mathbf{k},\sigma} = \frac{\hbar^2 k^2}{2m} \Psi_{\mathbf{k},\sigma},$$

(?) check if chi-und-sigma exist

where the eigenvectors are normalized plane waves of the form:

$$\Psi_{\mathbf{k},\sigma}(\mathbf{r}) = \frac{1}{\sqrt{\Omega}} e^{i\mathbf{k}\cdot\mathbf{r}} \chi_\sigma.$$

The idea is to use periodic boundary conditions to quantize the momentum, in particular we use the **Born-von Karman boundary conditions**, which state that the wavefunction must be periodic in the volume of the system, so that we have:

$$k_x = \frac{2\pi}{L_x} n_x, \quad k_y = \frac{2\pi}{L_y} n_y, \quad k_z = \frac{2\pi}{L_z} n_z,$$

so that, with the idea to pass to the continuum limit, we can express the sum over the momentum as an integral, so that we have to consider $L_i \to \infty$, such that

$$\sum_{\mathbf{k}} \to \frac{\Omega}{(2\pi)^3} \int \mathrm{d}^3\mathbf{k},$$

where the measure is dimensionless, since the factor Ω has dimensions of volume and the factor $(2\pi)^3$ has dimensions of volume in momentum space, so that the ratio is dimensionless. This is a very important step, since it allows us to pass from a discrete sum to a continuous integral, which is much easier to handle in the thermodynamic limit.

Now we can order the states according to their momenta and spin:

$$|\mathbf{k}_1, \uparrow\rangle, |\mathbf{k}_1, \downarrow\rangle, |\mathbf{k}_2, \uparrow\rangle, |\mathbf{k}_2, \downarrow\rangle, \ldots$$

and in this way we can order the eigenvalues of the non-interacting Hamiltonian from the lowest to the highest, so that we have:

$$\epsilon_{\mathbf{k}_1} < \epsilon_{\mathbf{k}_2} < \ldots$$

Thus if we have N electrons, we can fill the lowest N states according to the Pauli principle, so that we have a **Fermi sea** of electrons, which is the ground state of the non-interacting Hamiltonian, and it is given by:

$$|FS\rangle = \hat{c}^\dagger_{\mathbf{k}_{N/2},\uparrow} \hat{c}^\dagger_{\mathbf{k}_{N/2},\downarrow} \cdots \hat{c}^\dagger_{\mathbf{k}_2,\uparrow} \hat{c}^\dagger_{\mathbf{k}_2,\downarrow} \hat{c}^\dagger_{\mathbf{k}_1,\uparrow} \hat{c}^\dagger_{\mathbf{k}_1,\downarrow} |0\rangle,$$

where $|0\rangle$ is the vacuum state, which is the state with no particles, and the operators $\hat{c}^\dagger_{\mathbf{k},\sigma}$ are the creation operators that create an electron with momentum $\mathbf{k}$ and spin σ. The Fermi sea is a very important state, since it is the ground state of the non-interacting Hamiltonian, and it is also a very good approximation to the ground state of the interacting Hamiltonian, since the interactions are weak compared to the kinetic energy.

The fermi energy is the energy of the highest occupied state in the Fermi sea, and it is associated to the Fermi momentum, which is the momentum of the highest occupied state, so that we have:

$$E_F = \frac{\hbar^2 k_F^2}{2m}, \quad k_F = \frac{1}{\hbar}\sqrt{2mE_F},$$

from which we can extract the Fermi velocity, which is the velocity of the electrons at the Fermi surface, so that we have:

$$v_F = \frac{\hbar k_F}{m}.$$

Now how do we connect the the microscopic parameters of the model, such as the density of electrons, to the Fermi energy and the Fermi momentum? We can express the density of electrons as:

$$n = \frac{N}{\Omega} = \frac{1}{\Omega}\langle FS|\,\hat{N}\,|FS\rangle = \frac{1}{\Omega}\sum_{\mathbf{k},\sigma}\langle FS|\,\hat{n}_{\mathbf{k},\sigma}\,|FS\rangle\,,$$

where now we can express the sum over the momentum as an integral, so that we have:[2]

$$n = \frac{1}{\Omega}\frac{\Omega}{(2\pi)^3}\sum_\sigma\int \mathrm{d}^3\mathbf{k}\,\langle FS|\,\hat{n}_{\mathbf{k},\sigma}\,|FS\rangle = \frac{1}{(2\pi)^3}\sum_\sigma\int \mathrm{d}^3\mathbf{k}\,\theta(k_F - k) = \frac{2}{(2\pi)^3}\int_0^{k_F} k^2\mathrm{d}k\int_{-1}^{1}\mathrm{d}\cos\theta\int_0^{2\pi}\mathrm{d}\phi = \frac{1}{3\pi^2}k_F^3,$$

where the factor of 2 comes from the spin degeneracy, since we have two spin states for each momentum state, and the integral is over the volume of the Fermi sphere, which is given by:

$$\int_0^{k_F} k^2\mathrm{d}k\int_{-1}^{1}\mathrm{d}\cos\theta\int_0^{2\pi}\mathrm{d}\phi = \frac{4\pi}{3}k_F^3,$$

so that we can express the Fermi momentum as a function of the density as:

$$k_F = (3\pi^2 n)^{1/3},$$

which is a very important relation, since it allows us to connect the microscopic parameters of the model, such as the density of electrons, to the Fermi energy and the Fermi momentum, which are the parameters that determine the properties of the system.

Example (Some reasonable values for the parameters of the model). Let us consider a typical lattice, for example a cubic lattice with lattice constant $a = 2\,\text{Å}$, which is a typical value for the lattice constant of a metal. We are considering one electron per unit cell, which contains one atom in the center of the unit cell. Thus we are modelling one electron in a volume $\Omega = 8a^3 \sim 8(10\times 10^{-10}\,\mathrm{m}^3) \sim 10^{-29}\,\mathrm{m}^3$.

We can consider copper, which has only one electron in the outer shell, as the other transition metals have more than one electron in the outer shell, so that we can model it as a free electron gas:

$$\mathrm{Cu}: \quad 3d^{10}4s^1, \quad n_{Cu} = 8.47\times 10^{28}\,\mathrm{m}^{-3}.$$

Thus we can compute the Fermi momentum and energy as: **(?)** check temperature

$$k_F = 13.6\,\mathrm{nm}^{-1}, \quad E_F \sim 7\,\mathrm{eV} \sim 81\,600\,\mathrm{K}.$$

[2] The discretization goes to the continuum and we loose the PBC as we defined them, making just the right integration.......

TODO: put graph of energy density

The Fermi surface is very resilient to temperature, since the Fermi energy is much larger than the thermal energy at room temperature, which is of the order of $\sim 25\,\mathrm{meV}$, so that we can safely assume that the Fermi surface is well defined at room temperature, and that the properties of the system are determined by the electrons at the Fermi surface.

If we now compute the typical velocity of the electrons at the Fermi surface, we get:

$$v_F = \frac{\hbar k_F}{m} \sim 1.5 \times 10^6\,\mathrm{m\,s^{-1}},$$

which is a very high velocity, but only a fraction of the speed of light, so that we can safely assume that the electrons are non-relativistic, and that the non-relativistic quantum mechanics is a good approximation to describe their behavior.

TODO: put image of copper and sodium fermi surfaces

In the surface there are holes: it is a periodic lattice, so that the Fermi surface is not a perfect sphere, but it has some distortions due to the periodic potential of the lattice, which can lead to the formation of holes in the Fermi surface, which are regions where the density of states is very low, and which can have important consequences for the properties of the system, such as the transport properties and the response functions.the surface is connected to the Brillouin zone, which is the primitive cell of the reciprocal lattice, and it can have different shapes depending on the symmetry of the lattice, so that in some cases it can have a very complex shape, which can lead to interesting phenomena such as nesting and van Hove singularities.

Let us come back to the non interacting Hamiltonian of the electron gas, which is given by:

$$H^{(0)}_{jel} = \sum_{\mathbf{k}\sigma} \frac{\hbar^2 k^2}{2m} \hat{c}^\dagger_{\mathbf{k},\sigma} \hat{c}_{\mathbf{k},\sigma},$$

then the spctrum of this Hamiltonian is given by:

$$E^{(0)} = \langle FS| \hat{H}^{(0)}_{jel} |FS\rangle = \sum_{\mathbf{k}\sigma} \frac{\hbar^2 k^2}{2m} \langle FS| \hat{c}^\dagger_{\mathbf{k},\sigma} \hat{c}_{\mathbf{k},\sigma} |FS\rangle,$$

where we can express the sum over the momentum as an integral, and we recognize that the integrand is the occupation number, so that we have:

$$\begin{aligned} E^{(0)} &= \frac{2\hbar^2}{m} \frac{\Omega}{(2\pi)^3} \sum_\sigma \int \mathrm{d}^3\mathbf{k}\, \mathbf{k}^2 \theta(k_F - |\mathbf{k}|) = \dots \\ &= \frac{\Omega}{5\pi^2} \frac{\hbar^2 k_F^5}{2m} = \dots = \frac{3}{5} \Omega \frac{k_F^3}{3\pi^2} E_F, \end{aligned}$$

so that in the end $E^{(0)} = \frac{3}{5} N E_F$, which is the energy of the non-interacting Fermi gas, and it is a very important result, since it gives us a reference point to compare with the energy of the interacting system, and to understand the effects of the interactions on the properties of the system. It makes sense that the energy is proportional to the number of particles and to $\frac{3}{5} E_F$, since the energy of the highest occupied state is E_F, and the average energy of the occupied states is $\frac{3}{5} E_F$, which is a consequence of the fact that the density of states is proportional to the square root of the energy.

In the condensed matter community, this problem was central in the sixties and seventies, since it was the first example of a system of interacting fermions that could be solved exactly, and it was used as a benchmark for many-body perturbation theory and other techniques used to study interacting systems. The idea was now to define a universal and dimensionless parameter that characterizes the density of the system, and to use it to understand the properties of the system as a function of the density. This parameter is called the **Wigner-Seitz radius**, and it is defined as:

$$r_s, \quad n = \frac{1}{\frac{4}{3}\pi(r_s a_0)^3},$$

where the Bohr length a_0 is the length scale that characterizes the size of the electron wavefunction, r_s length instead is the length scale that characterizes the average distance between the electrons, so that it represents 1 electron per sphere of radius a_0 if $r_s = 1$. Otherwise for $r_s \to 0$ it represents a very high density, since we have many electrons in a small volume, while for $r_s \to \infty$ it represents a very low density, since we have few electrons in a large volume. Thus the Wigner-Seitz radius is a very important parameter, since it allows us to characterize the properties of the system as a function of the density, and to understand the effects of the interactions on the properties of the system. It is an inverse measure for density, and it is a very useful parameter since we can write

$$\frac{E^{(0)}}{N} = \frac{3}{5}E_F = \frac{3}{5}\frac{\hbar^2 k_F^2}{2m} = \frac{3}{5}\frac{1}{2}\frac{\hbar^2}{me^2}e^2\left(\frac{a_0^2}{a_0^2}k_F^2\right) = \frac{3}{5}\frac{a_0}{2}\frac{e^2}{a_0^2}k_F^2,$$

where if we compute

$$\frac{1}{n} = \frac{4}{3}\pi(r_s a_0)^3 = \frac{3\pi^2}{k_F} \to r_s = \left(\frac{9\pi}{4}\right)^{1/3}\frac{1}{k_F a_0},$$

so that we can express the energy per particle as a function of the Wigner-Seitz radius as:

$$\frac{E^{(0)}}{N} = \frac{3}{5}\left(\frac{9\pi}{4}\right)^{\frac{2}{3}}\underbrace{\frac{e^2}{2a_0}}_{\text{Rydberg}}\frac{1}{r_s^2} = \frac{2.211\ \text{Ry}}{r_s^2},$$

which shows that the energy per particle of the non-interacting Fermi gas scales as $1/r_s^2$, which is a very important result, since it gives us a reference point to compare with the energy of the interacting system, and to understand the effects of the interactions on the properties of the system as a function of the density.

TODO: graph of energy per particle as a function r-und-s.

Remeber that the Rydberg energy is the energy scale that characterizes the strength of the Coulomb interaction, and it is given by: $1\ \text{Ry} = 13.6\,\text{eV}$.

2.3 | I Order Perturbation Theory: Hartree-Fock

The first important term from perturbation theory is the exchange energy term, which is basically due to the undistinguishability of the electrons, and it represents the energy cost of having indistinguishable particles, since the electrons must obey the Pauli principle, which means that they cannot occupy the same state, so that they must have different momenta or different spins, which leads to a reduction of the energy of the system, since the electrons can avoid each other more effectively.

It is interactions that make the system more stable, even the ones just in the .....

$$\frac{E^{(1)}}{N} = \frac{\langle FS| \hat{V}_{\text{el-el}} |FS\rangle}{N} = \frac{1}{2\Omega N} \sum_{\sigma_1,\sigma_2} \sum_{\mathbf{k}_1,\mathbf{k}_2,\mathbf{q}\neq 0} \left(\frac{4\pi e^2}{q^2}\right) \langle FS| \hat{c}^\dagger_{\mathbf{k}_1+\mathbf{q},\sigma} \hat{c}^\dagger_{\mathbf{k}_2-\mathbf{q},\sigma'} \hat{c}_{\mathbf{k}_2,\sigma'} \hat{c}_{\mathbf{k}_1,\sigma} |FS\rangle \,,$$

where the only possibility for this term to be non-vanishing is that the operators $\hat{c}^\dagger_{\mathbf{k}_1+\mathbf{q},\sigma}$ and $\hat{c}^\dagger_{\mathbf{k}_2-\mathbf{q},\sigma'}$ create particles in the Fermi sphere, while the operators $\hat{c}_{\mathbf{k}_2,\sigma'}$ and $\hat{c}_{\mathbf{k}_1,\sigma}$ destroy particles in the Fermi sea, so that we have:

- $\mathbf{q} = 0$, which means that the two particles have the same momentum, so that we would create and destroy the same particles, which is not possible since we are dealing with fermions, and indeed this term is neglected directly by the summation over $\mathbf{q} \neq 0$.
- $\mathbf{q} = \mathbf{k}_2 - \mathbf{k}_1$ with $\sigma_1 = \sigma_2$, which means that we create a particle with momentum $\mathbf{k}_2$ and we destroy a particle with momentum $\mathbf{k}_1$, so that we have a non-vanishing contribution to the energy, which is called the exchange energy, since it is due to the exchange of particles between the two states.

TODO: add feynman diagram of the direct interaction and of the exchange interaction with right momentum and spin labels.

So the two particles scatter and after the coulomb interaction they exchange their momenta, but they keep the same spin, since the interaction is spin-independent, so that we have basically an exchange of particles between the two states, which is a consequence of the indistinguishability of the electrons, and it leads to a reduction of the energy of the system, since the electrons can avoid each other more effectively.

These two terms are called the direct and exchange terms, or the **Hartree** and **Fock** terms, and they are the first two terms in the perturbation theory expansion of the energy of the system.

TODO: oister diagram for exchange and direct interaction + comments on oister

Now we have to compute the exchange energy, which can be expressed as an integral:

$$\langle FS| \hat{c}^\dagger_{\mathbf{k}_1+\mathbf{q},\sigma} \hat{c}^\dagger_{\mathbf{k}_2-\mathbf{q},\sigma'} \hat{c}_{\mathbf{k}_2,\sigma'} \hat{c}_{\mathbf{k}_1,\sigma} |FS\rangle = \delta_{\sigma_1,\sigma_2} \delta_{\mathbf{k}_1+\mathbf{q},\mathbf{k}_2} \langle FS| \hat{c}^\dagger_{\mathbf{k}_1+\mathbf{q},\sigma_1} \hat{c}^\dagger_{\mathbf{k}_1,\sigma_1} \hat{c}_{\mathbf{k}_1+\mathbf{q},\sigma_1} \hat{c}_{\mathbf{k}_1,\sigma_1} |FS\rangle$$

where the delta functions come from the fact that we have to create and destroy the same particles, so that we have to have the same momentum and the same spin, and the expectation value can be computed using the anticommutation relations of the fermionic operators, so that we have:

$$= -\delta_{\sigma_1,\sigma_2} \delta_{\mathbf{k}_1+\mathbf{q},\mathbf{k}_2} \langle FS| \hat{n}_{\mathbf{k}_1+\mathbf{q},\sigma_1} \hat{n}_{\mathbf{k}_1,\sigma_1} |FS\rangle = -\delta_{\sigma_1,\sigma_2} \delta_{\mathbf{k}_1+\mathbf{q},\mathbf{k}_2} \theta(k_F - |\mathbf{k}_1 + \mathbf{q}|) \theta(k_F - |\mathbf{k}_1|),$$

where the minus sign comes from the anticommutation of the fermionic operators, and the theta functions come from the fact that we have to create and destroy particles in the Fermi sea, so that we have to have momenta that are less than the Fermi momentum. Thus the exchange energy can be expressed as:

$$\frac{E^{(1)}}{N} = -\frac{1}{2\Omega N} \sum_{\sigma} \sum_{\mathbf{k},\mathbf{q}\neq 0} \left(\frac{4\pi e^2}{q^2}\right) \theta(k_F - |\mathbf{k} + \mathbf{q}|) \theta(k_F - |\mathbf{k}|),$$

where we have renamed the momentum $\mathbf{k}_1$ to $\mathbf{k}$ for simplicity. Now we can express the sum over the momentum as an integral

$$\sum_{\mathbf{k},\mathbf{q}\neq 0} \to \frac{\Omega^2}{(2\pi)^6}\int \mathrm{d}^3\mathbf{k}\int \mathrm{d}^3\mathbf{q},$$

but having two theta functions in the integrand makes it a bit more complicated, since we have to consider the intersection of two spheres in momentum space, which is the region where both theta functions are equal to 1, so that we have to compute the volume of the intersection of two spheres of radius k_F centered at $\mathbf{k}$ and $\mathbf{k}+\mathbf{q}$.

TODO: add graph of the intersection of the two spheres in momentum space.

We have to notice that if $\mathbf{q} > 2k_F$ then the intersection of the two spheres is empty, so that the exchange energy vanishes in this case. In the overlapping case, the volume of the intersection of the two spheres can be computed as a spherical cap: having fixed $\mathbf{q}$ (which can vary in all directions but only for a limited range of module) we can integrate over the angles of $\mathbf{k}$ and then over the module of $\mathbf{k}$. So we can express the momenta in spherical coordinates as:

$$\mathbf{q} = (q\,|\,\theta_q\,|\,\phi_q), \quad \mathbf{k} = (k\,|\,\theta_k\,|\,\phi_k),$$

where the integtration limits of the

$$\theta_K^{min} = 0,\ \text{to}\ k_F\cos(\theta_k^{max}) = \frac{q}{2}, \quad \Longrightarrow \frac{q}{2k_F} < \cos(\theta_k^{max}) < 1,$$

since the two spheres intersect only if the distance between their centers is less than the sum of their radii, which means that we have to have $q < 2k_F$, and for the other piece of the sphere we have:

$$k^{max} = k_F, \quad k^{min}\cos(\theta_k) = \frac{q}{2}, \quad \Longrightarrow \frac{q}{2\cos(\theta_k)} < k < k_F,$$

so that the volume of the intersection of the two spheres can be expressed as:

$$\begin{aligned}\frac{E^{(1)}}{N} =& -\frac{2}{2\Omega N}\frac{\Omega}{(2\pi)^3}\int_{-1}^{1}\mathrm{d}\cos(\theta_q)\int_0^{2\pi}\mathrm{d}\phi_q\int_0^{2k_F}\mathrm{d}q\,q^2\times\\ &\times\frac{\Omega}{(2\pi)^3}\int_{\frac{q}{2k_F}}^{1}\mathrm{d}\cos(\theta_k)\int_0^{2\pi}\mathrm{d}\phi_k\int_{\frac{q}{2\cos(\theta_k)}}^{k_F}\mathrm{d}k\,k^2\times 2\left(\frac{4\pi e^2}{q^2}\right),\end{aligned}$$

where the first factor of 2 comes from the spin degeneracy, since we have two spin states for each momentum state, while the second factor of 2 comes from the fact that we have two equivalent sphere caps, since the intersection of the two spheres is symmetric with respect to the plane that bisects the line that connects their centers. Furthermore, ϕ is not constraint on the integral since it is like a rotation around the axis that connects the centers of the two spheres, so that we can integrate it out.

... conti...

Continuing the computation, we can express write the enormous integral as a function of k_F:

$$\frac{E^{(1)}}{N} = -\frac{2^7\pi^3e^2\Omega^2}{2^7\Omega N\pi^6}I(k_F) = -\frac{e^2\Omega}{\pi^3 N}I(k_F),$$

where the integral $I(k_F)$ is given by:

$$
\begin{aligned}
I(k_f) &= \int_0^{2k_F} \mathrm{d}q\, q^2 \int_{\frac{q}{2k_F}}^{1} \mathrm{d}\cos(\theta_k) \int_{\frac{q}{2\cos(\theta_k)}}^{k_F} \mathrm{d}k\, k^2 \\
&= \int_0^{2k_F} \mathrm{d}q\, q^2 \int_{\frac{q}{2k_F}}^{1} \mathrm{d}\cos(\theta_k) \frac{1}{3}\left(k_F^3 - \left(\frac{q}{2\cos(\theta_k)}\right)^3\right) \\
&= \int_0^{2k_F} \mathrm{d}q\, q^2 \frac{1}{3}\left[k_F^3 \cos\theta_k - \frac{q^3}{8}\left(-\frac{1}{2\cos^2\theta_k}\right)\right]_{\frac{q}{2k_F}}^{1} \\
&= \int_0^{2k_F} \mathrm{d}q\, q^2 \frac{1}{3}\left[k_F^3 - \frac{q}{2}k_F^2 + \frac{q^3}{8}\left(\frac{1}{2} - \frac{2}{q^2}k_F^2\right)\right] \\
&= \int_0^{2k_F} \mathrm{d}q\, q^2 \frac{1}{3}\left[k_F^3 - \frac{3q}{4}k_F^2 + \frac{q^3}{16}\right] \\
&= \frac{1}{3}\left[2k_F^4 + \dots\right] = \frac{1}{4}k_F^4,
\end{aligned}
$$

where we integrated q starting from 0 even if the integral is not well defined in this limit, since the integrand is not well defined for $q \to 0$, but we can regularize it by introducing a small cutoff ϵ, or anyway considering that in 0 the measure is null.

Thus in the end we can express the exchange energy as:

$$
\frac{E^{(1)}}{N} = -\frac{e^2\Omega}{\pi^3 N}\frac{1}{4}k_F^4 = -\frac{e^2}{4\pi^3}\frac{\Omega}{N}k_F^4,
$$

where we can express the ratio Ω/N the inverse of the density while adding a_0/a_0 as

$$
\frac{E^{(1)}}{N} = -\frac{e^2}{2a_0}(a_0 k_F)\frac{k_F^3}{2\pi^3}\frac{1}{n} = (-1\mathrm{Ry})\left[\left(\frac{9\pi}{4}\right)^{1/3}\frac{1}{r_s}\right]\frac{3}{2\pi} = -\frac{0.916\ \mathrm{Ry}}{r_s},
$$

where the first term is the Rydberg energy, which is the energy scale that characterizes the strength of the Coulomb interaction, while the second term is a dimensionless function of the Wigner-Seitz radius, which characterizes the density of the system; the last one comes from the fact that we can express the Fermi momentum as a function of the density as $k_F = (3\pi^2 n)^{1/3}$, and the density as a function of the Wigner-Seitz radius as $n = \frac{1}{\frac{4}{3}\pi(r_s a_0)^3}$, so that we can express the Fermi momentum as a function of the Wigner-Seitz radius.

TODO: checkkkkk

Thus the final result for the energy per particle of the interacting Fermi gas at first order in perturbation theory is given by:

$$
\frac{E}{N} = \frac{E^{(0)} + E^{(1)}}{N} = \frac{2.211\ \mathrm{Ry}}{r_s^2} - \frac{0.916\ \mathrm{Ry}}{r_s},
$$

which shows that the energy per particle of the interacting Fermi gas scales as $1/r_s^2$ at high density, and as $1/r_s$ at low density, which is a very important result, since it gives us a reference point to compare with the energy of the non-interacting system, and to understand the effects of the interactions on the properties of the system as a function of the density.

TODO: add potential graph

It is cricial the difference in scaling power, since:

- at **high density**, the kinetic energy dominates over the potential energy, since the kinetic energy scales as $1/r_s^2$ while the potential energy scales as $1/r_s$, so that the system behaves like a non-interacting Fermi gas, and the properties of the system are determined by the kinetic energy.

- at **low density**, the potential energy dominates over the kinetic energy, since the potential energy scales as $1/r_s$ while the kinetic energy scales as $1/r_s^2$, so that the system behaves like a classical plasma, and the properties of the system are determined by the interactions between the electrons.

The minimum of the energy per particle is given by:

$$\frac{\mathrm{d}}{\mathrm{d}r_s}\left(\frac{E}{N}\right) = 0 \implies r_s^* \sim 4.83, \quad \frac{E^*}{N} \sim -0.095 \text{ Ry} \sim -1.29\,\text{eV},$$

which is a very interesting result, since it shows that there is a minimum of the energy per particle at a finite value of the Wigner-Seitz radius, which means that there is a stable phase of the system at this density, and it also shows that the interactions between the electrons can lead to a reduction of the energy of the system, which is a consequence of the exchange interaction.

For real materials we find

- Sodium: Na, $1s^2\,2s^2\,2p^6\,3s^1$, has $r_s \sim 3.93$, which is close to the minimum of the energy per particle $\frac{E}{N} \sim -1.13\,\text{eV}$, so that it is a good metal, and it has a high density of states at the Fermi level, which leads to good conductivity.
- For metals usually we have $r_s \sim 2.5 - 6$, which means that they are in the regime where the interactions are important, but not dominant, so that they behave like a Fermi liquid, and their properties are determined by the interactions between the electrons, but they still have a well-defined Fermi surface and a high density of states at the Fermi level, which leads to good conductivity.

This result shows that the electron gas is stable when the repulsive interactions between the electrons are taken into account, and there is no need in confining the gas in some external potential in order to stabilize it, since the interactions themselves can lead to a stable phase of the system. This is a very important result, since it shows that the electron gas can be a good model for real materials such as metals, and it also shows that the interactions between the electrons can lead to interesting phenomena such as superconductivity and magnetism, which are not present in the non-interacting Fermi gas.

At the end of the course we will see that th next order in perturbation theory gives us a correction to the energy that scales as $\log(r_s)$, which is a very weak correction, but it is still important to understand the properties of the system at low density: in particular, in order to get the correct behavior of the energy per particle at low density, we will need to include an infinite number of terms in the perturbation theory expansion in order to get all the terms composing the logarithmic correction; stopping at any finite order will give us a correction that diverges.

In first quantization formalism, with plane waves, we can obtain the same result, which is the Hartree-Fock energy of the system; historically all the terms after the first order were called **correlation energy**, since they are due to the correlations between the electrons that are not captured by the Hartree-Fock approximation. The correlation energy is a very important quantity, since it gives us a measure of the strength of the interactions between the electrons, and it can lead to interesting phenomena such as superconductivity and magnetism, which are not present in the non-interacting Fermi gas.

2.4 | II Order Perturbation Theory

Since the first order perturbation theory gives us a good approximation to the energy of the system, we can now try to compute the second order correction to the energy. In second order we need to take into account the expectation value of the interaction term in the states that are not occupied in the Fermi sea, which are called virtual excitations:

$$\frac{E^{(2)}}{N} = \frac{1}{N} \sum_{|\nu\rangle \neq |FS\rangle} \frac{|\langle\nu| \hat{V}_{\text{el-el}} |FS\rangle|^2}{E^{(0)} - E^{(0)}_\nu} = \frac{1}{N} \sum_{|\nu\rangle \neq |FS\rangle} \frac{\langle FS| \hat{V}_{\text{el-el}} |\nu\rangle \langle\nu| \hat{V}_{\text{el-el}} |FS\rangle}{E^{(0)} - E^{(0)}_\nu},$$

which represent the contribution to the energy of the virtual excitations of the system, which are states that are not occupied in the Fermi sea, but they can be excited by the interactions between the electrons, so that they can contribute to the energy of the system even if they are not occupied in the ground state.

The interaction term can be expressed as:

$$\hat{V}_{\text{el-el}} = \frac{1}{2\Omega} \sum_{\sigma_1,\sigma_2} \sum_{\mathbf{k}_1,\mathbf{k}_2,\mathbf{q}\neq 0} \left(\frac{4\pi e^2}{q^2}\right) \hat{c}^\dagger_{\mathbf{k}_1+\mathbf{q},\sigma} \hat{c}^\dagger_{\mathbf{k}_2-\mathbf{q},\sigma'} \hat{c}_{\mathbf{k}_2,\sigma'} \hat{c}_{\mathbf{k}_1,\sigma},$$

so that we have to compute the matrix elements of this operator between the Fermi sea and the virtual excitations: thus the first two ladder operators $\hat{c}^\dagger_{\mathbf{k}_1+\mathbf{q},\sigma} \hat{c}^\dagger_{\mathbf{k}_2-\mathbf{q},\sigma'}$ have to create particles out of the Fermi sea, which seams a bit strange.

But how should these states look like? Let us write

$$|\nu\rangle = \Theta(|\mathbf{k}_1 + \mathbf{q}| - k_F)\Theta(k_F - |\mathbf{k}_1|)\Theta(|\mathbf{k}_2 - \mathbf{q}| - k_F)\Theta(k_F - |\mathbf{k}_2|)\hat{c}^\dagger_{\mathbf{k}_1+\mathbf{q},\sigma} \hat{c}^\dagger_{\mathbf{k}_2-\mathbf{q},\sigma'} \hat{c}_{\mathbf{k}_2,\sigma'} \hat{c}_{\mathbf{k}_1,\sigma} |FS\rangle,$$

since we have to annihilate particles in the Fermi sea and create particles out of the Fermi sea, so that we have to have momenta that are less than the Fermi momentum for the annihilation operators, and momenta that are greater than the Fermi momentum for the creation operators.

TODO: add graph of the virtual excitations out of the Fermi sea.

To restore FS the only two processes that preserve the conservation of momentum are the following:

- **Direct interaction**: where the two particles scatter exchanging a momenta $\mathbf{q}$, which then is reabsorbed by the same two particles, so that we have a direct interaction between the two particles, which is a second order process since we have to exchange the momentum twice.
- **Exchange interaction**: where the two particles scatter exchanging a momenta $\mathbf{q}$, which then is reabsorbed by the same two particles, in a way that they exchange their momenta, so that we have an exchange interaction between the two particles, which is also a second order process since we have to exchange the momentum twice.

TODO: add graph of the direct and exchange interactions at second order, both feynman and oister.

Inside the sphere the particles can only exchange position in phase space or in the end come back to their own starting state, since the Pauli principle forbids them to occupy the same state of other fermions in the FS.

2.4.1 | Direct Interaction

The contribution of the direct interaction to the energy can be expressed as:

$$\frac{E^{(2)}_d}{N} = \sum_{\mathbf{q}\neq\mathbf{0}} \sum_{\mathbf{k}_1,\mathbf{k}_2} \sum_{\sigma_1,\sigma_2} \left(\frac{1}{2\Omega}\frac{4\pi e^2}{q^2}\right) \frac{1}{E^{(0)} - E_\nu}$$
$$\times \Theta(|\mathbf{k}_1 + \mathbf{q}| - k_F)\Theta(k_F - |\mathbf{k}_1|)\Theta(|\mathbf{k}_2 - \mathbf{q}| - k_F)\Theta(k_F - |\mathbf{k}_2|),$$

where again the theta functions come from the fact that we have to create particles out of the Fermi sea and destroy particles in the Fermi sea, restricting the domain.

[considerations on magnitude of terms]

when $\mathbf{q} \to 0$, we get

$$(V_q)^2 \sim \frac{1}{q^4},$$

and

$$E^{(0)} - E_\nu \propto - \left[|\mathbf{k}_1 - \mathbf{q}|^2 - |\mathbf{k}_1|^2 + |\mathbf{k}_2 - \mathbf{q}|^2 - |\mathbf{k}_2|^2 \right] \frac{\hbar^2}{2m} \sim |\mathbf{k}_1|^2 + |\mathbf{q}|^2 + 2\mathbf{k}_1 \cdot \mathbf{q} - |\mathbf{k}_1| \cdots \sim \mathbf{q}$$

which is linear in $\mathbf{q}$.

[...]

where now the theta functions resctrict the domain in a different way: the second one is equal, the first one is requesting that the particle remains outside the FS. Thus we have to select the intersection and take the difference between the two spheres, which is a bit more complicated to compute, but euristically as $\mathbf{q}$ increases the intersection of the two spheres decreases, so that we have a smaller and smaller contribution to the energy, which is a consequence of the fact that the particles can avoid each other more effectively as they exchange more momentum, since they can explore a larger portion of the phase space.

Thus, ignoring the prefactors, we can express the contribution of the direct interaction to the energy as:

$$\frac{E_d^{(2)}}{N} \sim \int_0 \mathrm{d}q\, q^2 \, \frac{1}{q^4} \, \frac{1}{q} \, q \, q = \int_0 \mathrm{d}q \, \frac{1}{q} \sim \log(q)_0 \to \infty,$$

where the first factor of q^2 comes from the measure of the integral, the second factor of $1/q^4$ comes from the square of the Coulomb interaction, the third factor of $1/q$ comes from the energy denominator, while the last two factors of q come from the theta functions that restrict the domain of integration, since they restrict the domain to a region that scales as q for small q.

Thus we have a logarithmic divergence in the limit $q \to 0$, which is a consequence of the long-range nature of the Coulomb interaction, and it is a very important result, since it shows that we cannot treat the interactions between the electrons as a perturbation at low density, since we have a divergence in the energy at second order in perturbation theory, which means that we have to include an infinite number of terms in the perturbation theory expansion in order to get a finite result for the energy of the system at low density.

2.4.2 | Exchange Interaction

The exchange interaction is a bit more complicated to compute, but it does not present any divergence:

$$\frac{E_e^{(2)}}{N} \sim \sum_{\mathbf{q}} \frac{1}{q^2} \frac{1}{|\mathbf{k}_1 - \mathbf{k}_2 + \mathbf{q}|^2},$$

which will give us a finite result, but the direct interaction will give us a preponential factor that diverges, so that the exchange interaction will give us a correction to the energy that is negligible compared to the direct interaction at low density, since the direct interaction will give us a logarithmic divergence, while the exchange interaction will give us a finite correction. Thus in the end we are forced to compute an infinite number of divergences in ordeer to obtain a finite result, if we want to proceed with perturbation theory.

This problem arises from the long range nature of the Coulomb interaction, which makes the perturbation theory expansion diverge at low density. But we will consider other methods to compute the energy of the system at low density, which will give us a finite result for the energy of the system at low density, and it will also give us a better understanding of the properties of the system at low density, which are not captured by perturbation theory. It is time to move onto many-body techniques that will allow us to compute the energy of the system at low density, and to understand the properties of the system at low density, which are not captured by perturbation theory.

We are neglecting the screening of the other electrons, which is a very important effect that will allow us to get a finite result for the energy of the system at low density. The screening of the other electrons will reduce the strength of the Coulomb interaction at low density, so that we will have a screened Coulomb interaction that will give us a finite result.

3 Many Body Theory

Many body theory is a powerful framework for understanding the behavior of systems with a large number of interacting particles. In condensed matter physics, it is essential for describing phenomena such as superconductivity, magnetism, and the electronic properties of materials. This theory was developed to address the complexities that arise when dealing with systems of many particles, where the perturbation theory of non-interacting particles fails to capture the essential physics. This theory was developed in the 1957-1960s by pioneers such as John Bardeen, Leon Cooper, and Robert Schrieffer, who formulated the BCS theory of superconductivity. The many body theory has since been extended and refined to include various interactions and correlations, making it a cornerstone of modern condensed matter physics.

In this chapter, we will explore the fundamental concepts and techniques of many body theory, including Green's functions, Feynman diagrams, and the random phase approximation. We will also discuss how these tools are used to analyze the properties of interacting electron systems and to understand phenomena such as collective excitations and phase transitions.

Time Dependencies in Quantum Theories

Firstly we need to speak about time dependencies in quantum theories. Since we are dealing with particles being created and destroyed after some kind of propagation, we need to understand how to deal with time dependencies in quantum mechanics.

Schrodinger picture. In the Schrodinger picture, the state vectors evolve in time while the operators are time-independent. The time evolution of a state vector $|\psi\rangle$ is governed by the Schrodinger equation:

$$\dots$$

Heisenberg picture. In the Heisenberg picture, the operators evolve in time while the state vectors are time-independent. The time evolution of an operator A is given by:

$$\dots$$

we can thus derive an equation of motion for the operators in the Heisenberg picture, which is known as the Heisenberg equation of motion:

$$\begin{aligned}\hat{A}_H(t) &= e^{i\hat{H}t}A_S e^{-i\hat{H}t},\\ \frac{\mathrm{d}}{\mathrm{d}t}\hat{A}_H(t) &= \frac{\mathrm{d}}{\mathrm{d}t}\left(e^{i\hat{H}t}A_S e^{-i\hat{H}t}\right) = i\hat{H}e^{i\hat{H}t}A_S e^{-i\hat{H}t} - e^{i\hat{H}t}A_S i\hat{H}e^{-i\hat{H}t}\\ &= i\hat{H}\hat{A}_H(t) - i\hat{A}_H(t)\hat{H} = i[\hat{H}, \hat{A}_H(t)].\end{aligned}$$

Interaction (Dirac) picture. Sometimes we have an hamiltonian which is composed of a time independent part and a time independent part, such as:

$$\hat{H} = \hat{H}_0 + \hat{V}(t).$$

In this case, we can define the interaction picture, where both the state vectors and the operators have time dependence. The operators in the interaction picture evolve according to the free Hamiltonian $\hat{H}_0$ (or the quadratic part of the Hamiltonian), while the state vectors evolve according to the interaction part $\hat{V}(t)$:

$$A_I(t) = e^{i\hat{H}_0 t} A_S e^{-i\hat{H}_0 t},$$

$$|\psi_I(t)\rangle = e^{i\hat{H}_0 t} |\psi_S\rangle e^{-i\hat{H}_0 t}.$$

The interaction picture is particularly useful in many body theory, as it allows us to treat the interactions perturbatively while still accounting for the time evolution of the system. It makes evident the connection between the free and interacting theories, and is often used in the formulation of Feynman diagrams and perturbation theory.

Heisenberg - Dirac bridge. We can also derive a connection between the Heisenberg and Dirac pictures. Starting from the definition of the Heisenberg picture, we can add identity operators to express the Heisenberg operator in terms of the interaction picture:

$$\begin{aligned} A_H(t) = e^{i\hat{H}t} A_S e^{-i\hat{H}t} &= e^{i\hat{H}t} e^{i\hat{H}_0(t)t} e^{-i\hat{H}_0(t)t} A_S e^{-i\hat{H}_0(t)t} e^{i\hat{H}_0(t)t} e^{-i\hat{H}t} \\ &= U^\dagger(t,0) A_I(t) U(t,0), \end{aligned}$$

where we have defined the time evolution operator in the interaction picture as:

$$U(t,0) = e^{i\hat{H}_0 t} e^{-i\hat{H}t} = e^{i\hat{H}_0 t} e^{-i(\hat{H}_0 + \hat{V}(t))t} = e^{-i\int_0^t \mathrm{d}t' \hat{V}_I(t')}.$$

This evolution operator contains only the interaction part of the Hamiltonian, and this highlights the fact that the interaction picture is designed to evolve the state vectors according to the interaction while keeping the operators evolving according to the free Hamiltonian.

Pay attention:

$$e^{i\hat{H}_0 t} e^{-i\hat{H}t} \neq e^{i\hat{V}t},$$

and this is the kernel of many body theory, since the interaction part of the Hamiltonian does not commute with the free part normally, and thus we cannot simply factorize the time evolution operator into a product of exponentials. This non-commutativity is what gives rise to the rich physics of interacting systems, and it is what makes many body theory both challenging and fascinating.

Let us take the partial time derivative of the evolution operator:

$$\begin{aligned} \frac{\partial}{\partial t}\hat{U}(t,0) &= \dots \\ &= \dots \\ &= -i\hat{V}_I(t)\hat{U}(t,0). \end{aligned}$$

Now the question is: how do we solve this equation? We can formally integrate it to get:

$$\hat{U}(t,0) - \hat{U}(0,0) = -i\int_0^t \mathrm{d}t' \hat{V}_I(t')\hat{U}(t',0).$$

This is an integral equation for the evolution operator, and it can be solved iteratively by substituting the expression for $\hat{U}(t',0)$ back into the integral. This leads to a series expansion known as the **Dyson series**:

$$\hat{U}(t,0) = 1 - i\int_0^t \mathrm{d}t'\hat{V}_I(t') + (-i)^2\int_0^t \mathrm{d}t'\int_0^{t'} \mathrm{d}t''\hat{V}_I(t')\hat{V}_I(t'') + \dots$$

This series expansion allows us to compute the evolution operator perturbatively, and it forms the basis for many of the techniques used in many body theory, such as Feynman diagrams and perturbation theory. Each term in the Dyson series corresponds to a different order of interaction, and by summing over all possible interactions, we can capture the full dynamics of the system.

If we take a look at the domain of integration in the Dyson series, we can see that it is ordered in time, with $t' > t''$. This time ordering is crucial for ensuring causality in the theory, and it is often represented using the time-ordering operator $\hat{T}$, which arranges operators in chronological order. The Dyson series can thus be rewritten using the time-ordering operator knowing that

$$\int_0^t \mathrm{d}t'\hat{V}_I(t')\int_0^{t'} \mathrm{d}t''\hat{V}_I(t'') = \frac{1}{2}\int_0^t \mathrm{d}t'V_I(t')\int_0^{t'} \mathrm{d}t''\hat{V}_I(t'') + \frac{1}{2}\int_0^t \mathrm{d}t''\hat{V}_I(t'')\int_0^{t''} \mathrm{d}t'\hat{V}_I(t')$$

where now we can introduce the Heaviside step function $\theta(t)$ to enforce the time ordering of the operators (changing accordingly the integration domain) as

$$= \frac{1}{2}\int_0^t \mathrm{d}t'V_I(t')\int_0^t \mathrm{d}t''\hat{V}_I(t'')\theta(t'-t'') + \frac{1}{2}\int_0^t \mathrm{d}t''\hat{V}_I(t'')\int_0^t \mathrm{d}t'\hat{V}_I(t')\theta(t''-t').$$

Thus now we can write the Dyson series in a compact form using the time-ordering operator:

$$= \frac{1}{2}\int_0^t \mathrm{d}t'\int_0^t \mathrm{d}t''\left(\hat{V}_I(t')\hat{V}_I(t'') + \hat{V}_I(t'')\hat{V}_I(t')\right) = \frac{1}{2}\int_0^t \mathrm{d}t'\int_0^t \mathrm{d}t''\hat{T}_t\left(\hat{V}_I(t')\hat{V}_I(t'')\right),$$

where we have used the definition of the time-ordering operator to combine the two terms into a single integral: the term with the smaller time argument is placed to the right of the term with the larger time argument. Let us make a quick example.

Example (Time ordering of n operators). Consider the case of three operators $\hat{V}_I(t_1)$, $\hat{V}_I(t_2)$, and $\hat{V}_I(t_3)$. The time-ordered product of these operators is given by all the combinations of the operators ordered according to their time arguments:

$$\hat{T}\left(\hat{V}_I(t_1)\hat{V}_I(t_2)\hat{V}_I(t_3)\right) = \frac{1}{3!}\left(V_I(t_1)\hat{V}_I(t_2)\hat{V}_I(t_3)\theta(t_1-t_2)\theta(t_2-t_3) + \dots\right).$$

This means that in the time-ordered product, the operator with the latest time argument (in this case, t_3) is placed to the left, while the operator with the earliest time argument (in this case, t_1) is placed to the right. This ordering ensures that the physical processes described by these operators occur in the correct chronological sequence.

With this new formal way of expressing the ordered integrals, we can now write the Dyson series in a compact form as

$$U(t,0) = \sum_{n=0}^{\infty}\frac{(-i)^n}{n!}\int_0^t \mathrm{d}t_1\cdots\int_0^t \mathrm{d}t_n\hat{T}_t\left(\hat{V}_I(t_1)\cdots\hat{V}_I(t_n)\right).$$

This expression can be recognized as the expansion of the exponential function, and thus we can write the evolution operator formally in a compact form as

$$\hat{U}(t,0) = \hat{T}\exp\left(-i\int_0^t \mathrm{d}t'\hat{V}_I(t')\right).$$

This compact expression encapsulates the entire perturbative expansion of the evolution operator, and it is a fundamental result in many body theory, even if in practice we never compute the evolution operator in this form, but rather we use the Dyson series expansion to compute specific terms in the perturbation theory (i.e. we stop at a certain order in the expansion).

3.1 | Green's Functions

In this tractation, we have to distinguish the theory of bosons and fermions, since the statistics of the particles will affect the properties of the Green's functions. For simplicity, we will focus on the case of bosons, but the generalization to fermions is straightforward and will be discussed at the end of this section.

3.1.1 | Single Particle Green's Function

The single particle Green's function, also known as the propagator, is a fundamental quantity in many body theory that describes the propagation of a single particle in an interacting system. It is defined as the time-ordered expectation value of the field operators:[1]

$$\mathcal{G}^R(\mathbf{r},\sigma,t;\,\mathbf{r}',\sigma',t') = -i\theta(t-t')\langle[\hat{\Psi}_\sigma(\mathbf{r},t),\,\hat{\psi}^\dagger_{\sigma'}(\mathbf{r}',t')]\rangle.$$

This Green's function encodes information about the energy spectrum, the density of states, and the response of the system to external perturbations. It can be used to compute various physical quantities, such as the spectral function, the self-energy, and the effective mass of quasiparticles.

Let us study the formula introduced above:

- The field operators $\hat{\Psi}_\sigma(\mathbf{r},t)$ and $\hat{\psi}^\dagger_{\sigma'}(\mathbf{r}',t')$ are the annihilation and creation operators for particles with spin σ at position $\mathbf{r}$ and time t, and for particles with spin σ' at position $\mathbf{r}'$ and time t', respectively. Since they depend on time, they are in the Heisenberg picture, and thus they evolve according to the full Hamiltonian of the system, including interactions.

- The "^{R}" superscript denotes that this is the retarded Green's function, which means that it describes the response of the system to a perturbation applied at time t' and observed at a later time t. The retarded Green's function is causal, meaning that it vanishes for $t < t'$, which is enforced by the Heaviside step function $\theta(t-t')$.

- The expectation value $\langle\cdots\rangle$ is taken with respect to the ground state of the system, or more generally, with respect to the thermal equilibrium state at a given temperature. This is a **thermal average**:

  $$\langle\cdots\rangle = \frac{1}{Z}\mathrm{Tr}\left(e^{-\beta H}\cdots\right) = \frac{1}{Z}\sum_n \langle n|\, e^{-\beta H}\cdots|n\rangle = \frac{1}{Z}\sum_n e^{-\beta E_n}\langle n|\cdots|n\rangle\,,$$

  where Z is the partition function and β is proportional to the inverse of temperature. We usually don't know the exact spectrum of the complete Hamiltonian, indeed usually only the energy of the ground state, i.e. set to 0, contributes significantly to the sum, and thus we can approximate the thermal average as

  $$\langle\cdots\rangle \approx \langle 0|\cdots|0\rangle\,,$$

  where $|0\rangle$ is the ground state of the system. This approximation is valid at low temperatures, where the contributions from excited states are negligible compared to the ground state.

[1] For fermions it will be the same but with anticommutators.

Let us compute the Green's function at zero temperature for fermions, where we can use the approximation of the thermal average as the expectation value in the ground state $|GS\rangle$:

$$\mathcal{G}^R(\mathbf{r},\sigma,t;\,\mathbf{r}',\sigma',t') = -i\theta(t-t')\,\langle GS|\,\{\hat{\Psi}_\sigma(\mathbf{r},t),\,\hat{\Psi}^\dagger_{\sigma'}(\mathbf{r}',t')\}\,|GS\rangle\,,$$

where we are creating a particle at time t' and position $\mathbf{r}'$ and then we are annihilating it at a later time t and position $\mathbf{r}$ (retarded GF). This Green's function describes the amplitude for a particle to propagate from $(\mathbf{r}',t')$ to $(\mathbf{r},t)$ in the presence of interactions. It contains information about the energy spectrum of the system, the density of states, and the response to external perturbations.

This objects is foundamentally an amplitude for a particle to propagate from one point in space and time to another, without taking into account all the infinite possible interactions that can happen in between, and thus it is a very useful quantity to understand the behavior of the system. It is also important to note that the Green's function measures the probability to destroy the previously created particle, in exactely the same state in which it was created.

We are using the commutator and not only the product of the two operators because we want to ensure that the Green's function is properly symmetrized (or antisymmetrized for fermions) and to ensure causality: it can be showed that the commutator (or anticommutator) is the interesting expectation value that captures the correct physical behavior of the system and of the operators, but we will not spend time on this point, since it is not crucial for the understanding of the many body theory.

Remember the structure of the field operators as a sum over quantum states:

$$\hat{\Psi}_\sigma(\mathbf{r},t) = \sum_\nu \phi_\nu(\mathbf{r})\hat{c}_{\alpha,\sigma}(t) = \sum_k \frac{1}{\sqrt{\Omega}} e^{i\mathbf{k}\cdot\mathbf{r}}\hat{c}_{\mathbf{k},\sigma}(t),$$

which can be inserted into the definition of the Green's function to express it in terms of the creation and annihilation operators in momentum space of the first quantization:

$$\begin{aligned}\mathcal{G}^R(\mathbf{r},\sigma,t;\,\mathbf{r}',\sigma',t') &= -i\theta(t-t')\,\langle GS|\,\{\hat{\Psi}_\sigma(\mathbf{r},t),\,\hat{\Psi}^\dagger_{\sigma'}(\mathbf{r}',t')\}\,|GS\rangle = \\ &= -i\theta(t-t')\,\langle GS|\left\{\sum_k \phi_\nu(\mathbf{r})\hat{c}_{\mathbf{k},\sigma}(t),\,\sum_{k'}\phi_{\nu'}(\mathbf{r}')\hat{c}^\dagger_{\mathbf{k}',\sigma'}(t')\right\}|GS\rangle = \\ &= -i\theta(t-t')e^{i(\mathbf{k}\cdot\mathbf{r}-\mathbf{k}'\cdot\mathbf{r}')}e^{-i(\epsilon_k t-\epsilon_{k'}t')} = -ie^{-i(\epsilon_k(t-t'))}e^{i(\mathbf{k}\cdot(\mathbf{r}-\mathbf{r}'))},\end{aligned}$$

TODO: Write correct ecxpression.

which in the momentum space can be expressed as

$$\mathcal{G}^R(\mathbf{k},\sigma,t;t') = -i\theta(t-t')\langle\{\hat{c}_{\mathbf{k},\sigma}(t),\,\hat{c}^\dagger_{\mathbf{k},\sigma}(t')\}\rangle,$$

since the Hamiltonian is diagonal in momentum space, and thus the Green's function is also diagonal in momentum space. Let us use this to compute a free electron Green's function.

Example (Green's function for free electrons). Consider a system of free electrons described by the Hamiltonian

$$\hat{H} = \hat{H}_0 = \sum_{\mathbf{k},\sigma}\xi_{\mathbf{k}}\hat{c}^\dagger_{\mathbf{k},\sigma}\hat{c}_{\mathbf{k},\sigma},$$

where $\xi_{\mathbf{k}}$ is the energy of an electron with momentum $\mathbf{k}$:

$$\xi_{\mathbf{k}} = \epsilon_{\mathbf{k},\sigma} - \mu = \frac{\hbar^2k^2}{2m} - \mu.$$

The Green's function for this system can be computed using the definition:

$$\mathcal{G}^R(\mathbf{k},\sigma,t;t') = -i\theta(t-t')\langle\{\hat{c}_{\mathbf{k},\sigma}(t),\,\hat{c}^\dagger_{\mathbf{k},\sigma}(t')\}\rangle.$$

Since the Hamiltonian is diagonal in momentum space and we are in the Heisenberg picture, we can express the time evolution of the creation and annihilation operators as

$$\frac{\mathrm{d}}{\mathrm{dt}}\hat{c}_{\mathbf{k},\sigma}(t) = i\left[\hat{H}, \hat{c}_{\mathbf{k},\sigma}(t)\right] = ie^{i\hat{H}t}\left[\hat{H}, \hat{c}_{\mathbf{k},\sigma}\right]e^{-i\hat{H}t},$$

where expanding the commutator we get

$$\left[\hat{H}, \hat{c}_{\mathbf{k},\sigma}\right] = \sum_{\mathbf{k}',\sigma'} \xi_{\mathbf{k}'}\left[\hat{c}^{\dagger}_{\mathbf{k}',\sigma'}\hat{c}_{\mathbf{k}',\sigma'},\, \hat{c}_{\mathbf{k},\sigma}\right] = -\xi_{\mathbf{k}}\hat{c}_{\mathbf{k},\sigma},$$

where we have expanded the commutator and used the anticommutation relations of the fermionic operators $[AB, C] = A\{B, C\} - \{A, C\}B$. Thus we have

$$\frac{\mathrm{d}}{\mathrm{dt}}\hat{c}_{\mathbf{k},\sigma}(t) = -i\xi_{\mathbf{k}}\hat{c}_{\mathbf{k},\sigma}(t) = -i\xi_{\mathbf{k}}e^{i\hat{H}t}\hat{c}_{\mathbf{k},\sigma}e^{-i\hat{H}t},$$

showing that the fermionic annihilation operator for a non interacting system evolves in time as the previous expression, and thus we can write the Green's function as **TODO:** correct expression.

$$\begin{aligned}\mathcal{G}^R(\mathbf{k}, \sigma, t; t') &= -i\theta(t-t')\langle\{e^{i\hat{H}t}\hat{c}_{\mathbf{k},\sigma}e^{-i\hat{H}t},\, e^{i\hat{H}t'}\hat{c}^{\dagger}_{\mathbf{k},\sigma}e^{-i\hat{H}t'}\}\rangle \\ &= -i\theta(t-t')e^{-i\xi_{\mathbf{k}}(t-t')}.\end{aligned}$$

We found the Fermi function, since the thermal everage of the number operator $\hat{n}_{\mathbf{k},\sigma} = \hat{c}^{\dagger}_{\mathbf{k},\sigma}\hat{c}_{\mathbf{k},\sigma}$ for fermions is given by the Fermi-Dirac distribution:

$$\langle\hat{c}^{\dagger}_{\mathbf{k},\sigma}\hat{c}_{\mathbf{k},\sigma}\rangle = \frac{1}{Z}\mathrm{Tr}\left(e^{-\beta H}(\hat{c}^{\dagger}_{\mathbf{k},\sigma}\hat{c}_{\mathbf{k},\sigma})\right) = \frac{1}{Z}\frac{\sum_{n_{\mathbf{k}}=0,1}\hat{n}_{\mathbf{k}}e^{-\beta\xi_{\mathbf{k}}\hat{n}_{\mathbf{k}}}}{\sum_{n_{\mathbf{k}}=0,1}e^{-\beta\xi_{\mathbf{k}}\hat{n}_{\mathbf{k}}}} = \frac{1}{e^{\beta\xi_{\mathbf{k}}}+1} = n_F(\xi_{\mathbf{k}}).$$

Going back to the Green's function, we can also express the other thermal average as $1 - \hat{n}_{\mathbf{k}}$, using the anticommutation relations of the fermionic operators, and thus we can write the Green's function of a free electron system as

$$\mathcal{G}^R(\mathbf{k}, \sigma, t; t') = -i\theta(t-t')e^{-i\xi_{\mathbf{k},\sigma}(t-t')}, \quad \propto e^{-\frac{t}{\tau}}.$$

For an interacting hamiltonian, the Green's function will have a more complicated form, and it will contain information about the interactions and correlations in the system. The longer the time difference $t - t'$, the more interactions and correlations will be taken into account in the Green's function, and thus it will deviate more from the free electron case.

Thus the idea is to arrive and compute the thermal average for the interacting system, which is a very hard task, and thus we will use perturbation theory to compute the Green's function for the interacting system, starting from the free electron Green's function as the zeroth order approximation. This will allow us to capture the effects of interactions and correlations in a systematic way, and to understand how they modify the properties of the system compared to the non-interacting case.

Instead of time, usually the central quantity of interest is the Fourier transform of the Green's function in frequency space, which is given by

$$\mathcal{G}^R(\mathbf{k}, \sigma, \omega) = \int_{-\infty}^{\infty}\mathrm{d}te^{i\omega t}\mathcal{G}^R(\mathbf{k}, \sigma, t; 0) = \int_{-\infty}^{\infty}\mathrm{d}te^{i\omega t}(-i)\theta(t)e^{-i\xi_{\mathbf{k}}t}e^{-\delta t},$$

where we have used the Fourier transform of the Heaviside step function and the exponential function; we set $t' = 0$ since it just corresponds to a shift in the time variable, and thus it does not affect the physics of the problem. Furthermore we added a small imaginary part δ to the exponent to ensure convergence of the integral, which is a common technique in many body theory to handle the singularities that can arise in the Green's function. The result of this integral, after the integration of the delta, is

$$\mathcal{G}^R(\mathbf{k},\sigma,\omega) = -i\int_0^\infty \mathrm{d}t e^{i(\omega-\xi_{\mathbf{k},\sigma}+i\delta)t} = -i\left[\frac{e^{i(\omega-\xi_{\mathbf{k},\sigma}+i\delta)t}}{i(\omega-\xi_{\mathbf{k},\sigma}+i\delta)}\right]_0^\infty = \frac{1}{\omega-\xi_{\mathbf{k},\sigma}+i\delta},$$

where the importance of the small exponential factor $e^{-\delta t}$ is evident, since it ensures that the integral converges and that the Green's function has the correct analytic properties in the complex frequency plane. The poles of the Green's function in the complex frequency plane correspond to the energy levels of the system, and the imaginary part of the poles gives information about the lifetime of the quasiparticles in the system.

Every time we Fourier transform the Green's function, we are changing the representation of the Green's function from the time domain to the frequency domain, and thus we are changing the way we look at the physics of the problem. The δ term in the Green's function is crucial for ensuring that the Green's function has the correct analytic properties: it is a slight shift in the complex frequency plane that ensures that the poles of the Green's function are located in the correct half-plane, which is essential for maintaining causality and for correctly describing the response of the system to external perturbations.

We found $\mathcal{G}_0^R(\mathbf{k},\sigma,\omega)$ the retarded Green's function for a free electron system, which is the starting point for our perturbative expansion

$$\mathcal{G}_0^R(\mathbf{k},\sigma,\omega) = \frac{1}{\omega-\xi_{\mathbf{k},\sigma}+i\delta}.$$

3.1.2 | Spectral Function

Associated to this green's function, we can also define the **spectral function**, which is given by the imaginary part of the Green's function:

$$A(\mathbf{k},\sigma) = -\frac{1}{\pi}\mathrm{Im}\mathcal{G}_0^R(\mathbf{k},\sigma,\omega) = -\frac{1}{\pi}\mathrm{Im}\left(\frac{1}{\omega-\xi_{\mathbf{k},\sigma}+i\delta}\right),$$

where we know

$$\frac{1}{x+i\delta} = \mathcal{P}\left(\frac{1}{x}\right) - i\pi\delta(x),$$

where $\mathcal{P}$ denotes the Cauchy principal value, and thus we can write the spectral function as

$$A(\mathbf{k},\sigma) = -\frac{1}{\pi}\mathrm{Im}\left(\frac{1}{\omega-\xi_{\mathbf{k},\sigma}+i\delta}\right)\frac{\omega-\xi_{\mathbf{k},\sigma}-i\delta}{\omega-\xi_{\mathbf{k},\sigma}-i\delta}$$

where we multiplied and divided by the denominator with a changed sign in front of $i\delta$, and computing the imaginary part we get

$$A(\mathbf{k},\sigma) = -\frac{1}{\pi}\mathrm{Im}\left(\frac{\omega-\xi_{\mathbf{k},\sigma}-i\delta}{(\omega-\xi_{\mathbf{k},\sigma})^2+\delta^2}\right) = \frac{1}{\pi}\left(\frac{\delta}{(\omega-\xi_{\mathbf{k},\sigma})^2+\delta^2}\right) \xrightarrow{\delta\to 0} \frac{1}{\pi}\frac{\delta}{(\omega-\xi_{\mathbf{k},\sigma})^2+\delta^2},$$

since the lorentzian function $\frac{1}{\pi}\frac{\delta}{x^2+\delta^2}$ converges to the Dirac delta function $\delta(x)$ in the limit $\delta \to 0$, we can write the spectral function as

$$A(\mathbf{k},\sigma) = \delta(\omega-\xi_{\mathbf{k},\sigma}),$$

which means that the spectral function for a free electron system is a delta function centered at the energy $\xi_{\mathbf{k},\sigma}$. This indicates that the energy levels of the system are sharply defined, and that the quasiparticles have an infinite lifetime (since there is no broadening of the spectral function).

Thus the only possible energy for a particle with momentum $\mathbf{k}$ and spin σ is $\xi_{\mathbf{k},\sigma}$, and the spectral function tells us that the density of states at this energy is infinite, while it is zero at all other energies. This is a characteristic feature of non-interacting systems, where the energy levels are discrete and sharply defined. In contrast, in interacting systems, the spectral function will generally have a more complex structure that reflect the effects of interactions and correlations in the system.

If we now add a term in the Green's function to account for the interactions in the system: a time exponential decay term $e^{-\frac{t-t'}{\Gamma}}$, where Γ takes into account the interactions and correlations in the system, and thus it gives a finite lifetime to the quasiparticles in the system. The Green's function for an interacting system can thus be written as

$$\mathcal{G}^R(\mathbf{k},\sigma,t;t') = -i\theta(t-t')e^{-i\xi_{\mathbf{k},\sigma}(t-t')}e^{-\frac{t-t'}{\Gamma}},$$

where the exponential decay term $e^{-\frac{t-t'}{\Gamma}}$ captures the effects of interactions and correlations in the system, and it leads to a broadening of the spectral function, which indicates that the energy levels of the system are no longer sharply defined, and that the quasiparticles have a finite lifetime. The Fourier transform of this Green's function will give us a more complex expression for the Green's function in frequency space, which will contain information about the interactions and correlations in the system. The spectral function for this interacting system can be computed as

$$A(\mathbf{k},\sigma) = -\frac{1}{\pi}\mathrm{Im}\mathcal{G}^R(\mathbf{k},\sigma,\omega) = \frac{1}{\pi}\frac{(1/\Gamma)}{(\omega-\xi_{\mathbf{k},\sigma})^2+(1/\Gamma)^2},$$

where we have used the Fourier transform of the Green's function with the exponential decay term, and we can see how the interactions and correlations in the system lead to a broadening of the spectral function, which indicates that the energy levels of the system are no longer sharply defined, and that the quasiparticles have a finite lifetime. The width of the spectral function is given by $1/\Gamma$, which is inversely proportional to the lifetime of the quasiparticles: the shorter the lifetime, the broader the spectral function, and vice versa. This illustrates how interactions and correlations in the system can lead to a finite lifetime for quasiparticles, and how this is reflected in the properties of the Green's function and the spectral function.

TODO: Add image of parabula with delta functions and lorentzian functions.

Thus given the interaction we need a machinery capable of computing Γ, in order to obtain the Green's function for the interacting system, and thus to understand the properties of the system in the presence of interactions and correlations. This is where perturbation theory and Feynman diagrams come into play, as they provide a systematic way to compute the effects of interactions on the Green's function and to extract physical quantities such as Γ from the theory. In a certain sense we are measuring how many states are available for the particle to decay into, and thus how many interactions it can have with other particles in the system, which is what gives rise to the finite lifetime of the quasiparticles in the system. Thus we can generalize the Densitiy of States (DeS) to the case of an interacting system, by using the spectral function, which is given by

$$\mathrm{DeS}(\omega) = \sum_{\mathbf{k},\sigma} A(\mathbf{k},\sigma,\omega),$$

where the spectral function $A(\mathbf{k},\sigma,\omega)$ captures the effects of interactions and correlations in the system, and thus it gives us a more accurate picture of the density of states in the presence of interactions. The density of states for an interacting system will generally have a more complex structure than the non-interacting case, and it will reflect the effects of interactions and correlations in the system, such as the broadening of energy levels and the emergence of new features in the density of states due to interactions.

Experiments in angular resolved photoemission spectroscopy (ARPES) can directly measure the spectral function $A(\mathbf{k}, \sigma, \omega)$ of a material, providing insights into the electronic structure and interactions in the system. By analyzing the ARPES data, we can extract information about the energy dispersion, the lifetime of quasiparticles, and the effects of interactions and correlations in the material. This makes the spectral function a crucial quantity for understanding the physics of condensed matter systems and for interpreting experimental results.

3.2 | Imaginary Time Formalism

Let us assume $t > 0$, then we can write the Green's function as

$$\mathcal{G}^R(\mathbf{k}, \sigma, t) = -i \left\langle \{\hat{c}_{\mathbf{k},\sigma}(t),\, \hat{c}^\dagger_{\mathbf{k},\sigma}\} \right\rangle = \dots$$

where we have used the definition of the Green's function and the time evolution of the annihilation operator in the Heisenberg picture. Now we are interested in computing the **correlation function** for an interacting system, which is given by

$$\begin{aligned}\mathcal{C}_{AB}(t, t') = \langle \hat{A}(t)\hat{B}(t')\rangle &= -\frac{1}{Z}\mathrm{Tr}\left(e^{-\beta H}\hat{A}(t)\hat{B}(t')\right)\\ &= -\frac{1}{Z}\mathrm{Tr}e^{-\beta H}\hat{U}(0,t)\hat{A}(t)\hat{U}(t,0)\hat{U}(0,t')\hat{B}(t')\hat{U}(t',0),\end{aligned}$$

where we have inserted the identity operator in the form of the time evolution operator $\hat{U}(t, 0)$ and $\hat{U}(0, t)$ to express the operators in the Heisenberg picture in terms of the operators in the Schrödinger picture. Now we can use the definition of the time evolution operator to write

$$\hat{A}(t) = e^{iH_0t}\hat{A}e^{-iH_0t}.$$

Now the idea is to express the time evolution operator in the interaction picture, which allows us to separate the effects of the non-interacting part of the Hamiltonian from the effects of the interactions. Furthermore from the Dyson series expansion of the time evolution operator, we can express

$$\hat{U}(t, 0) = \hat{T}\exp\left(-i\int_0^t \mathrm{d}t'\hat{V}_I(t')\right).$$

The trick is to substitute the time with a complex counterpart:

$$t \to -i\tau,$$

where τ is a real variable that we will call imaginary time. This substitution allows us to transform the time evolution operator into a form that is more amenable to perturbative calculations, and it also allows us to connect the Green's function to the partition function of the system, which is a central quantity in statistical mechanics. We can write the operators as

$$A(\tau) = e^{H\tau}Ae^{-H\tau},$$

...

and for the time evolution operator we have

$$\hat{U}(\tau, 0) = \hat{T}\exp\left(-\tau H_0 + \tau\hat{H}\right),$$

ao that if we want to compute the evolution operator in the interaction picture, we can write

$$\frac{\mathrm{d}}{\mathrm{d}\tau}\hat{U}_I(\tau, 0) = -i\hat{V}_I(\tau)\hat{U}_I(\tau, 0),$$

and we can compute the time evolution operator in the interaction picture using the Dyson series expansion as

$$\hat{U}_I(\tau, 0) = \hat{T}_\tau\exp\left(-i\int_0^\tau \mathrm{d}\tau'\hat{V}_I(\tau')\right) = 1-i\int_0^\tau \mathrm{d}\tau'\hat{V}_I(\tau')+(-1)^2\int_0^\tau \mathrm{d}\tau'\int_0^{\tau'} \mathrm{d}\tau''\hat{V}_I(\tau')\hat{V}_I(\tau'')+\dots$$

where we have used the definition of the time-ordering operator in imaginary time, which orders the operators according to their imaginary time arguments, with the operator with the largest imaginary time argument placed to the left. This allows us to express the time evolution operator in a form that is more suitable for perturbative calculations, and it also allows us to connect the Green's function to the partition function of the system, which is a central quantity in statistical mechanics.

TODO: Check the sign in the exponentials.

If we operate the substitution $\tau = \beta$ we get

$$U(\beta, 0) = e^{\beta H_0} e^{-\beta H} \implies e^{-\beta H} = e^{-\beta H_0} \hat{U}(\beta, 0).$$

We thus found an expression for the density matrix of the system in terms of the time evolution operator in the interaction picture, which allows us to connect the Green's function to the partition function of the system, and thus to compute the Green's function using perturbation theory. This is a crucial step in many body theory, as it allows us to compute physical quantities such as the Green's function and the spectral function in the presence of interactions and correlations, but in the expression is teh free hamiltonian to appear.

The interacting correlation function can thus be expressed as (assuming $\tau > \tau'$):

$$\begin{aligned} C_{AB}(\tau, \tau') &= -\left\langle \hat{A}(\tau) \hat{B}(\tau') \right\rangle \\ &= -\frac{1}{Z} \mathrm{Tr} e^{-\beta H} \hat{U}(0, \tau) \hat{A}(\tau) \hat{U}(\tau, 0) \hat{U}(0, \tau') \hat{B}(\tau') \hat{U}(\tau', 0) \\ &= -\frac{1}{Z} \mathrm{Tr} e^{-\beta H_0} \hat{U}(\beta, 0) \hat{U}(0, \tau) \hat{A}(\tau) \hat{U}(\tau, 0) \hat{U}(0, \tau') \hat{B}(\tau') \hat{U}(\tau', 0), \end{aligned}$$

where we have used the expression for the density matrix in terms of the time evolution operator in the interaction picture, and we can now use the previously derived expression for the time evolution operator in the interaction picture to express the correlation function in a form that is more suitable for perturbative calculations. This allows us to compute the Green's function and other physical quantities in the presence of interactions and correlations, using perturbation theory and Feynman diagrams to systematically account for the effects of interactions in the system. Thus

$$C_{AB}(\tau, \tau') = -\frac{1}{Z} \mathrm{Tr} e^{-\beta H_0} \hat{U}(\beta, \tau) \hat{A}(\tau) \hat{U}(\tau, \tau') \hat{B}(\tau') \hat{U}(\tau', 0),$$

and we will see how the imaginary time τ is limited in the range $0 > \tau < \beta$, and the same for τ'. But let us assume it for now. This expression written as this is *tau-ordered*.

We can proceed noting how sign do not change if we reorder some operators, so that

$$\begin{aligned} C_{AB}(\tau, \tau') &= -\frac{1}{Z} \mathrm{Tr} e^{-\beta H_0} \hat{T}_\tau \left\{ \hat{U}(\beta, \tau) \hat{U}(\tau, \tau') \hat{U}(\tau', 0) \hat{A}(\tau) \hat{B}(\tau') \right\} \\ &= -\frac{1}{Z} \mathrm{Tr} e^{-\beta H_0} \hat{T}_\tau \left\{ \hat{U}(\beta, 0) \hat{A}(\tau) \hat{B}(\tau') \right\}, \end{aligned}$$

This is a crucial step in the derivation, as it allows us to express the correlation function in a form that is more suitable for perturbative calculations, and it also allows us to connect the Green's function to the partition function of the system, which is a central quantity in statistical mechanics. The final expression for the correlation function in imaginary time is thus

$$C_{AB}(\tau, \tau') = -\frac{1}{Z} \mathrm{Tr} e^{-\beta H_0} \hat{T}_\tau \left\{ \hat{U}(\beta, 0) \hat{A}(\tau) \hat{B}(\tau') \right\},$$

which is a trace with respect to the non-interacting Hamiltonian H_0 of the time-ordered product of the time evolution operator in the interaction picture and the operators A and B in imaginary time (in Dirac picture, thus evolving with the non-interacting Hamiltonian). But as a normalization, we have the inverse of the partition function, which is still accouinting for the interactions

$$Z = \mathrm{Tr}\, e^{-\beta H} = \mathrm{Tr}\, e^{-\beta H_0} \hat{U}(\beta, 0).$$

If we multiply and derive for the non interacting partition function

$$C_{AB}(\tau, \tau') = -\frac{Z_0}{Z_0} \frac{1}{Z} \mathrm{Tr} e^{-\beta H_0} \hat{T}_\tau \left\{ \hat{U}(\beta, 0) \hat{A}(\tau) \hat{B}(\tau') \right\},$$

we can rewrite the correlation function as

$$C_{AB}(\tau, \tau') = -\frac{\left\langle \hat{U}(\beta, 0) \hat{A}(\tau) \hat{B}(\tau') \right\rangle_0}{\left\langle \hat{U}(\beta, 0) \right\rangle_0},$$

where the $\frac{1}{Z_0}$ added let us write the thermal average at the numerator with respect to the non-interacting Hamiltonian H_0, while the inverse of $\frac{Z_0}{Z}$ let us write the thermal average at the denominator with respect to the non-interacting Hamiltonian H_0 as well.

This is a crucial step in the derivation, as it allows us to express the correlation function in a form that is more suitable for perturbative calculations, and it also allows us to connect the Green's function to the partition function of the system, which is a central quantity in statistical mechanics. The only price we are forced to pay is that we have to compute the time evolution operator in the interaction picture, which is given by the Dyson series expansion, and thus we have to compute the effects of interactions in a perturbative way, using Feynman diagrams to systematically account for the effects of interactions in the system. This is the essence of many body theory, and it allows us to compute physical quantities such as the Green's function and the spectral function in the presence of interactions and correlations, which are crucial for understanding the physics of condensed matter systems.

Remark (Tau ordering for fermions). *Let us define the tau ordering for fermions (since for bosons the sign does not change) as*

$$[\dots]$$

for three operators $C(\tau_1)$, $C(\tau_2)$ and $C(\tau_3)$ we would have had

$$[\dots]$$

We had $\tau > \tau'$ such that

$$\langle C(\tau) C(\tau') \rangle \to U(\beta, 0) U(0, \tau) \hat{C}(\tau) U(\tau, 0) U(0, \tau') \hat{B}(\tau') U(\tau', 0),$$

and remember that $U(\tau, \beta) = e^{-i \int_\tau^\beta V(\tau) \mathrm{d}\tau}$, and in the interaction integral we usually have an even number of ladder operators, since every time we have an annihilation operator we have a creation operator, so that the sign does not change if we reorder the time evolution operators (it follows from the anticommutation relations of the ladder operators).

When instead we want to swap the two operators $C(\tau)$ and $C(\tau')$ we have to consider their anticommutation relations, so that we have a sign change. The tau ordering only assures the correct ordering of the time evolution operators, but it does not account for the sign change due to the anticommutation relations of the fermionic operators.

3.2.1 | Correlation Functions in Imaginary Time

In the imaginary time formalism, we can define the correlation function for two operators A and B as

$$\mathcal{C}_{AB}(\tau,\tau') = -\left\langle \hat{T}_\tau \left\{ \hat{A}(\tau)\hat{B}(\tau') \right\} \right\rangle,$$

which would be the same for bosonic or fermionic operators. We can compute further the correlation function as

$$\mathcal{C}_{AB}(\tau,\tau') = -\theta(\tau-\tau') \left\langle \hat{A}(\tau)\hat{B}(\tau') \right\rangle \mp \theta(\tau'-\tau) \left\langle \hat{B}(\tau')\hat{A}(\tau) \right\rangle,$$

where the minus sign is for bosonic operators and the plus sign is for fermionic operators, and we have used the definition of the tau ordering operator to express the correlation function in terms of the Heaviside step function θ and the thermal averages of the operators A and B in imaginary time. This expression allows us to compute the correlation function in imaginary time, which is a crucial quantity in many body theory, as it allows us to compute physical quantities such as the Green's function and the spectral function in the presence of interactions and correlations, using perturbation theory and Feynman diagrams to systematically account for the effects of interactions in the system.

We use imaginary time to put on equal foot the time evolution and the thermal average: the exponentials appearing in the time evolution and in the thermal average are of the same form (the imaginary unit vanished), and thus we can use the same techniques to compute both quantities, which allows us to connect the Green's function to the partition function of the system, which is a central quantity in statistical mechanics. It is just more convenient to work with imaginary time, since it allows us to use the same techniques to compute both the time evolution and the thermal average, and control the convergence of the series: with imaginary exponentials we have oscillations that are difficult to manage, but with real exponentials we have a more straightforward convergence behavior; then we can analytically continue the results to real time if we are interested in computing physical quantities in real time, such as when comparing experimental results with theoretical predictions, since most experiments are performed in real time.

If we assume $-\beta < \tau < 0$ and $\tau' = 0$, we can write the correlation function as

$$\mathcal{C}_{AB}(\tau,0) = \mp \left\langle \hat{B}(0)\hat{A}(\tau) \right\rangle = \mp \frac{1}{Z} \operatorname{Tr}\Big(e^{-\beta \hat{H}} \hat{B}(0) e^{\tau \hat{H}} \hat{A} e^{-\tau \hat{H}} \Big),$$

but since the trace is cyclic we can write

$$\mathcal{C}_{AB}(\tau,0) = \mp \frac{1}{Z} \operatorname{Tr}\Big(e^{\tau \hat{H}} \hat{A} e^{-\tau \hat{H}} e^{-\beta \hat{H}} \hat{B}(0) \Big) = \mp \frac{1}{Z} \operatorname{Tr}\Big(e^{-\tau \hat{H}} \hat{A} e^{-(\tau+\beta)\hat{H}} \hat{B}(0) \Big) = \pm \mathcal{C}_{AB}(\tau+\beta,0),$$

thus the correlation function is periodic in imaginary time with period β for bosonic operators, and antiperiodic in imaginary time with period β for fermionic operators.

TODO: add plot of periodical correlation functions.

We can clearly notice how periodically with β on the imaginary time axis the correlation function repeats itself, and how the sign changes for fermionic operators, while it does not change for bosonic operators. It will be clear in the future why they have a similar plot, for now we can just notice the periodicity and the antiperiodicity of the correlation function in imaginary time. Also an interesting feature of this correlation functions is that the jump at $\tau = 0$ is related to the commutation or anticommutation relations of the operators A and B, and thus it is 1 for both bosonic and fermionic operators, but the sign of the jump is changes.

3.2.2 | Imaginary Frequencies

Since the correlation function $\mathcal{C}_{AB}(\tau, 0)$ is periodic in imaginary time, we can express it as a Fourier series in imaginary time, which allows us to express the correlation function in terms of its Fourier coefficients, which are called the *Matsubara frequencies*. The Fourier series expansion of the correlation function in imaginary time is given by

$$\mathcal{C}_{AB}(\tau, 0) = \frac{1}{\beta}\sum_{n=-\infty}^{+\infty} e^{-i\pi n\frac{\tau}{\beta}}\tilde{\mathcal{C}}_{AB}(n) = \begin{cases} \dfrac{1}{\beta}\displaystyle\sum_{n=-\infty}^{+\infty} e^{-i\pi(2n)\frac{\tau}{\beta}}\tilde{\mathcal{C}}_{AB}(n), & \text{for bosonic operators,} \\ \dfrac{1}{\beta}\displaystyle\sum_{n=-\infty}^{+\infty} e^{-i\pi(2n+1)\frac{\tau}{\beta}}\tilde{\mathcal{C}}_{AB}(n), & \text{for fermionic operators.} \end{cases}$$

Indeed if we were to commpute these transforms at $\tau + n'\beta$ we would have the same expression with or without the sign change: it automatically enforces the statistics of the considered particles.

Usually the last expression is written in a different form:

$$\mathcal{C}_{AB}(\tau, 0) = \begin{cases} \dfrac{1}{\beta}\displaystyle\sum_{n=-\infty}^{+\infty} e^{-i\omega_n\tau}\tilde{\mathcal{C}}_{AB}(\omega_n), & \text{for bosonic operators,} \\ \dfrac{1}{\beta}\displaystyle\sum_{n=-\infty}^{+\infty} e^{-i\omega_n\tau}\tilde{\mathcal{C}}_{AB}(\omega_n), & \text{for fermionic operators,} \end{cases}$$

where we have defined the **Matsubara frequencies** as

$$\begin{cases} \omega_n = 2n\pi/\beta, & \text{for bosonic operators,} \\ \omega_n = (2n+1)\pi/\beta, & \text{for fermionic operators.} \end{cases}$$

For fermions $\omega_n \neq 0$ for every value of n, while for bosons $\omega_n = 0$ for $n = 0$. We can show haw this is a direct consequence of the bose or fermi distributions, which are given by

$$n_F(\xi) = \frac{1}{e^{\beta\xi} + 1}, \qquad n_B(\xi) = \frac{1}{e^{\beta\xi} - 1},$$

where ξ is the imaginary energy of the state. We have to look out for every case in which the denominator of the distribution function can vanish, since this would lead to a divergence in the distribution function, and thus to a divergence in the correlation function.

For fermions for example we get

$$e^{-\beta\xi} = -1 = e^{i\pi}e^{2\pi n} \implies \xi = i\pi(2n+1)/\beta,$$

which are exactly the fermionic Matsubara frequencies. For bosons instead we get

$$e^{-\beta\xi} = 1 = e^{2\pi i n} \implies \xi = i2\pi n/\beta,$$

which are exactly the bosonic Matsubara frequencies. Thus the Matsubara frequencies are a direct consequence of the statistics of the particles, and they are crucial for understanding the behavior of the correlation function in imaginary time, and thus for computing physical quantities such as the Green's function and the spectral function in the presence of interactions and correlations, using perturbation theory and Feynman diagrams to systematically account for the effects of interactions in the system.

The temperature is the residue of the poles in the density of states.

Of course, if we know the imaginary energy correlation function, we can compute the correlation function in imaginary time by performing the inverse Fourier transform, which is given by

$$\tilde{\mathcal{C}}_{AB}(\omega_n) = \int_0^\beta \mathrm{d}\tau' e^{i\omega_n\tau'}\mathcal{C}_{AB}(\tau').$$

We have always treated many body physics at zero temperature, where ω_n becomes a continuous variable, but the imaginary time formalism allows us to treat many body physics at finite temperature, where ω_n is a discrete variable, and thus we can compute physical quantities.

3.2.3 | Matsubara Green's Function

As we have seen, the building block of many body theory is the imaginary time non-interacting correlation function, which is given by

$$\mathcal{G}_0(\mathbf{k},\sigma,\tau) = -\left\langle \hat{T}_\tau \left\{ \hat{c}_{\mathbf{k},\sigma}(\tau)\hat{c}^\dagger_{\mathbf{k},\sigma}(0) \right\} \right\rangle_0,$$

in the Heisemberg picture, where the operators evolve with the non-interacting Hamiltonian H_0

$$\frac{\mathrm{d}}{\mathrm{d}\tau}\hat{c}_{\mathbf{k},\sigma}(\tau) = [H_0, \hat{c}_{\mathbf{k},\sigma}(\tau)], \implies \begin{cases} \hat{c}_{\mathbf{k},\sigma}(\tau) = e^{-\xi_{\mathbf{k}}\tau}\hat{c}_{\mathbf{k},\sigma}, \\ \hat{c}^\dagger_{\mathbf{k},\sigma}(\tau) = e^{\xi_{\mathbf{k}}\tau}\hat{c}^\dagger_{\mathbf{k},\sigma}, \end{cases}$$

where we have no more imaginary unit in the exponentials, since we are working with imaginary time, and $H_0 = \sum_{\mathbf{k},\sigma} \xi_{\mathbf{k}}\hat{c}^\dagger_{\mathbf{k},\sigma}\hat{c}_{\mathbf{k},\sigma}$. Thus we can compute the non-interacting correlation function as

$$\begin{aligned} \mathcal{G}_0(\mathbf{k},\sigma,\tau) &= -\theta(\tau)\left\langle \hat{c}_{\mathbf{k},\sigma}(\tau)\hat{c}^\dagger_{\mathbf{k},\sigma}(0)\right\rangle_0 + \theta(-\tau)\left\langle \hat{c}^\dagger_{\mathbf{k},\sigma}(0)\hat{c}_{\mathbf{k},\sigma}(\tau)\right\rangle_0 \\ &= -\theta(\tau)e^{-\xi_{\mathbf{k}}\tau}\left\langle \hat{c}_{\mathbf{k},\sigma}\hat{c}^\dagger_{\mathbf{k},\sigma}\right\rangle_0 + \theta(-\tau)e^{-\xi_{\mathbf{k}}\tau}\left\langle \hat{c}^\dagger_{\mathbf{k},\sigma}\hat{c}_{\mathbf{k},\sigma}\right\rangle_0 \\ &= \left[-\theta(\tau)(1-n_F(\xi_{\mathbf{k}})) + \theta(-\tau)n_F(\xi_{\mathbf{k}})\right]e^{-\xi_{\mathbf{k}}\tau}, \end{aligned}$$

where we have used the definition of the non-interacting correlation function, the time evolution of the operators in the Heisenberg picture, and the definition of the Fermi distribution function $n_F(\xi) = \frac{1}{e^{\beta\xi}+1}$. We can also write the non-interacting correlation function in terms of the Matsubara frequencies as[2]

$$\begin{aligned} \mathcal{G}_0(\mathbf{k},\sigma,\omega_n) &= \int_0^\beta \mathrm{d}\tau e^{i\omega_n\tau}\mathcal{G}_0(\mathbf{k},\sigma,\tau) = -(1-n_F(\xi_{\mathbf{k}}))\int_0^\beta \mathrm{d}\tau e^{i\omega_n\tau}e^{-\xi_{\mathbf{k}}\tau} \\ &= -(1-\frac{1}{e^{\beta\xi}+1})\frac{e^{(i\omega_n-\xi_{\mathbf{k}})\beta}-1}{i\omega_n-\xi_{\mathbf{k}}} = -\frac{1}{e^{\beta\xi_{\mathbf{k}}}+1}\frac{e^{(i\omega_n-\xi_{\mathbf{k}})\beta}-1}{i\omega_n-\xi_{\mathbf{k}}} = \frac{1}{i\omega_n-\xi_{\mathbf{k}}}, \end{aligned}$$

[2]We have taken only one term since in the integral we are fixing $\tau > 0$.

where we used the fact that $e^{i\omega_n\beta} = -1$ for each n for fermions. If we compare the same expression in the real time:

$$\mathcal{G}_0(\mathbf{k}, \sigma, \omega) = \frac{1}{\omega - \xi_\mathbf{k} + i\delta},$$

we can note an heavy similarity between the two expressions, with the only difference being the imaginary unit in the denominator, which is a direct consequence of the fact that we are working with imaginary time instead of real time. Indeed we can pass from one expression to the other by performing the **analytical continuation**

$$i\omega_n \to \omega + i\delta,$$

which is a crucial step in many body theory, as it allows us to connect the results obtained in imaginary time to physical quantities that can be measured in real time, such as the spectral function and the density of states, which are crucial for understanding the physics of condensed matter systems. It is valid in general even if we presented only a specific case, but any correlation function in imaginary time can be analytically continued to real time, and thus we can compute physical quantities in real time starting from the results obtained in imaginary time, which is a powerful technique in many body theory.

Example (Feynman Diagrams). .... TODO: diagram

$$\sum_{i\omega_n} \frac{1}{i\omega_n + i\nu_n - \xi_{\mathbf{k}+\mathbf{q}}} \frac{1}{i\omega_n - \xi_\mathbf{k}},$$

but we will see...

3.3 | Wick's Theorem

Recalling the expansion in non interacting Green's functions, we have

$$\mathcal{C}_{AB}(\tau,\tau') = \frac{\left\langle \hat{T}_\tau\{\hat{U}(\beta,0)\hat{A}(\tau)\hat{B}(\tau')\}\right\rangle_0}{\left\langle \hat{U}(\beta,0)\right\rangle_0},$$

where from the definition of the time evolution operator, we have

$$\begin{aligned}\left\langle \hat{U}(\beta,0)\right\rangle_0 &= \left\langle \hat{T}_\tau \sum_{n=1}^{\infty}\frac{(-1)^n}{n!}\left[-\int_0^\beta d\tau_1\int_0^\beta \mathrm{d}\tau_2\cdots\int_0^\beta \mathrm{d}\tau_n \hat{V}_I(\tau_1)\hat{V}_I(\tau_2)\cdots\hat{V}_I(\tau_n)\right]\right\rangle_0\\ &= \langle 1\rangle_0 + \sum_{n=1}^{\infty}\frac{(-1)^n}{n!}\int_0^\beta d\tau_1\int_0^\beta \mathrm{d}\tau_2\cdots\int_0^\beta \mathrm{d}\tau_n\left\langle \hat{T}_\tau\{\hat{V}_I(\tau_1)\hat{V}_I(\tau_2)\cdots\hat{V}_I(\tau_n)\}\right\rangle_0,\end{aligned}$$

where each of the interaction potential is a product of a different number of creation and annihilation operators. Note how we are producing a sum of ....

Theorem 3.1 (Wick's Theorem)**.** *The expectation value of a time ordered product of creation and annihilation operators can be expressed as the sum of all possible contractions of the operators. We can write this as*

TODO: check computatioons

$$\begin{aligned}&\mathcal{G}_0^{(n)}((\nu_1,\tau_1),(\nu_2,\tau_2),\ldots,(\nu_n,\tau_n);(\nu_1',\tau_1'),(\nu_2',\tau_2'),\ldots,(\nu_n',\tau_n'))\\ &= (-1)^n\left\langle \hat{T}_\tau\{\hat{c}_{\nu_1}(\tau_1)\hat{c}_{\nu_2}(\tau_2)\cdots\hat{c}_{\nu_n}(\tau_n)\hat{c}^\dagger_{\nu_1'}(\tau_1')\hat{c}^\dagger_{\nu_2'}(\tau_2')\cdots\hat{c}^\dagger_{\nu_n'}(\tau_n')\}\right\rangle_0,\end{aligned}$$

where the right hand side can be expressed as the sum of all possible contractions of the operators, which can be written as

$$\mathcal{G}_0^{(n)}((\nu_1,\tau_1),\ldots,(\nu_n',\tau_n')) = \begin{pmatrix}\mathcal{G}_0(\nu_1,\tau_1;\nu_1',\tau_1') & \mathcal{G}_0(\nu_1,\tau_1;\nu_2',\tau_2') & \cdots & \mathcal{G}_0(\nu_1,\tau_1;\nu_n',\tau_n')\\ \mathcal{G}_0(\nu_2,\tau_2;\nu_1',\tau_1') & \mathcal{G}_0(\nu_2,\tau_2;\nu_2',\tau_2') & \cdots & \mathcal{G}_0(\nu_2,\tau_2;\nu_n',\tau_n')\\ \vdots & \vdots & \ddots & \vdots\\ \mathcal{G}_0(\nu_n,\tau_n;\nu_1',\tau_1') & \mathcal{G}_0(\nu_n,\tau_n;\nu_2',\tau_2') & \cdots & \mathcal{G}_0(\nu_n,\tau_n;\nu_n',\tau_n')\end{pmatrix}$$

where based on the statistics of the particles, we have the determinant for fermions and the permanent for bosons.

Example. For the two particle Green's function for non interacting fermions, we have

$$\begin{aligned}\mathcal{G}_0^{(2)} &= \left\langle \hat{T}_\tau\{\hat{c}_{\nu_1}(\tau_1)\hat{c}_{\nu_2}(\tau_2)\hat{c}^\dagger_{\nu_2'}(\tau_2')\hat{c}^\dagger_{\nu_1'}(\tau_1')\}\right\rangle_0\\ &= \mathcal{G}_0(\nu_1,\tau_1;\nu_1',\tau_1')\mathcal{G}_0(\nu_2,\tau_2;\nu_2',\tau_2') - \mathcal{G}_0(\nu_1,\tau_1;\nu_2',\tau_2')\mathcal{G}_0(\nu_2,\tau_2;\nu_1',\tau_1')\\ &= \det\begin{pmatrix}\mathcal{G}_0(\nu_1,\tau_1;\nu_1',\tau_1') & \mathcal{G}_0(\nu_1,\tau_1;\nu_2',\tau_2')\\ \mathcal{G}_0(\nu_2,\tau_2;\nu_1',\tau_1') & \mathcal{G}_0(\nu_2,\tau_2;\nu_2',\tau_2')\end{pmatrix}\\ &= \left\langle \hat{T}_\tau\{\hat{c}_{\nu_1}(\tau_1)\hat{c}^\dagger_{\nu_1'}(\tau_1')\}\right\rangle_0\left\langle \hat{T}_\tau\{\hat{c}_{\nu_2}(\tau_2)\hat{c}^\dagger_{\nu_2'}(\tau_2')\}\right\rangle_0 - \left\langle \hat{T}_\tau\{\hat{c}_{\nu_1}(\tau_1)\hat{c}^\dagger_{\nu_2'}(\tau_2')\}\right\rangle_0\left\langle \hat{T}_\tau\{\hat{c}_{\nu_2}(\tau_2)\hat{c}^\dagger_{\nu_1'}(\tau_1')\}\right\rangle_0,\end{aligned}$$

which are exactly the *exchange energy* and the *direct energy* of the two particle Green's function: it's the Hartree-Fock term for the two particle Green's function.

TODO: check everything

3.4 | Interacting Green's Function

The interating green's function

$$\begin{aligned}\mathcal{G}(\sigma_b,\, \mathbf{r}_b,\, \tau_b,\, \sigma_a,\, \mathbf{r}_a,\, \tau_a) &= -\left\langle \hat{T}_\tau \hat{\Psi}_{\sigma_b}(\mathbf{r}_b, \tau_b)\hat{\Psi}_{\sigma_a}(\mathbf{r}_a, \tau_a)\right\rangle \\ &= -\frac{\left\langle \hat{T}_\tau \{\hat{U}(\beta, 0)\hat{\Psi}_{\sigma_b}(\mathbf{r}_b, \tau_b)\hat{\Psi}_{\sigma_a}(\mathbf{r}_a, \tau_a)\}\right\rangle_0}{\left\langle \hat{U}(\beta, 0)\right\rangle_0},\end{aligned}$$

[...]

$$\mathcal{G}(b, a) = -\frac{\sum_{n=0}^{\infty} \frac{(-1)^n}{n!} \int_0^\beta \mathrm{d}\tau_1 \cdots \int_0^\beta \mathrm{d}\tau_n \left\langle \hat{T}_\tau \hat{V}_{int}(\tau_1)\cdots \hat{V}_{int}(\tau_n)\hat{\Psi}(a)\hat{\Psi}(b)\right\rangle_0}{\sum_{n=0}^{\infty} \frac{(-1)^n}{n!} \int_0^\beta \mathrm{d}\tau_1 \cdots \int_0^\beta \mathrm{d}\tau_n \left\langle \hat{T}_\tau \hat{V}_{int}(\tau_1)\cdots \hat{V}_{int}(\tau_n)\right\rangle_0},$$

$$\int \mathrm{d}\tau_j \hat{V}_{int}(\tau_j) = \frac{1}{2}\sum_{\sigma_j\sigma_{j'}} \int\int \mathrm{d}r_j\, \mathrm{d}r_{j'} \int_0^\beta \mathrm{d}\tau_j \left(\hat{\Psi}^\dagger_{\sigma_j}(\mathbf{r}_j, \tau_j)\hat{\Psi}^\dagger_{\sigma_{j'}}(\mathbf{r}_{j'}, \tau_j)\frac{e^2}{|\mathbf{r}_j - \mathbf{r}_{j'}|}\hat{\Psi}_{\sigma_{j'}}(\mathbf{r}_{j'}, \tau_j)\hat{\Psi}_{\sigma_j}(\mathbf{r}_j, \tau_j)\right),$$

since the fields operators are in the Heisenberg picture we need time; it is important to notice that all these operators are expressed with resoect to the same τ_j.

We can usa a more compact notation, renaming the interacion potential

$$\hat{V}_{int}(j, j') = \frac{e^2}{|\mathbf{r}_j - \mathbf{r}_{j'}|},$$

and we can also use this notation for the field operators, where in the second field we have the two different indices appearing; we can thus add an integral over $\mathrm{d}\tau_{j'}$ with a $\delta(\tau_j, \tau_{j'})$, so that we get:

$$\int \mathrm{d}\tau_j \hat{V}_{int}(\tau_j) = \frac{1}{2}\sum_{\sigma_j\sigma_{j'}} \int\int \mathrm{d}r_j\, \mathrm{d}r_{j'} \int_0^\beta \mathrm{d}\tau_j \int_0^\beta \mathrm{d}\tau_{j'} \left(\hat{\Psi}^\dagger(j)\hat{\Psi}(j')\hat{V}_{int}(j, j')\hat{\Psi}(j)\hat{\Psi}(j')\right),$$

where the integration we added can restore the mixed indices in the second field operator.

We can now rewrite the first expression for the interacting green function as

$$\mathcal{G}(b, a) = -\frac{\sum_{n=0}^{\infty} \frac{(-1/2)^n}{n!} \int \mathrm{d}1\mathrm{d}1' \cdots \mathrm{d}n\mathrm{d}n' \hat{V}_{11'} \cdots \hat{V}_{nn'} \left\langle \hat{T}_\tau \hat{\Psi}^\dagger_1 \hat{\Psi}^\dagger_{1'} \hat{\Psi}_{1'} \hat{\Psi}_1 \cdots \hat{\Psi}^\dagger_n \hat{\Psi}^\dagger_{n'} \hat{\Psi}_{n'} \hat{\Psi}_n \hat{\Psi}_b \hat{\Psi}^\dagger_a\right\rangle_0}{\sum_{n=0}^{\infty} \frac{(-1/2)^n}{n!} \int \mathrm{d}1\mathrm{d}1' \cdots \mathrm{d}n\mathrm{d}n' \hat{V}_{11'} \cdots \hat{V}_{nn'} \left\langle \hat{T}_\tau \hat{\Psi}^\dagger_1 \hat{\Psi}^\dagger_{1'} \hat{\Psi}_{1'} \hat{\Psi}_1 \cdots \hat{\Psi}^\dagger_n \hat{\Psi}^\dagger_{n'} \hat{\Psi}_{n'} \hat{\Psi}_n\right\rangle_0},$$

where we have used the compact notation for the interaction potential and the field operators. We can now use Wick's theorem to express the expectation values in terms of non interacting green's functions, which are known, and thus we can express the interacting green's function in terms of non interacting green's functions, which are known. We can thus write

$$\mathcal{G}(b, a) = -\frac{\sum_{n=0}^{\infty} \frac{(-1/2)^n}{n!} \int \mathrm{d}1\mathrm{d}1' \cdots \mathrm{d}n\mathrm{d}n' \hat{V}_{11'} \cdots \hat{V}_{nn'} (-1)^{2n+1}\mathcal{G}_0(b, 1, 1', \ldots, n, n';\, a, 1, 1', n, n')}{\sum_{n=0}^{\infty} \frac{(-1/2)^n}{n!} \int \mathrm{d}1\mathrm{d}1' \cdots \mathrm{d}n\mathrm{d}n' \hat{V}_{11'} \cdots \hat{V}_{nn'} (-1)^{2n}\mathcal{G}_0(1, 1', \ldots, n, n';\, 1, 1', n, n')},$$

which can be computed as the enormous determinant e presented for Wick's theorem.

Let us compute the denominetor:

$$\begin{aligned}D = 1 - \frac{1}{2}\int \mathrm{d}1\mathrm{d}1' \hat{V}_{11'} &\begin{bmatrix} \mathcal{G}_0(1, 1) & \mathcal{G}_0(1, 1') \\ \mathcal{G}_0(1', 1) & \mathcal{G}(1', 1') \end{bmatrix} + \\ &+ \frac{1}{8}\int \mathrm{d}1\mathrm{d}1'\mathrm{d}2\mathrm{d}2' \hat{V}_{11'}\hat{V}_{22'}[\ldots].\end{aligned}$$

But it is very useful to represent these green's functions as Feynman Diagrams:

TODO: draw the diagrams

[*subtraction of the Hartree and Fock diagrams (dumbell with wiggle line in the middle minus oister with wiggle in the middle)*]

$\mathcal{G}_0(11) = -\left\langle \hat{\Psi}_1^\dagger \hat{\Psi}_1 \right\rangle$ which describes the annihilation and the consequent creation of a particle in the same 1 (position, spin, time)

There is a simple rule that gives us the exponent on the $(-1)^{\#}$, where $\# =$*number of fermionic loops in the diagram*, so that the Hartree diagram has $(-1)^2$ in front of himself, while Fock has just (-1).

We can notice how, even if the diagrams remains very simple, from the determinant in the second term of the expansion of the denominator we get 24 diagrams. Let us draw the more important ones:

TODO: draw the diagrams

[*double hartreee (dumbell wiggle), double fock (two oysters with one wiggle line each), hartree + fock, two oisters connected by two wiggle lines (in the vertexes)*]

and by countimg the loops we can decide the sign of each diagram, which is very important for the final result. Some of these diagrams are disconnected and can be recognized as the product of smaller diagrams, while others are more complicated and connected.

We will see that summing all of these diagrams we can obtain the interaction energy for our system, which is the main goal of this expansion. We can also obtain the self energy, which is the main quantity we are interested in, and which is the main quantity we will be interested in for the rest of the course. We can also obtain the spectral function, which is another important quantity that we will be interested in for the rest of the course. We can also obtain the density of states, which is another important quantity that we will be interested in for the rest of the course. We can also obtain the conductivity, which is another important quantity that we will be interested in for the rest of the course. We can also obtain the optical conductivity, which is another important quantity that we will be interested in for the rest of the course. We can also obtain the magnetic susceptibility, which is another important quantity that we will be interested in for the rest of the course. We can also obtain the specific heat, which is another important quantity that we will be interested in for the rest of the course. We can also obtain the compressibility, which is another important quantity that we will be interested in for the rest of the course. We can also obtain the superfluid density, which is another important quantity that we will be interested in for the rest of the course. We can also obtain the critical temperature, which is another important quantity that we will be interested in for the rest of the course. We can also obtain the critical exponents, which are another important quantity that we will be interested in for the rest of the course. We can also obtain the phase diagram, which is another important quantity that we will be interested in for the rest of the course. We can also obtain the renormalization group flow, which is another important quantity that we will be interested in for the rest of the course. We can also obtain many other quantities that we will be interested in for the rest of the course. [DELETE]

This infinite sum magically becomes a finite result, giving the *correlation energy* for the system (the whole sum without the hartree fock term).

Let us remine that

$$\mathcal{Z} = \operatorname{Tr} e^{-\beta H} = \operatorname{Tr} e^{-\beta H_0} \hat{U}(\beta, 0),$$

and since the logarithm of the partition function is the free energy, and we can recognize the infinite sum of digrams in that expression, we can infere (but we will compiute this result in the future) that pplying the logarithm to the previous expansion in some ways we remove the disconnected diarams and we are left withe the ... Note that the denominator is only bubbles, vacuum diagrams.

Going to the numerator we get:

$$N = \mathcal{G}_0(b,a) - \frac{1}{2}\int \mathrm{d}1\mathrm{d}1'\hat{V}_{11'} \begin{bmatrix} \mathcal{G}_0(b,a) & \mathcal{G}_0(b,1) & \mathcal{G}_0(b,1') \\ \mathcal{G}_0(1,a) & \mathcal{G}(1,1) & \mathcal{G}_0(1,1') \\ \mathcal{G}_0(1',a) & \mathcal{G}_0(1',1) & \mathcal{G}_0(1',1') \end{bmatrix} + \ldots$$

where the first term is just the free propagator from a to b, while the second term is the sum of all the diagrams with one interaction line:

[*all of the diagrams have a wiggle line between 1 and 1', hartree + free prop among a and b, oister + free propagator, a to 1 to b + wiggleline from 1 to 1' + self loop 1' to 1', a to 1 to 1' to b + wiggleline 1 to 1', as before but crossed propagators (a to 1' and 1 to b), a to 1' to b + wiggleline among 1' and 1 and self loop 1 to 1.*]

and the sign is again decided by the number of fermionic loops. Again we have both connected and disconnected diagrams, but continuing the computation it becomes clear that we have the same diagrams from the denominator multiplied by the free propagators. Thus we can factorize the numerator as the product of the connected diagrams times the denominator (both connected and disconnected from D) of this correlation function:

[(*connected diagrams for* N) $\times$ (*all diagrams of* D)]

Thus our full interacting Green's function can be expressed as an infinite sum of connected diagrams from the numerator:

$$\mathcal{G}(b,a) = \frac{N}{D} = \text{Sum of all connected single particle green's functions} =$$

[*free prop a to b, a to b + wiggle to self loop, other with loops connected to ab prop with wiggle lines*]

If by cutting only one fermionic line the diagram become disconnected, we can call them *one particle reducible*, while if cutting any fermionic line in the digram we are left with a different diagram but still connected, then the diagram is called *one particle irreducuble*. Now if we define Σ as the sum of all 1 particle irreducible diagrams without the external legs, we have found the **self-energy**.

Take the diagrams from the connected N which are just the self energy diagrams with the legs attached back, we can write the full green's function as

TODO: substitute Gcal 0 with the line with Gcal 0 written below.

$$\mathcal{G}(b,a) = \mathcal{G}_0 + \mathcal{G}_0\Sigma\mathcal{G}_0 + \mathcal{G}_0\Sigma\mathcal{G}_0\Sigma\mathcal{G}_0 + \ldots,$$

and in this way we can get all the connected diagrams we were looking for. We can rewrite this as

$$\mathcal{G}(b,a) = \mathcal{G}_0 + \mathcal{G}_0\Sigma\left[\mathcal{G}_0 + \mathcal{G}_0\Sigma\mathcal{G}_0 + \mathcal{G}_0\Sigma\mathcal{G}_0\Sigma\mathcal{G}_0 + \ldots\right] = \mathcal{G}_0 + \mathcal{G}_0\Sigma\mathcal{G}(b,a) = \mathcal{G}_0(b,a) + \mathcal{G}(b,l)\Sigma(l,j)\mathcal{G}(j,a),$$

since inthe parenthesis we could recognize the full interacting green's function.

Since we are deaking mostly with translationally invariant systems, we can obtain diagonal expressions for the green's functions and the self energy, so that we can write

$$\mathcal{G}_0(\mathbf{k}, i\omega) = \frac{1}{i\omega - \xi_{\mathbf{k}}},$$

then

$$\begin{aligned}\mathcal{G}(\mathbf{k}, i\omega_n) = \mathcal{G}_0(\mathbf{k}, i\omega_n)\Sigma(\mathbf{k}, i\omega_n)\mathcal{G}(\mathbf{k}, i\omega_n) &= \frac{\mathcal{G}_0(\mathbf{k}, i\omega_n)}{1 - \mathcal{G}_0(\mathbf{k}, i\omega_n)\Sigma(\mathbf{k}, i\omega_n)} \\ &= \frac{1}{\mathcal{G}_0^{-1}(\mathbf{k}, i\omega_n) - \Sigma(\mathbf{k}, i\omega_n)} = \frac{1}{i\omega_n - \xi_{\mathbf{k}} - \Sigma(\mathbf{k}, i\omega_n)},\end{aligned}$$

which is the famous **Dyson's equation**, which gives us the full interacting green's function in terms of the non interacting green's function and the self energy, which is the sum of all 1 particle irreducible diagrams without the external legs. This is a very important result, since it allows us to compute the full interacting green's function once we know the self energy, which is a much simpler quantity to compute than the full interacting green's function.

If in the denominator Σ is null, we are left withe the non interacting green's function, while if Σ is a constant, we are left with a shift of the chemical potential, which is the Hartree-Fock approximation. If Σ is a function of $\mathbf{k}$ and $i\omega_n$, we have a more complicated expression for the green's function, which can be computed once we know the self energy. Thus the self energy is a crucial quantity in the study of interacting systems, since it contains all the information about the interactions in the system

$$\mathcal{G}(\mathbf{k}, i\omega_n) = \frac{1}{i\omega_n - \xi_{\mathbf{k}} - \Sigma(\mathbf{k}, i\omega_n)} = \frac{1}{i\omega_n - \frac{\hbar^2 k^2}{2m} - \mathrm{Re}\Sigma(\mathbf{k}, i\omega_n) - i\mathrm{Im}\Sigma(\mathbf{k}, i\omega_n)},$$

where the real part of the self energy gives us a shift of the energy levels (the *band distortion*, so we can for example pass from a parabolic spectre of energy levels, given by the free particle dispersion relation, to some other shape), while the imaginary part gives us a finite lifetime for the quasiparticles (the one mediating the interactions), which is a very important result, since it tells us that the quasiparticles are not stable, but they have a finite lifetime, which is a consequence of the interactions in the system. We can also obtain the spectral function from the green's function, which is another important quantity that we will be interested in for the rest of the course. The spectral function gives us information about the density of states and the excitations in the system, which are crucial for understanding the physical properties of the system.

Thus we can see that Wick's theorem is a very powerful tool that allows us to compute all these quantities from the non interacting green's function and the self energy, which are much simpler quantities to compute than the full interacting green's function.

3.5 | Free Energy of an Interacting System

Can we still talk about the fermi sphere in the presence of interactions? The answer is yes, since the real part of the self energy gives us a shift of the energy levels, which will become zero at the level of the fermi energy. We are now going to address the problem of how interactions affect the energy levels.

Let us connect the partition function (encoding the statistics of the interacting system) to the free energy, which is the quantity we are interested in, since it gives us all the thermodynamic properties of the system:

$$F = E - TS,$$

from thermodynamics, where E is the internal energy and S is the entropy. Thus our goal is to define the entropy of an interacting system, which is a very complicated quantity to define, since we have to take into account all the interactions. Starting from the Boltzmann factor of each state

$$p_n = \frac{e^{-\beta E_n}}{Z}, \quad H\left|n\right\rangle = E_n \left|n\right\rangle, \quad \sum_n p_n = 1,$$

and remembering that the partition function is the trace of the density matrix, we can write

$$Z = \operatorname{Tr} e^{-\beta H} = \sum_n \left\langle n\right| e^{-\beta H} \left|n\right\rangle = \sum_n e^{-\beta E_n}.$$

We can define the entropy as

$$S = -k_B \sum_n p_n \ln p_n = -k_B \sum_n \frac{e^{-\beta E_n}}{Z} \ln\left(\frac{e^{-\beta E_n}}{Z}\right) = k_B \beta \sum_n p_n E_n + k_B \ln Z,$$

where the first term is the average energy of the system (since it is a sum over all states wheighted by the Boltzmann factors for each level), while the second term is the logarithm of the partition function, which is related to the free energy. Thus we can write

$$\begin{aligned} S &= \frac{1}{T}\langle E\rangle + k_B \ln Z, \\ \implies TS &= \langle E\rangle + \frac{1}{\beta}\ln Z, \\ \implies F &= E - TS = -k_B T \ln Z, \end{aligned}$$

which is a very important result, since it tells us that the free energy of the system is directly related to the partition function, which is the quantity we can compute from the green's functions and the self energy.

Let us recall that the density matrix is the evolution operator in imaginary time, so that we can write

$$F = -\frac{1}{\beta}\ln \operatorname{Tr} e^{-\beta H} = -\frac{1}{\beta}\ln \operatorname{Tr} e^{-\beta H_0}\hat{U}(\beta, 0),$$

where thus

$$\begin{aligned} U(\beta,0) &= \hat{T}_\tau e^{-\int_0^\beta \mathrm{d}\tau \hat{V}_{int}(\tau)} \\ &= 1 - \int_0^\beta \mathrm{d}\tau \hat{V}_{int}(\tau) + \frac{1}{2!}\int_0^\beta \mathrm{d}\tau_1 \int_0^\beta \mathrm{d}\tau_2 \hat{V}_{int}(\tau_1)\hat{V}_{int}(\tau_2) + \dots, \end{aligned}$$

If we define the non interacting partition function as $Z_0 = \operatorname{Tr} e^{-\beta H_0}$, then we can write

$$F = -\frac{1}{\beta}\ln\left(\frac{Z_0}{Z_0}\operatorname{Tr} e^{-\beta H_0}\hat{U}(\beta,0)\right) = -\frac{1}{\beta}\ln Z_0 + \frac{1}{\beta}\ln \operatorname{Tr}\left(e^{-\beta H_0}\hat{U}(\beta,0)\right).$$

This allows us to separate the free energy into a non-interacting part and an interacting correction:

$$F = F_0 - \frac{1}{\beta} \ln \left\langle \hat{U}(\beta, 0) \right\rangle_0 ,$$

and the change in free energy due to interactions is given by

$$\Delta F = F - F_0 = -\frac{1}{\beta} \ln \left\langle \hat{U}(\beta, 0) \right\rangle_0 ,$$

which is exactely the correlation energy of the system, which is the energy correction due to interactions beyond the Hartree-Fock approximation (the previously derived connected and disconnected vacuum diagrams). We can compute this quantity using the expansion of $\hat{U}(\beta, 0)$ and Wick's theorem, which allows us to express it in terms of non-interacting green's functions and self energies, which are much simpler quantities to compute than the full interacting green's function.

Thus we have a powerful tool to compute the free energy of the system, which gives us all the thermodynamic properties of the system. We can also obtain other quantities from the free energy, such as the specific heat, the compressibility, the magnetic susceptibility, etc. Thus we have a very powerful framework to study interacting systems and their thermodynamic properties.

We are going to show that computing the infinite diagrammatic sum of the energy corrections, we will get a finite result, which is the correlation energy of the system. We have to study the logarithm, since it will erase all the disconnected diagrams, which are the ones that give us the infinite contributions to the energy, while the connected diagrams will give us a finite contribution. The following can be formally proved, but we are going only to show a convincing argument for this result, which is the one that we will use in the rest of the course.

Any disconnected diagram can be written as the product of two or more connected diagrams, so that we can write

$$\langle U(\beta, 0)\rangle_0 = 1 + \frac{1}{2}D_1 + \frac{1}{2}D_2 + \frac{1}{8}D_1^2 - \frac{1}{8}D_1 D_2 - \frac{1}{8}D_2 D_1 - \frac{1}{8}D_2^2 + \dots ,$$

where D_1 is the Hartree diagram and D_2 is the Fock diagram, which are the two connected diagrams that we have at the first order of the expansion. Thus continuing the computation we can write

$$\begin{aligned}
\langle U(\beta, 0)\rangle_0 &= 1 + \frac{1}{2}D_1 + \frac{1}{2}(-\frac{1}{2}D_1)^2 + \frac{1}{2}D_2 + \frac{1}{2}(-\frac{1}{2}D_2)^2 - \frac{1}{4}D_1 D_2 + \dots \\
&= 1 + (-\frac{1}{2}D_1 + \frac{1}{2}D_2 + \dots) + \frac{1}{2}(-\frac{1}{2}D_1 + \frac{1}{2}D_2 + \dots)^2 + \dots \\
&= e^{\frac{1}{2}D_1 + \frac{1}{2}D_2},
\end{aligned}$$

where we can recognize the formal Taylor expansion of the exponential of the sum of all connected vacuum diagrams, which is a very important result, since it shows how, putting this result in the expression for the free energy, we get

$$\Delta F = -\frac{1}{\beta} \ln \left\langle \hat{U}(\beta, 0) \right\rangle_0 = -\frac{1}{\beta} \left(\sum \text{ connected vacuum diagrams } \right).$$

This result is proved in the **Linked Cluster Theorem**, which states that *the logarithm of the partition function can be expressed as the sum of all connected vacuum diagrams*, which are the ones that give us the finite contribution to the free energy, while the disconnected diagrams give us the infinite contributions, which are erased by the logarithm.

We will see that the free energy and the total energy, at zero temperature, are the same quantity, since we are at zero temperature, so that the entropy is zero.

[*frawing of ΔF as one over beta multiplied by the sum of connected vacuum diagrams*]

Not all the vacuum conected diagrams are finite, for example the third one diverges as $\frac{1}{q^4}$, while the third as ... and the fourth as $\frac{1}{q^6}$... In general the diagrams diverges as

$$\left(\frac{1}{q^2}\right)^n, \quad n = \text{ number of interaction lines,}$$

so that higher order diagrams diverge more and more.

3.5.1 | Random Phase Approximation

A first approximation to the correlation energy is given by the **Random Phase Approximation**, which consists in summing only the bubble diagrams, which are the ones that give us the most divergent contributions to the energy, so that we can write

$$\Delta F^{EPA}_{correlation} = -\frac{1}{\beta}\sum \text{ connected ring diagrams,}$$

where we put the underscript correlation since we always include the Hartree-Fock term in the free energy:

$$\Delta F = \Delta F_{Hartree-Fock} + \Delta F_{correlation}.$$

Each order add a bubble and an interaction line (Coulomb potential).

[*drawingsssss of n bubbles and n interaction lines, with the first one having two bubbles and two interaction lines, etc.*]

After recognizing the coulomb potential in the interaction line, we can call each bubble (they are all identical) as Π; since each diagram has n bubbles and n interaction lines, alternating between odd n and even n we have a different signs in the summation, so that we can write

$$\Delta F^{RPA}_{correlation} = -\frac{1}{2\beta}\sum_{\mathbf{q}}\sum_{i\nu_n}\sum_{n=2}^{\infty}\frac{(-1)^{n+1}}{n}\left[V_q\Pi_0(\mathbf{q},\, i\nu_n)\right]^n,$$

where the first sum is over the momentum, the second sum is over the imaginary frequencies, and the third sum is over the order of the diagram, which starts from 2 since we are not including the Hartree-Fock term. Then the term $(-1)^{n+1}$ comes from the sign of the fermionic loops and from the combinatorial factor of the diagram, $\frac{1}{2n}$ from the computation of the determinant in the interacting correlation function, while the term $[V_q\Pi_0(\mathbf{q},\, i\nu_n)]^n$ comes from the fact that each diagram has n bubbles and n interaction lines, so that we have n factors of $V_q\Pi_0(\mathbf{q},\, i\nu_n)$. The prefactor is computed after complicated combinatorial arguments, but it is not important for us, since we are interested in the final result of the summation, which is the correlation energy in the RPA approximation.

Considering the loops Π_0 we can understand how the incoming and outgoing momenta are related always q, they are identical, but it contributes in the sum over momenta and frequencies: we will need to compute it in the future, but it is important to notice that it is a function of the momentum and the frequency.

The final expression for the correlation energy in the RPA approximation is given by computing the infinite sum over n, which is a geometric series:

$$\sum_{n=1}^{\infty}\frac{(-1)^{n+1}}{n}x^n = \ln(1+x),$$

thus subtracting the term for $n = 1$, we get:

$$\Delta F^{RPA}_{correlation} = \frac{1}{2\beta} \sum_{\mathbf{q}} \sum_{i\nu_n} \left[\ln(1 + V_q \Pi_0(\mathbf{q},\, i\nu_n)) - V_q \Pi_0(\mathbf{q},\, i\nu_n) \right],$$

where we were able to obtain a finite result for the correlation energy after an infinite summation of the most divergent diagrams, which is a very important result, since it shows how we can obtain finite results from infinite summations of diagrams, the main goal of many body condensed matter physics. We will see that this result is a very good approximation for the correlation energy, which is the main quantity we are interested in for the rest of the course, since it gives us all the thermodynamic properties of the system, and it is also a very good approximation for the total energy of the system.

We said that $\Pi_0(q, i\nu_n)$ is a crucial quantity, since it is the building block of the diagrams considered in the RPA approximation, so that we will need to compute it.

[*Drawing of Pi bubble, with $k, i\omega_n - i\nu_n$ on the fermionic line below, and $k+q, i\omega_n$ on the fermionic line above*]

Thus the two lines are propagators given by

$$\begin{cases} \mathcal{G}_0(k, i\omega_n - i\nu_n) = \dfrac{1}{i\omega_n - -i\nu_n \xi_k}, \\ \mathcal{G}_0(k+q, i\omega_n) = \dfrac{1}{i\omega_n - \xi_{k+q}}, \end{cases}$$

and since the two lines are connected by a vertex, we have to sum over all the possible momenta and frequencies (we should make an excursus on the Feynman rules, but for now we will just state the result), so that we can write[3]

$$\begin{aligned} \Pi_0(q, i\nu_n) &= -\frac{2}{\beta V} \sum_{k, i\omega_n - i\nu_n} \mathcal{G}_0(k, i\omega_n) \mathcal{G}_0(k+q, i\omega_n) \\ &= -\frac{2}{\beta V} \sum_{k, i\omega_n} \frac{1}{i\omega_n - i\nu_n - \xi_k} \frac{1}{i\omega_n - \xi_{k+q}}, \end{aligned}$$

where V is the volume of the system, and the factor of 2 comes from the spin degeneracy. This is called the **Lindhard function**, which is a very important quantity in the study of interacting systems, since it gives us information about the response of the system to external perturbations, such as an external electric field or an external magnetic field.

We can now use a general technique to compute the sum over the fermionic frequencies, which is called the **Matsubara summation**, which is general and can be applied to any sum over fermionic frequencies, and it is based on the use of the residue theorem from complex analysis. Recalling the Matsubara frequencies for fermions, which are given by

$$i\omega_n = \frac{i\pi(2n+1)}{\beta},$$

thus we can rewrite

$$\Pi_0(q, i\nu_n) = -\frac{2}{V} \sum_k \frac{1}{\beta} \sum_{i\omega_n} \mathcal{F}(i\omega_n),$$

where the method we are going to implement will give us a way to compute the sum over the frequencies of this $\mathcal{F}(i\omega_n)$,

$$\mathcal{F}(z) = \frac{1}{z - i\nu_n - \xi_k} \frac{1}{z - \xi_{k+q}},$$

[3] We ignored the sum over the spin indices, since it is conserved and gives a degeneracy, and we added only a factor of 2 in front of the sum.

where $z = i\omega_n$. Since the Matsubara frequencies are the poles of the Fermi-Dirac distribution, we can use the residue theorem to compute the sum over the frequencies. Let us write

$$I = \lim_{R\to\infty} \oint \frac{\mathrm{d}z}{2\pi i} \mathcal{F}(z) n_F(z) = \sum \text{residues of } \mathcal{F}(z) n_F(z) = 0,$$

since the radius of the contour goes to infinity and the function goes to zero at infinity. The poles of the Fermi-Dirac distribution are given by the infinite number of Matsubara frequencies, located in the imaginary axis of the complex plane, so that we can write

[*drawing of complex plane, poles (matsubara and denominator of F) and circle*]

and in the final expression we have that $\frac{1}{\beta}$ is the residue of the Fermi-Dirac distribution at the Matsubara frequencies, while the term we are going to compute is the residue of $\mathcal{F}(z)$ at the poles of the denominator, which are given by $z_b = i\nu_n + \xi_k$ and $z_a = \xi_{k+q}$ (and z_n the Matsubara frequencies).

Thus we can write

$$I = 0 = \sum_{z_n} \text{Residue}(z_n) + \text{Residue}(z_a) + \text{Residue}(z_b),$$

where

$$\text{Residue}(z_n) = \lim_{z\to z_n} (z - z_n)\mathcal{F}(z) n_F(z) = \lim_{z\to z_n} (z - z_n)\mathcal{F}(z) \frac{1}{0 + \frac{\mathrm{d}}{\mathrm{d}z}[e^{\beta z} + 1]\Big|_{z=z_n} (z - z_n) + \dots},$$

where we just expanded the Fermi-Dirac distribution around the pole z_n and we used the fact that the residue of a function at a simple pole is given by the limit of the function multiplied by $(z - z_n)$ as z approaches z_n. We can stop to the first order in the expansion since we are interested in the limit $z \to z_n$ (and the $(z - z_n)$ term simplifies, wile the others are multiplied by $(z - z_n)$ which is going to zero in this limit), so that we can write

$$\text{Residue}(z_n) = \lim_{z\to z_n} \mathcal{F}(z) \frac{1}{\beta e^{\beta z_n}} = -\frac{1}{\beta} \mathcal{F}(z_n),$$

where the minus sign comes from the fact that $e^{\beta z_n} = e^{i\pi(2n+1)} = -1$. This is exactly the term we have in the expression for Π_0, so that in the end we will compute it as the other residue with sign changed.

The other two residues are given by

$$\begin{aligned} \text{Residue}(z_a) &= \lim_{z\to z_a} (z - z_a)\mathcal{F}(z) n_F(z) \\ = \lim_{z\to z_a} (z - z_a) \frac{1}{z - i\nu_n - \xi_k} \frac{1}{z - \xi_{k+q}} n_F(z) &= \frac{n_F(\xi_{k+q})}{\xi_{k+q} - i\nu_n - \xi_k}, \end{aligned}$$

for the pole z_a, while for the pole z_b we have

$$\begin{aligned} \text{Residue}(z_b) &= \lim_{z\to z_b} (z - z_b)\mathcal{F}(z) n_F(z) \\ = \lim_{z\to z_b} (z - z_b) \frac{1}{z - i\nu_n - \xi_k} \frac{1}{z - \xi_{k+q}} n_F(z) &= \frac{n_F(i\nu_n + \xi_k)}{i\nu_n + \xi_k - \xi_{k+q}}. \end{aligned}$$

Thus we can write

$$I = 0 = -\frac{1}{\beta} \sum_{z_n} \mathcal{F}(z_n) - \frac{n_F(\xi_{k+q})}{i\nu_n + \xi_k - \xi_{k+q}} + \frac{n_F(i\nu_n + \xi_k)}{i\nu_n + \xi_k - \xi_{k+q}},$$

where we can notice how in the last term

$$n_F(i\nu_n + \xi_k) = n_F(\xi_k),$$

since the fermi distribution is periodic in the imaginary axis with period $i\nu_n$, so that we can write

$$n_F(z) = \frac{1}{e^{\beta z}+1}, \quad e^{\beta(i\nu_n+\xi_k)} = e^{\beta i\nu_n}e^{\beta\xi_k} = e^{i\pi(2n+1)}e^{\beta\xi_k} = -e^{\beta\xi_k}.$$

Example. Let us show how a fermionic closed loop gives us

$$loop = \frac{1}{\beta}\sum_{i\omega_n}\frac{1}{i\omega_n - \xi_k} = n_F(\xi_k),$$

since... [...]

Thus we can write

$$\frac{1}{\beta}\sum_{i\omega_n}\mathcal{F}(i\omega_n) = \frac{n_F(\xi_k) - n_F(\xi_{k+q})}{i\nu_n + \xi_k - \xi_{k+q}},$$

and we have finally found an expression for the **Lindhard function** in the Matsubara formalism, which is

$$\Pi_0(q, i\nu_n) = \frac{2}{V}\sum_{\mathbf{k}}\frac{n_F(\xi_k) - n_F(\xi_{k+q})}{i\nu_n + \xi_k - \xi_{k+q}} \to \frac{2}{(2\pi)^3}\int \mathrm{d}^3k\frac{n_F(\xi_k) - n_F(\xi_{k+q})}{i\nu_n + \xi_k - \xi_{k+q}},$$

which is a very important result, since it gives us the building block of the diagrams considered in the RPA approximation, which is a very good approximation for the correlation energy of the system.

In QFT when treating loops in the diagrammatic expansion, usually we have to introduce a **cutoff** in the momentum integration, since the loops give us divergent contributions to the energy; in condensed matter physics we do not have this problem, since the momentum integration is always limited by the size of the Brillouin zone, which is a consequence of the **lattice structure** of the system: we have always *periodic boundary conditions*, so that the momentum is quantized and limited by the size of the Brillouin zone, which gives us a natural cutoff for the momentum integration.

Note how the numerator of the Lindhard function is the difference of two Fermi-Dirac distributions, which gives us a non zero contribution only when we are in a region of the momentum space near the Fermi surface: this functions are both either 0 or 1, so that the difference is zero unless we are in a region where one of them is 1 and the other is 0, which happens only near the Fermi surface. So the main contribution to the correlation energy in the RPA approximation comes from the region near the Fermi surface, which is a very important result: the low energy excitations of the system are the ones that give us the main contribution to the correlation energy (the interactions are more effective near the Fermi surface, where we have a high density of states).

Recall then the expression for the correlation energy in the RPA approximation:

$$\Delta F^{RPA}_{correlation} = \frac{1}{2\beta}\sum_{\mathbf{q}}\sum_{i\nu_n}\left[\ln\left(1 + \frac{4\pi}{q^2}\Pi_0(\mathbf{q},\, i\nu_n)\right) - \frac{4\pi}{q^2}\Pi_0(\mathbf{q},\, i\nu_n)\right] = \text{ sum of connected ring diagrams},$$

now we are interested in the behavior of this quantity in the limit of small q (transfer momentum), which is the limit that gives us the most divergent contributions to the energy: if we destroy a particle inside the fermi sphere (near the surface) we can only create a particle outside the fermi sphere (near the surface), so that we have a small momentum transfer q between the two states.

All the processes are dominated by small momentum transfers, which is a very important result, since it tells us that the main contribution to the correlation energy comes from the region near the Fermi surface, where we have a high density of states and where the interactions are more effective.

Thus we compute the expansion of the numerator of the Lindhard function for small q, which is given by

$$n_F(\xi_{k+q}) = n_F(\xi_k) + \mathbf{q}\cdot\nabla_k n_F(\xi_k) + O(q^2),$$

which implies that

$$n_F(\xi_k) - n_F(\xi_{k+q}) = n_k - n_{k+q} \sim -\mathbf{q}\cdot\nabla_k n_k = -\mathbf{q}\cdot\nabla_k\theta(k_F - k) = \mathbf{q}\cdot\hat{k}\delta(k-k_F) = q\cos\theta\delta(k-k_F),$$

where we used the fact that the derivative of the theta functions has a minus sign and the delta function, and we defined θ as the angle between the momentum $\mathbf{k}$ and the momentum transfer $\mathbf{q}$. The nominator of the Lindhard function for small q is given by[4]

$$\xi_{k+q} - \xi_k = \frac{1}{2}\left[(\mathbf{k}+\mathbf{q})^2 - \mathbf{k}^2\right] = \frac{1}{2}(2kq\cos\theta + q^2) \sim kq\cos\theta,$$

since q is small and we stop to the first order in q, so that we can write

$$\Pi_0(q, i\nu_n) = \frac{2}{(2\pi)^3}\int \mathrm{d}^3\mathbf{k}\frac{n_F(\xi_k) - n_F(\xi_{k+q})}{i\nu_n + \xi_k - \xi_{k+q}} \sim \frac{2}{(2\pi)^3}\int \mathrm{d}^3 k\frac{q\cos\theta\delta(k-k_F)}{i\nu_n - kq\cos\theta},$$

where the measure can be written as $\mathrm{d}^3\mathbf{k} = k^2\sin\theta\mathrm{d}k\mathrm{d}\theta\mathrm{d}\phi$, so that we can write

$$\Pi_0(q, i\nu_n) \sim \frac{2}{(2\pi)^3}\int_0^\infty k^2\mathrm{d}k\int_{-1}^1 \mathrm{d}\cos\theta\int_0^{2\pi}\mathrm{d}\phi\frac{q\cos\theta\delta(k-k_F)}{i\nu_n - kq\cos\theta},$$

which can be rewritten with a change of variable as

$$\Pi_0(q, i\nu_n) = \frac{k_F^2}{2\pi^2}\int_{-1}^1 \mathrm{d}\mu\frac{q\mu}{i\nu_n - v_F\mu q} = \frac{k_F^2}{2\pi^2}\int_{-1}^1\mathrm{d}\mu\frac{1}{v_F}\frac{\mu}{\frac{i\nu_n}{v_F q} - \mu},$$

and v_F is the Fermi velocity, which is given by $v_F = \frac{k_F}{m}$ (in atomic units $v_F = k_F$), cancelling out the square of the Fermi momentum in the numerator, we get

$$\Pi_0(q, i\nu_n) = \frac{k_F}{2\pi^2}\int_{-1}^1\mathrm{d}\mu\frac{\mu}{\frac{i\nu_n}{v_F q} - \mu} = \frac{k_F}{\pi^2}\left[-1 + \frac{i\nu_n}{2v_F q}\ln\left(\frac{i\nu_n + v_F q}{i\nu_n - v_F q}\right)\right],$$

where we can write the logarithm of an imaginary number as a series of sines and cosines, which in the end gives an arcotangent, so that we can write

$$\Pi_0(q, i\nu_n) = \frac{k_F}{\pi^2}\left[-1 + \frac{\nu_n}{v_F q}\arctan\left(\frac{v_F q}{\nu_n}\right)\right].$$

Now we can look at the limit for small frequencies, where the Lindhard function goes to a constant, which is given by

$$\Pi_0(q, 0) = -\frac{k_F}{\pi^2} = -D(\epsilon_F) \sim \frac{1}{r_s},$$

where $D(\epsilon_F)$ is the density of states at the Fermi energy, which is a very important quantity in the study of interacting systems, since it gives us information about the number of states available for the particles to occupy at the Fermi energy, which is crucial for understanding the physical

[4] We are using atomic units, so $\hbar = m = 1$, so that the dispersion relation is given by $\xi_k = \frac{k^2}{2}$.

properties of the system. The parameter r_s is a dimensionless parameter that characterizes the strength of the interactions in the system.

TODO: check this last part.

[... screening ...]

Since at zero temperature we are not summing over a discrete set of frequencies, but we are integrating over a continuous variable

$$i\nu_n = i\frac{2\pi n}{\beta} \xrightarrow{T\to 0} \dots$$

then the summation in the expression for the correlation energy becomes an integral, so that we can write

$$\Delta F^{RPA}_{correlation} = \frac{1}{2}\int \frac{\mathrm{d}^3\mathbf{q}}{(2\pi)^3/V}\int_{-\infty}^{\infty}\frac{\mathrm{d}\nu}{2\pi}\left[\ln(1+V_q\Pi_0(\mathbf{q},\, i\nu)) - V_q\Pi_0(\mathbf{q},\, i\nu)\right],$$

where the volume factor appeared from

$$\frac{1}{V}\sum_q \to \int \frac{\mathrm{d}^3\mathbf{q}}{(2\pi)^3},$$

and since it was not present in the original expression, we have to put it in the denominator of the measure of the integral. Now we have to substitute the expression for the Lindhard function, which was given by

$$\Pi_0(q, i\nu) = \frac{k_F}{\pi^2}\Phi(\frac{\nu}{v_F q}), \quad \Phi(x) = -1 + \frac{x}{2}\ln\left(\frac{x+1}{x-1}\right),$$

so that we can write

$$\Delta F^{RPA}_{correlation} = \frac{1}{2}\int \frac{\mathrm{d}^3\mathbf{q}}{(2\pi)^3/V}\int_{-\infty}^{\infty}\frac{\mathrm{d}\nu}{2\pi}\left[\ln\left(1+\frac{q_{TF}^2}{q^2}\Phi(\frac{\nu}{v_F q})\right) - \frac{q_{TF}^2}{q^2}\Phi(\frac{\nu}{v_F q})\right],$$

where we defined the Thomas-Fermi wavevector as

$$q_{TF}^2 = \frac{4k_F}{\pi},$$

which is a very important quantity in the study of interacting systems, since it gives us information about the screening of the interactions in the system, which is crucial for understanding the physical properties of the system. The Thomas-Fermi wavevector is a measure of the strength of the screening in the system, and it is related to the density of states at the Fermi energy.

Renaming also

$$\frac{\nu}{v_F q} = u, \quad \nu = u v_F q, \quad \mathrm{d}\nu = v_F q \mathrm{d}u,$$

we can write

$$\frac{\Delta F^{RPA}_{correlation}}{V} = \frac{1}{2}\int \frac{\mathrm{d}^3\mathbf{q}}{(2\pi)^3}\int_{-\infty}^{\infty}\frac{\mathrm{d}u}{2\pi}v_F q\left[\ln\left(1+\frac{q_{TF}^2}{q^2}\Phi(u)\right) - \frac{q_{TF}^2}{q^2}\Phi(u)\right],$$

and now going to polar coordinates in the momentum space,

$$\mathrm{d}^3\mathbf{q} = 4\pi q^2 \mathrm{d}q,$$

and renaming also the momentum as

$$q = q_{TF}y, \quad q^3\mathrm{d}q = q_{TF}^4 y^3 \mathrm{d}y,$$

we can write

$$\frac{\Delta F^{RPA}_{correlation}}{V} = \mathcal{C} v_F q_{TF}^4 \int_0^\infty y^3 \mathrm{d}y \int_{-\infty}^{\infty} \mathrm{d}u \left[\ln\left(1 + \frac{\Phi(u)}{y^2}\right) - \frac{\Phi(u)}{y^2} \right],$$

where $\mathcal{C}$ is a numerical constant that we are not interested in, now the idea is to break the y integral in three parts:

$$\begin{cases} 0 \le y \le 1 \, : \, 0 \le q \le q_{TF}, \\ 1 \le y \le \dfrac{1}{r_s} \, : \, q_{TF} \le q \le k_F, \\ \dfrac{1}{r_s} \le y \le \infty \, : \, k_F \le q \le \infty, \end{cases}$$

where the first two region will give us a finite contribution to the correlation energy, while the last one will need to be controlled since the domain of integration goes to infinity, which...

$$\Delta F^{RPA}_{correlation} \sim \text{constant} \cdot \int_1^{k_F/q_{TF}} \frac{1}{y} \mathrm{d}y \sim \text{constant} \cdot \ln\left(\frac{k_F}{q_{TF}}\right) \sim \text{constant} \cdot \ln\left(\frac{1}{\sqrt{r_s}}\right),$$

since

$$k_F \sim \frac{1}{r_s}, \quad q_{TF}^2 \sim k_F \sim \frac{1}{r_s}.$$

This was one of the most important results in the history of many body physics, since it was the first time that a finite result for the correlation energy of the electron gas was obtained, which is a very important quantity in the study of interacting systems, since it gives us information about the energy correction due to interactions beyond the Hartree-Fock approximation, which is crucial for understanding the physical properties of the system. The RPA approximation gives us a very good approximation for the correlation energy, which is a very important result, since it shows how we can obtain finite results from infinite summations of diagrams, which is the main goal of many body condensed matter physics.

Thus in the end

$$\frac{\Delta F^{RPA}}{V} = \frac{2.21}{r_s^2} - \frac{0.911}{r_s} + A \ln(r_s) + B + O(r_s),$$

which is the **Gellman formula** for the correlation energy of the electron gas, where the first term is the kinetic energy, the second term is the exchange energy, and the last two terms are the correlation energy in the RPA approximation, which is a very good approximation for the correlation energy of the electron gas.

3.5.2 | Long Range Coulomb Interactions

Fermi Surface

3.6 | Phonons

The phonons are the other important quasi-particles in the system. We aim to derive the ...

The easyiest way to derive phonons is to start from a line of atoms with a lattice constant a labeled by n. Thu $t_n = na$ is the position of the n-th atom. We will thenote with u_n the displacement of the n-th atom from its equilibrium position.

[*Draw a picture of the line of atoms with the displacements*]

Atoms in the system interact with their neighbors, and the energy of the system depends on the displacements of the atoms. Then we want to write down the Hamiltonian of the system as a function of the displacements u_n:

$$E_0(\{u_n\}) = E_0(\{u_n = 0\}) + \sum_n \frac{\partial E_0}{\partial u_n}\Big|_{\{u_n=0\}} u_n + \frac{1}{2}\sum_{n,m} \frac{\partial^2 E_0}{\partial u_n \partial u_m}\Big|_{\{u_n=0\}} u_n u_m + \cdots$$

where we have expanded the energy of the system in a Taylor series around the equilibrium position. The first term is just a constant while the second vanishes because we are at the equilibrium position. Therefore, the leading term is the third term, which is quadratic in the displacements. We can then write the Hamiltonian as:

$$E_0(\{u_n\}) = E_0(\{u_n = 0\}) + \frac{1}{2}\sum_{n,n'} D_{nn'} u_n u_{n'},$$

where we have defined the interaction force costants matrix $D_{nn'} = \frac{\partial^2 E_0}{\partial u_n \partial u_{n'}}\Big|_{\{u_n=0\}}$. This matrix has some useful properties:

1. The matrix $D_{nn'}$ is real, which ensures that the energy is real.
2. The matrix $D_{nn'}$ is symmetric, i.e., $D_{nn'} = D_{n'n}$ if the system is invariant under atoms translations $t_n - t_{n'} = t_m - t_{m'}$.
3. The force acting on the n-th atom is given by $F_n = -\frac{\partial E_0}{\partial u_n} = -\sum_{n'} D_{nn'} u_{n'}$. Therefore, if we sum all the entries on the n-th row or column of the matrix $D_{nn'}$, we get $\sum_{n'} D_{nn'} = 0$ because the total force acting on the system must be zero for configurations in which all the atoms are displaced by the same amount, i.e., $u_n = u$ for all n.

Since we are considering a line of atoms with only nearest neighbor interactions, the matrix $D_{nn'}$ has the following form:

$$D_{n,n+1} = -C, \quad D_{n-1,n} = C, \quad D_{nn} = 2C, \quad D_{nm} = 0 \text{ for } |n-m| > 1.$$

Thus the energy of the system can be written as:

$$E_0(\{u_n\}) = E_0(\{0\}) + \frac{C}{2}\sum_n (2u_n^2 - u_n u_{n+1} - u_{n-1}u_n) = E_0(\{0\}) + \frac{C}{2}\sum_n (u_n - u_{n+1})^2,$$

where the proof of the last equality is left as an exercise for the reader. Thus we have found the enrgy for a system of N coupled harmonic oscillators, which is the Hamiltonian of the phonons. If we close the chain of atoms into a ring, we have periodic boundary conditions, which means that $u_{N+1} = u_1$.

The phonons are the normal modes of the system, which can be found by diagonalizing the matrix $D_{nn'}$. The found basis of phonons is called the normal mode basis, and the phonons are the quanta of the normal modes.

Until now we made a classical treatment of the phonons, but we can also quantize the phonons, obtaining an hamiltonian for n coupled quantum harmonic oscillators. The quantization of the phonons is done by promoting the displacements u_n and their conjugate momenta p_n to operators that satisfy the canonical commutation relations $[\hat{u}_n, \hat{P}_m] = i\hbar\delta_{nm}$. The resulting Hamiltonian is given by:

$$\hat{H} = \sum_n \frac{1}{2M}\hat{P}_n^2 + \frac{1}{2}C\sum_n \left[2\hat{u}_n^2 - \hat{u}_n\hat{u}_{n+1} - \hat{u}_{n-1}\hat{u}_n\right],$$

where M is the mass of the atoms. We now want to diagonalize this Hamiltonian, which is equivalent to finding the normal modes of the system. We can do this by performing a Fourier transform of the displacements and momenta:

$$\hat{P}_q = \frac{1}{\sqrt{N}} \sum_n e^{-iqt_n} \hat{P}_n, \quad \hat{u}_q = \frac{1}{\sqrt{N}} \sum_{t_n} e^{-iqt_n} \hat{u}_n, \quad [\hat{u}_q, \hat{P}_{q'}] = i\hbar\delta_{q,q'}.$$

Let us now convert tge Hamiltonian in the Fourier space. The displacement term can be converted with the inverse of te previous Fourier transform, which is given by:

$$\sum_n \hat{u}_n^2 = \frac{1}{N} \sum_n \sum_{qq'} e^{i(q+q')t_n} \hat{u}_q \hat{u}_{q'} = \sum_{qq'} \left[\frac{1}{N} \sum_n e^{i(q+q')t_n} \right] \hat{u}_q \hat{u}_{q'} = \sum_q \hat{u}_q \hat{u}_{-q},$$

and for the squared momentum term we have almost the same expression, but with the momenta instead of the displacements:

$$\sum_n \hat{P}_n^2 = \sum_q \hat{P}_q \hat{P}_{-q}.$$

The last term is a bit more tricky, but we can use the following trick:

$$\sum_n \hat{u}_n \hat{u}_{n+1} = \frac{1}{N} \sum_n \sum_{qq'} e^{i(q+q')t_n + iq'a} \hat{u}_q \hat{u}_{q'} =$$

$$\sum_{qq'} \left[\frac{1}{N} \sum_n e^{i(q+q')t_n} \right] e^{iq'a} \hat{u}_q \hat{u}_{q'} = \sum_q e^{iqa} \hat{u}_q \hat{u}_{-q}.$$

Thus the Hamiltonian in the Fourier space is given by:

$$\hat{H} = \sum_q \frac{1}{2M} \hat{P}_q \hat{P}_{-q} + \frac{C}{2} \sum_q \hat{u}_q \hat{u}_{-q} (2 - e^{iqa} - e^{-iqa}) = \sum_q \frac{1}{2M} \hat{P}_q \hat{P}_{-q} + 2C \sum_q \hat{u}_q \hat{u}_{-q} (1 - \cos(qa)),$$

which can be rewritten as:

$$\hat{H} = \frac{1}{2} \sum_q \left[\frac{1}{M} \hat{P}_q \hat{P}_{-q} + M\omega_q^2 \hat{u}_q \hat{u}_{-q} \right], \quad M\omega_q^2 = C(2 - e^{iqa} - e^{-iqa}) = 4C \sin^2\left(\frac{qa}{2}\right).$$

We have found the Hamiltonian of a system of decoupled harmonic oscillators, which is the Hamiltonian of the phonons. The phonon frequencies are given by $\omega_q = 2\sqrt{\frac{C}{M}} \left|\sin\left(\frac{qa}{2}\right)\right|$, which is the dispersion relation of the phonons. The phonons are the quanta of the normal modes of the system, and they are bosonic particles. If we plot the dispersion relation of the phonons, we can see that it has a linear behavior for small values of q, while for higher values of q it becomes periodic.

[*Draw the dispersion relation of the phonons*]

A parabolic dispersion relation is a fingerprint of a free particle, while a linear dispersion relation indicates a collective mode. Therefore, the phonons are collective modes of the system, which are not free particles.

The typical energy of the electrons on the fermi surface is of the order of electronvolts, but confronting this energy with the typical energy of the phonons, which is of the order of millielectronvolts, we can see that the phonons are much slower than the electrons. This is due to the mass of the ions which is much larger than the mass of the electrons. The **Debye energy**, which is the maximum energy of the phonons, is of the order of millielectronvolts, which is much smaller than the typical energy of the electrons on the fermi surface.

From the equation of sound waves, we can see that the speed of sound is given by $v_s = \sqrt{\frac{Ca^2}{M}}$, which is similar to the speed of the phonons for small values of q. Phonons at small values of q

are called acoustic phonons, while phonons at higher values of q are called optical phonons. The acoustic phonons are the ones that are responsible for the sound waves in the system, while the optical phonons are the ones that are responsible for the optical properties of the system.

As in second quantization, where we defined the operators ...

$$\hat{H} = \frac{1}{2}\frac{P^2}{2m} + \frac{1}{2}m\omega^2 x^2 \to \hat{H} = \hbar\omega\left(\hat{a}^\dagger\hat{a} + \frac{1}{2}\right),$$

we can make use of a symilar canonical transformation to define the creation and annihilation operators for the phonons, which are defined by:

$$\hat{u}_q = \sqrt{\frac{\hbar}{2M\omega_q}}(\hat{b}_q + \hat{b}^\dagger_{-q}), \quad \hat{P}_q = i\sqrt{\frac{\hbar M\omega_q}{2}}(\hat{b}^\dagger_{-q} - \hat{b}_q),$$

which is endowed with the usual commutation relations $[\hat{b}_q, \hat{b}^\dagger_{q'}] = \delta_{q,q'}$. With this transformation, the Hamiltonian of the phonons can be rewritten as:

$$\hat{H} = \sum_q \hbar\omega_q\left(\hat{b}^\dagger_q\hat{b}_q + \frac{1}{2}\right),$$

which is the second quantized version of the non interacting phonons hamiltonian. The phonons are bosonic particles, which means that they can occupy the same state, and they are the quanta of the normal modes of the system.

3.6.1 | Electron and Phonons

If we consider a system of electrons and phonons, we can write the Hamiltonian of the system as:

$$\hat{H} = \sum_{k\sigma}\xi_{k\sigma}\hat{c}^\dagger_{k\sigma}\hat{c}_{k\sigma} + \sum_q \hbar\omega_q\left(\hat{b}^\dagger_q\hat{b}_q + \frac{1}{2}\right) + H_{\text{e-ph}},$$

basically the sum of the two free hamiltonians plus the **electron-phonon interaction hamiltonian** (term responsible for resistivity and conductivity of the material). $\xi_{k\sigma} = 2t\cos(ka)$ is the energy of the electron with momentum k and spin σ on the lattice, and g_q is the coupling constant between the electrons and the phonons.

[*Draw the dispersion relation of the electrons and the phonons in lattice momentum space*]

The lattice in which we are considering the electrons and the phonons is the same, so that the momenta k and q are defined in the same Brillouin zone (different letters are used just to distinguish the momenta of the electrons from the momenta of the phonons); the only difference is the order of magnitude of the momenta, as we saw.

Imagine the electrons as a spread out homogeneus charge, if we take a small volume element of the system, we can see

$$\mathrm{d}q = -en(r)\mathrm{d}V,$$

where $n(r)$ is the density of electrons at position r.

[*Draw of system and volume element and coordinates*]

We know the interaction potential among electrons and ions in a static lattice to be given by

$$\hat{V}_{\text{e-ions}} = \sum_j^N \int \mathrm{d}^3\mathbf{r}(-e)n(\mathbf{r})V_{\text{ions}}(\mathbf{r} - \mathbf{R}_j - \mathbf{u}_j),$$

where the ion potential is given by

$$V_{\text{ions}}(\mathbf{r}-\mathbf{R}_j-\mathbf{u}_j) \sim V_{\text{ions}}(\mathbf{r}-\mathbf{R}_j) - \nabla V_{\text{ions}}(\mathbf{r}-\mathbf{R}_j)\cdot\mathbf{u}_j,$$

which inserted in the previous equation gives

$$V_{\text{e-ions}} = \sum_j^N \int \mathrm{d}^3\mathbf{r}(-e)n(\mathbf{r})V_{\text{ions}}(\mathbf{r}-\mathbf{R}_j) + \sum_j^N \int \mathrm{d}^3\mathbf{r}(-e)n(\mathbf{r})\nabla V_{\text{ions}}(\mathbf{r}-\mathbf{R}_j)\cdot\mathbf{u}_j,$$

where in the first term the nuclei are fixed in their equilibrium positions (we have no dependence upon the $\mathbf{u}_j$ displacements), while in the second term the nuclei are displaced from their equilibrium positions. The first term is somehow the Born-Oppenheimer approximation (considering fixed nuclei we can solve the electronic problem exactly), while in the second term the electron-phonon interaction is encoded (it contains both displacements and electron density).

If we second consider only the second term, we can give an idea of how to second quantize the electron-phonon interaction. Starting from the electron density, we can write it in second quantization as

$$n(\mathbf{r}) = \sum_\sigma \hat{\Psi}^\dagger_\sigma(\mathbf{r})\hat{\Psi}_\sigma(\mathbf{r}),$$

where the field operators can be expanded in the basis given by the plane waves as

$$\hat{\Psi}_\sigma(\mathbf{r}) = \frac{1}{\sqrt{V}}\sum_k e^{ik\cdot r}\hat{c}_{k\sigma}, \quad \hat{\Psi}^\dagger_\sigma(\mathbf{r}) = \frac{1}{\sqrt{V}}\sum_k e^{-ik\cdot r}\hat{c}^\dagger_{k\sigma}.$$

Thus the electron density can be rewritten as

$$n(\mathbf{r}) = \frac{1}{V}\sum_{kk'\sigma} e^{i(k-k')\cdot r}\hat{c}^\dagger_{k\sigma}\hat{c}_{k'\sigma}.$$

Thus the electron-phonon interaction can be rewritten as

$$\hat{H}_{\text{e-ph}} = -\sum_j^N \int \mathrm{d}^3\mathbf{r}(-e)n(\mathbf{r})\nabla V_{\text{ions}}(\mathbf{r}-\mathbf{R}_j)\cdot\mathbf{u}_j = \frac{1}{V}\sum_{k\sigma}\sum_q g_q\hat{c}^\dagger_{k+q,\sigma}\hat{c}_{k,\sigma}(\hat{b}_q+\hat{b}^\dagger_{-q}),$$

where all the constants have been absorbed in the **electron-phonon coupling constant** g_q:

$$g_q = ie\frac{\sqrt{N}}{\sqrt{2M\omega_q}}qV_q,$$

where V_q is the Fourier transform of the ion potential: it can be the coulomb potential, but it can also be a screened coulomb potential, which is more realistic in a metal.

We can rewrite the full Hamiltonian of the system as

$$\hat{H} = \sum_{k\sigma}\xi_{k\sigma}\hat{c}^\dagger_{k\sigma}\hat{c}_{k\sigma} + \sum_q \hbar\omega_q\left(\hat{b}^\dagger_q\hat{b}_q+\frac{1}{2}\right) + \frac{1}{V}\sum_{k\sigma}\sum_q g_q\hat{c}^\dagger_{k+q,\sigma}\hat{c}_{k,\sigma}(\hat{b}_q+\hat{b}^\dagger_{-q}),$$

which is the Hamiltonian of a system of electrons and phonons with electron-phonon interaction.

The electron-phonon interaction describes the process in which an electron with momentum k and spin σ scatters to a state with momentum $k+q$ and the same spin σ by emitting or absorbing a phonon with momentum q.

[*Draw process of absorption of phonon and exited electron + inverse (-q) (emission)*]

We are thus considering both the processes of absorption and emission of phonons, which are responsible for the resistivity and conductivity of the material. The electron-phonon interaction is also responsible for the superconductivity in the material, which will be discussed in future.

We can now take a step further: if we consider two electrons in the system, they can interact with each other by exchanging a phonon. This process is called **phonon-mediated electron-electron interaction**, and it is responsible for the superconductivity in the material. Basically, one electron can emit a phonon, which is then absorbed by another electron, and this process can be repeated multiple times, each times with a transfer of momentum q and energy ω_q. The effective interaction between the two electrons is given by the sum of all the possible processes of emission and absorption of phonons, which can be represented by a Feynman diagram. This is possible only in a lattice, because in a free space the phonons do not exist, and therefore the electrons cannot interact through phonon exchange.

[*Draw graph of e-e phonon mediated interaction*]

The interaction mediated by the phonons is retarded with respect to the interaction mediated by the photons, which is almost instantaneous (order of c). This is due to the fact that the phonons are much slower than the electrons, as we saw before, and this velocity will be dictated by the phonon dispersion relation through the phonon energy ω_q. The retarded nature of the phonon-mediated interaction is responsible for the superconductivity in the material, which will be discussed in future.

We will have to define the phonon propagator which mediates the interaction between the electrons: in the case of electrons interacting through photons, we have the following coupling constant

$$V_{phot} = e\frac{4\pi}{q^2}e,$$

while in the case of electrons interacting through phonons, we have to consider the electron-phonon "potential"

$$V_{el-phon}(q, i\nu_n) = \frac{1}{V}|g_q|^2 D_0(q, i\nu_n),$$

which is the interacting part of the hamiltonian, where $D_0(q, i\nu_n)$ is the phonon propagator, which is what we want to compute. Calling $\hat{A}_q = (\hat{b}_q + \hat{b}^\dagger_{-q})$, then we can define the phonon propagator as

$$D_0(q, i\nu_n) = -\left\langle T_\tau \hat{A}_q(\tau)\hat{A}_{-q}(0)\right\rangle_0,$$

which can be computed remembering that we are considering states in the interaction picture, and that the time evolution of the operators is given by the free hamiltonian of the phonons. Let us compute firstly the operator $\hat{A}(q,\tau)$:

$$\begin{aligned} A(q,\tau) &= e^{\tau\hat{H}_{phon}}A_q e^{-\tau\hat{H}_{phon}} = e^{\tau H_{phon}}(\hat{b}_q + \hat{b}^\dagger_{-q})e^{-\tau\hat{H}_{phon}} \\ &= e^{\tau H_{phon}}\hat{b}_q e^{-\tau\hat{H}_{phon}} + e^{\tau\hat{H}_{phon}}\hat{b}^\dagger_{-q}e^{-\tau\hat{H}_{phon}} = e^{-\tau\hbar\omega_q}\hat{b}_q + e^{\tau\hbar\omega_q}\hat{b}^\dagger_{-q}, \end{aligned}$$

where the phonon hamiltonian is given by $\hat{H}_{phon} = \sum_q \hbar\omega_q\left(\hat{b}^\dagger_q\hat{b}_q + \frac{1}{2}\right)$ (we need to consider the non interacting part and the non fermionic part since we are basically treating free bosons). With this expression for the operator $A(q,\tau)$, we can now compute the phonon propagator in the imaginary time formalism:

$$\begin{aligned} D_0(q,\tau) = -\left\langle T_\tau A(q,\tau)A(-q,0)\right\rangle_0 &= -\left\langle A(q,\tau)A(-q,0)\right\rangle_0 \quad \text{for } \tau > 0 \\ &= -\left\langle (\hat{b}_q(\tau) + \hat{b}^\dagger_{-q}(\tau))(\hat{b}_{-q}(0) + \hat{b}^\dagger_q(0))\right\rangle_0 \\ &= -\left\langle (e^{-\tau\hbar\omega_q}\hat{b}_q + e^{\tau\hbar\omega_q}\hat{b}^\dagger_{-q})(\hat{b}_{-q} + \hat{b}^\dagger_q)\right\rangle_0 \end{aligned}$$

where the signo in ω_q is positive even when associated to $\hat{b}^\dagger_{-q}$ since the dispersion relation of the phonons is an even function of the momentum, i.e., $\omega_q = \omega_{-q}$. The last expression can be computed remembering that we are considering the non interacting part of the hamiltonian, and that the phonons are bosonic particles, which means that they can occupy the same state. The final result is given by

$$D_0(q,\tau) = -\left[\left\langle \hat{b}_q \hat{b}_q^\dagger \right\rangle_0 e^{-\tau\hbar\omega_q} + \left\langle \hat{b}^\dagger_{-q} \hat{b}_{-q} \right\rangle_0 e^{\tau\hbar\omega_q}\right],$$

where from four terms only two remains, due to orthogonality of the states. The remaining two terms can be computed remembering that the phonons are bosonic particles, which means that they can occupy the same state, and that the average number of phonons in a state with energy $\hbar\omega_q$ is given by the Bose-Einstein distribution $n_B(\omega_q) = \frac{1}{e^{\beta\hbar\omega_q}-1}$. Thus we have

$$D_0(q,\tau) = -\left[(n_B(\omega_q)+1)e^{-\tau\hbar\omega_q} + n_B(\omega_q)e^{\tau\hbar\omega_q}\right],$$

and now we can finally compute the phonon propagator in the Matsubara frequency space by performing a Fourier transform of the previous expression:

$$\begin{aligned}
D_0(q,i\nu_n) = \int_0^\beta \mathrm{d}\tau e^{i\nu_n\tau} D_0(q,\tau) &= \int_0^\beta \mathrm{d}\tau e^{i\nu_n\tau}\left[-(n_B(\omega_q)+1)e^{-\tau\hbar\omega_q} - n_B(\omega_q)e^{\tau\hbar\omega_q}\right], \\
&= -(n_B(\omega_q)+1)\int_0^\beta \mathrm{d}\tau e^{(i\nu_n-\hbar\omega_q)\tau} - n_B(\omega_q)\int_0^\beta \mathrm{d}\tau e^{(i\nu_n+\hbar\omega_q)\tau} \\
&= -(n_B(\omega_q)+1)\frac{e^{(i\nu_n-\hbar\omega_q)\beta}-1}{i\nu_n-\hbar\omega_q} - n_B(\omega_q)\frac{e^{(i\nu_n+\hbar\omega_q)\beta}-1}{i\nu_n+\hbar\omega_q}.
\end{aligned}$$

We now that the matsubara frequencies for bosons are given by $\nu_n = 2n\pi/\beta$, which means that $e^{i\nu_n\beta} = 1$ (for fermions this would be -1). With this expression, we can simplify the previous result as

$$\begin{aligned}
D_0(q,i\nu_n) &= -(n_B(\omega_q)+1)\frac{e^{-\beta\hbar\omega_q}-1}{i\nu_n-\hbar\omega_q} - n_B(\omega_q)\frac{e^{\beta\hbar\omega_q}-1}{i\nu_n+\hbar\omega_q} \\
&= -(n_B(\omega_q)+1)\frac{1-e^{-\beta\hbar\omega_q}}{i\nu_n-\hbar\omega_q} - \frac{1}{e^{\beta\hbar\omega_q}-1}\frac{e^{\beta\hbar\omega_q}-1}{i\nu_n+\hbar\omega_q},
\end{aligned}$$

which in the end will give us the final result for the phonon propagator in the Matsubara frequency space:

$$D_0(q,i\nu_n) = \frac{1}{i\nu_n-\hbar\omega_q} - \frac{1}{i\nu_n+\hbar\omega_q} = \frac{2\hbar\omega_q}{(i\nu_n)^2-(\hbar\omega_q)^2}.$$

where we can see that the phonon propagator has poles at the phonon frequencies ω_q and $-\omega_q$, which is a signature of the fact that the phonons are bosonic particles. The phonon propagator is a key quantity in the theory of electron-phonon interactions, as it describes how the phonons mediate the interaction between the electrons. Comparing it to the fermionic green's function in the matsubara frequency space, we can see that the phonon propagator has a different structure:

$$\begin{cases}
D_0(q,i\nu_n) = \dfrac{2\hbar\omega_q}{(i\nu_n)^2-(\hbar\omega_q)^2} & \text{for bosons,} \\
G_0(k,i\omega_n) = \dfrac{1}{i\omega_n-\xi_k} & \text{for fermions,}
\end{cases}$$

mainly due to the fact that we derived it from an interacting term which presented two terms with different signs. The phonon propagator is a key quantity in the theory of electron-phonon interactions, as it describes how the phonons mediate the interaction between the electrons.

Using this propagator, we can now compute the effective interaction between the electrons mediated by the phonons...

$\Sigma(k,i\omega_n) = [$ *draw of single line with one phonon propagator and two electron-phonon vertices* $]$

$$= \int \mathrm{d}q \sum_{i\nu_n} |g_q|^2 \mathcal{G}_0(k-q, i\omega_n - i\nu_n) D_0(q, i\nu_n),$$

[...]

Thus now we can consider two different kinds of interactions between the electrons: the direct interaction mediated by the photons, which is given by the coulomb potential, and the retarded interaction mediated by the phonons, which is given by the electron-phonon coupling constant g_q and the phonon propagator $D_0(q, i\nu_n)$. The competition between these two interactions is responsible for different phenomena in the material, such as superconductivity, charge density waves, and so on. The phonon-mediated interaction is attractive, while the photon-mediated interaction is repulsive, and this competition can lead to different phases of the material depending on the strength of the interactions and the temperature.

Thus now we have two different interactions between the electrons:

$$V_{eff}(q, i\nu_n) = V_{phot} + V_{el-phon} = e\frac{4\pi}{q^2}e + \frac{1}{V}|g_q|^2 D_0(q, i\nu_n),$$

which is the effective interaction between the electrons mediated by both the photons and the phonons. The competition between these two interactions is responsible for different phenomena in the material, such as superconductivity, charge density waves, and so on. The phonon-mediated interaction is attractive, while the photon-mediated interaction is repulsive, and this competition can lead to different phases of the material depending on the strength of the interactions and the temperature.

Thus

$V_{eff} =$ [*draw of effective interaction mediated by both photons and phonons*]

where the general formula for the electron-phonon coupling constant is given by

$$g_q = ie\frac{\sqrt{N}}{\sqrt{2M\hbar\omega_q}} q \frac{4\pi Ze}{q^2},$$

which in the specific case of the **jellium model** can be rewritten as

$$g_q^2 = \frac{\Omega}{2} V_c(q),$$

where Ω is the characteristic frequency of the system, improperly letting us write $\delta\rho^+(t) \propto e^{i\Omega t}\delta\rho$. This formula can be computed as...

[...]

Putting this result back in the expression for the effective interaction, we can see that the effective interaction between the electrons mediated by both the photons and the phonons is given by

$$V_{eff} = V_c(q) + \frac{\Omega}{2} V_c(q) D_0(q, i\nu_n) = V_c(q)\left[1 + \frac{\Omega}{2} D_0(q, i\nu_n)\right] = V_c(q)\left[1 + \frac{\Omega}{2}\frac{2\Omega}{(i\nu_n)^2 - \Omega^2}\right].$$

Since this result is specific for the jellium model, where we do not have a real lattice, thus the specific frequency of the system will be constant (which then won't depend on the momentum q), and we can rewrite the previous result as

$$V_{eff} = V_c(q)\left[1 + \frac{\Omega^2}{(i\nu_n)^2 - \Omega^2}\right] = V_c(q)\frac{(i\nu_n)^2}{(i\nu_n)^2 - \Omega^2},$$

which is the effective interaction between the electrons mediated by both the photons and the phonons in the jellium model. We can see that for frequencies ν_n much smaller than Ω, the effective interaction is attractive, while for frequencies ν_n much larger than Ω, the effective interaction is repulsive. This is a signature of the fact that the phonon-mediated interaction is attractive, while the photon-mediated interaction is repulsive, and this competition can lead to different phases of the material depending on the strength of the interactions and the temperature.

If we consider the analytical continuation of the effective interaction to the real frequency axis, we can see that the effective interaction has a pole at the frequency Ω, which is a signature of the fact that the phonons are bosonic particles, and that they mediate the interaction between the electrons. The effective interaction can be rewritten as

$$V_{eff}(q, i\nu_n) = V_c(q)\frac{\nu^2}{\nu^2 - \Omega^2} \xrightarrow{an.cont.} V^R_{eff}(q,\nu) = V_c(q)\frac{\nu^2}{\nu^2 - \Omega^2 + i\delta},$$

and if we try to plot it as a function of the real frequency ν, we can see that it has a peak at the frequency Ω, which is a sort of resonance. The effective interaction is attractive (negative) for frequencies ν much smaller than Ω, while it is repulsive (positive) for frequencies ν much larger than Ω, which is a signature of the competition between the phonon-mediated interaction and the photon-mediated interaction.

[*Draw plot of effective interaction as a function of real frequency*]

If we try to imagine the physical meaning of this result, we can imagine electrons on a thin layer ($\sim$ 10 eV) of the fermi surface, which can interact with each other by exchanging phonons with energy Ω of the order of millielectronvolts. For frequencies ν much smaller than Ω, the effective interaction is attractive, which means that the electrons can form Cooper pairs and become superconducting. For frequencies ν much larger than Ω, the effective interaction is repulsive, which means that the electrons cannot form Cooper pairs and remain in a normal state. The competition between these two interactions is responsible for the superconductivity in the material, which will be discussed in future.

[*Draw of fermi surface and order of energies involved*]

We are used to think that two particles that interact with each other through a repulsive interaction cannot form a bound state, but in this case we have an effective interaction that is attractive for frequencies ν much smaller than Ω. This is just the many body justification of the repulsive and attractive nature of the photon-mediated and phonon-mediated interactions, respectively.

Polarizability of the lattice as an electron passes by the lattice, which is the physical origin of the electron-phonon interaction. The electron can polarize the lattice, creating a distortion in the lattice, which can then interact with another electron, leading to an effective attraction between the electrons mediated by the phonons. This is the physical picture behind the electron-phonon interaction, and it is responsible for different phenomena in the material, but depends on the time scales of this polarization and the entity of the interaction: the two electrons has to interact at energies near the energy of the lattice vibration. The effective interaction, depending from energy scale and momentum, can be attractive or repulsive. In the case of an attractive interaction, the system can be unstable and undergo a phase transition to a new state of matter. The nature of this new state depends on the specific form of the effective interaction and the symmetries of the system. In other cases, it can lead to charge density waves, spin density waves, or other types of ordered states. The study of effective interactions is crucial for understanding the rich variety of phases and phenomena observed in condensed matter systems.

The divergence is not very physical and it can be removed by considering the finite size of the system or by introducing a cutoff in the momentum space.

$$V_{eff}(q, i\nu_n) = Draw$$

[*Draw of sum of wiggle and spring to obtain zigzag to represent the effective interaction*]

Now we can consider the rpa approximation, which is a way to sum up an infinite series of diagrams in perturbation theory. In the rpa approximation, we consider the bubble diagrams, which represent the polarization of the system due to the interactions. The effective interaction in the rpa approximation can be written as:

$V^{eff}_{RPA}(q, i\nu_n) =$ [*zigzag + zigazag-bubble-zigzag + zigzag-bubble-zigzag-bubble-zigzag + ... = boldzigzag*]

remembering

$$V^{eff}(q, i\nu_n) = V_c(q)\frac{(i\nu_n)^2}{(i\nu_n)^2 - \Omega^2}$$

we call all that series a bold zigzag, and we can write it as:

$$(1 + [\textit{zigzag-bubble}])\, V^{eff}_{RPA}(q, i\nu_n) = [\textit{zigzag}] = V^{eff}(q, i\nu_n),$$

TODO: last mat. passage is on paper, check if you want to add it here

since it is a sort of geometric series, we can write it as:

$$V^{eff}_{RPA}(q, i\nu_n) = \frac{V^{eff}(q, i\nu_n)}{1 - V^{eff}(q, i\nu_n)\Pi_0(q, i\nu_n)} = \frac{V_c(q)}{1 - V_c(q)\Pi_0(q, i\nu_n)}\frac{(i\nu_n)^2}{(i\nu_n)^2 - \frac{\Omega^2}{1 - V_c(q)\Pi_0(q, i\nu_n)}},$$

where we can recognize the renormalized phonon frequency:

$$\frac{\Omega}{\sqrt{1 - V_c(q)\Pi_0(q, i\nu_n)}},$$

and the screened Coulomb interaction:

$$\frac{V_c(q)}{1 - V_c(q)\Pi_0(q, i\nu_n)},$$

and now we can rewrite the effective interaction in the rpa approximation as:

$$\ldots$$

and since the renormalized phonon frequency has an imaginary part, it can be used to remove the divergence of the effective interaction, and it can lead to a finite value of the effective interaction at low energy and momentum.

[*Draw of effective potential with rpa approximation, in which the curve do not goes to infinity and omegaq*]

The main message is that the effective interaction between two electrons in a lattice, mediated by phonons, can be attractive at low energy and momentum, and it can lead to the formation of Cooper pairs and superconductivity. The rpa approximation is a powerful tool to study the effective interaction in condensed matter systems, and it can provide insights into the mechanisms of superconductivity and other phenomena.

The new frequency

$$\omega_q = \frac{\Omega}{\sqrt{1 - \frac{4\pi}{q^2}\Pi_0(q, i\nu_n)}} \simeq \frac{\Omega}{\sqrt{1 - \frac{4\pi}{q^2}\mathcal{N}(0)}} \xrightarrow{q\to 0} \frac{\Omega}{\sqrt{1 + \frac{q^2_{TF}}{q^2}}} \sim \Omega\frac{q}{q_{TF}},$$

where we have introduced the Thomas-Fermi wavevector:

$$q^2_{TF} = 4\pi\mathcal{N}(0),$$

and we can see that the phonon frequency goes to zero as $q \to 0$. This means also that the frequancy goes linearly with q at small q.

3.7 | Cooper pairs and superconductivity

[...]

[*Draw of sphere of metal, we add two extra electrons with opposite momenta and spin on top of it, forming a narrow shell (dotted circle) with energy proportional to frequency of*]

fermi energy $\sim$ 10eV, phonon frequency $\sim$ 0.01eV, so the shell is very narrow, and the electrons in this shell can form Cooper pairs due to the attractive interaction mediated by phonons. The Cooper pairs can condense into a superconducting state at low temperatures, leading to zero resistance and other remarkable properties of superconductors. The study of Cooper pairs and superconductivity is a fundamental topic in condensed matter physics, and it has important implications for both fundamental science and technological applications.

In order to form a cooper pair, we do not need only a finite attractive interaction, but we also need a finite density of states at the Fermi level, in order to have a finite number of electrons that can form Cooper pairs. There is a fermi sphere, and we add two extra electrons with opposite momenta and spin on top of it, forming a narrow shell with energy proportional to the frequency of the phonons. Then we can consider the attractive potential mediated by phonons.

From a first quantized perspective we want to solve the Schrodinger equation for two electrons interacting via the effective interaction mediated by phonons. The Schrodinger equation can be written as:

$$\left[\frac{\hat{P}_1^2}{2m}+\frac{\hat{P}_2^2}{2m}+U(\hat{r}_1,\hat{r}_2)\right]\psi(r_1,\sigma_1;r_2,\sigma_2)=E\psi(r_1,\sigma_1;r_2,\sigma_2),$$

we wnat to find the bound state of two electrons, and get the energy of the bound state.

We can factorize the wavefunction as:

$$\psi(r_1,\sigma_1;r_2,\sigma_2)=\phi(r_1,r_2)\chi(\sigma_1,\sigma_2),$$

as usual, in a radial and angular part. The spinorial component can be either a singlet or a triplet, and the spatial component can be either symmetric or antisymmetric. The spinor is given by

$$\chi(\sigma_1,\sigma_2)=\begin{cases}(s=0) & \frac{1}{\sqrt{2}}\left(|\uparrow\downarrow\rangle-|\downarrow\uparrow\rangle\right) \qquad \text{singlet}\\ (s=1) & \begin{cases}|\uparrow\uparrow\rangle\\ \frac{1}{\sqrt{2}}\left(|\uparrow\downarrow\rangle+|\downarrow\uparrow\rangle\right)\\ |\downarrow\downarrow\rangle\end{cases}\end{cases}$$

The spinorial part can be either symmetric or antisymmetric, and the spatial part can be either symmetric or antisymmetric. The total wavefunction must be antisymmetric under the exchange of the two electrons, so if the spinorial part is a singlet (antisymmetric), the spatial part must be symmetric, and vice versa.

The radial part of the wavefunction can be written as:

$$\phi(r_1,r_2)=\sum_{\mathbf{k}}g(\mathbf{k})\frac{e^{i\mathbf{k}\cdot\mathbf{r}_1}}{\sqrt{V}}\frac{e^{-i\mathbf{k}\cdot\mathbf{r}_2}}{\sqrt{V}}=\sum_{\mathbf{k}}g(\mathbf{k})\frac{e^{i\mathbf{k}\cdot(\mathbf{r}_1-\mathbf{r}_2)}}{V},$$

and in order to infer the symmetry of the spatial part, we have to exploit the nature of the summation constant $g(\mathbf{k})$. If $g(\mathbf{k})$ is an even function of $\mathbf{k}$, then the spatial part is symmetric, and if $g(\mathbf{k})$ is an odd function of $\mathbf{k}$, then the spatial part is antisymmetric.

Then the total wavefunction can be written as:

$$\psi(r_1,\sigma_1;r_2,\sigma_2) = \sum_{\mathbf{k}} g(\mathbf{k}) \frac{e^{i\mathbf{k}\cdot(\mathbf{r}_1-\mathbf{r}_2)}}{V}\chi^S,$$

where χ^S is the spinorial part, which can be either a singlet ($S = 0$) or a triplet ($S = 1$). If we exchange the two particles, we get a minus sign on the exponential (we exchanged $\mathbf{r}_1$ and $\mathbf{r}_2$), but we need to add a condition to the function $g(\mathbf{k})$ in order to have the total wavefunction antisymmetric:

$$\begin{aligned} S = 0) \quad & g(\mathbf{k}) = g(-\mathbf{k}), \\ S = 1) \quad & g(\mathbf{k}) = -g(-\mathbf{k}), \end{aligned}$$

so if the spinorial part is a singlet, the spatial part must be symmetric, and if the spinorial part is a triplet, the spatial part must be antisymmetric.

The symmetry of the spatial part will become pretty relevant in a moment, when we discuss...

We put the two electrons in a narrow shell around the Fermi surface, such that from this follows a constraint:

$$g(\mathbf{k}) = 0 \quad \text{if} \quad E_{\mathbf{k}} = \frac{\hbar^2 k^2}{2m} < E_F, g(\mathbf{k}) = 0 \quad \text{if} \quad E_{\mathbf{k}} = \frac{\hbar^2 k^2}{2m} > E_F + \hbar\omega_D,$$

where we know the Debye frequency $\omega_D \ll E_F$ to be orders of magnitude smaller than the Fermi energy, so the shell is very narrow.

Let us now define

$$U_{kk'} = \frac{1}{V}\int \mathrm{d}^3r_1\mathrm{d}^3r_2\, e^{i(\mathbf{k}-\mathbf{k}')\cdot(\mathbf{r}_1-\mathbf{r}_2)}U(\mathbf{r}_1,\mathbf{r}_2),$$

and we can write the Schrodinger equation as:

$$\left[-\frac{\hbar^2}{2m}(\nabla_1^2+\nabla_2^2) + U(\mathbf{r}_1,\mathbf{r}_2)\right]\cdots = 0,$$

such that

$$\sum_{k'}\left[2\frac{\hbar^2|\mathbf{k}'|^2}{2m} + U(\mathbf{r}_1,\mathbf{r}_2) - E\right] g(\mathbf{k}')\frac{1}{V}e^{i(\mathbf{k}'-\mathbf{k})\cdot(\mathbf{r}_1-\mathbf{r}_2)}\chi^S = 0,$$

and if we now integrate over $\mathbf{r}_1$ and $\mathbf{r}_2$, we get:

$$\int\int \mathrm{d}^3r_1\mathrm{d}^3r_2 \sum_{k'}\left[2\frac{\hbar^2|\mathbf{k}'|^2}{2m} + U(\mathbf{r}_1,\mathbf{r}_2) - E\right] g(\mathbf{k}')\frac{1}{V}e^{i(\mathbf{k}'-\mathbf{k})\cdot(\mathbf{r}_1-\mathbf{r}_2)}\chi^S = 0,$$

and it will still be zero if the term in square brackets is zero, so we can write:

$$\begin{aligned} &\sum_{k'}\left[2E_{k'} - E\right] g(k') \int\int \mathrm{d}^3r_1\mathrm{d}^3r_2\, \frac{1}{V}e^{i(\mathbf{k}'-\mathbf{k})\cdot(\mathbf{r}_1-\mathbf{r}_2)} \\ &+ \sum_{k'} g(k') \int\int \mathrm{d}^3r_1\mathrm{d}^3r_2\, U(\mathbf{r}_1,\mathbf{r}_2)\frac{1}{V}e^{i(\mathbf{k}'-\mathbf{k})\cdot(\mathbf{r}_1-\mathbf{r}_2)} = 0, \end{aligned}$$

which has to be true for any χ^S, and we can recognize the Fourier transform of the potential $U_{kk'}$, while the first term gives basically a delta function, so we can write:

$$(2E_k - E)g(\mathbf{k}) + \sum_{k'} U_{kk'}g(\mathbf{k}') = 0,$$

which is an equation for the coefficients $g(\mathbf{k})$, which we know to be nonzero only in a narrow shell around the Fermi surface:

$$E_F < E_k,\, E_{k'} < E_F + \hbar\omega_D.$$

In this narrow shell, we can approximate the potential $U_{kk'}$ as a constant, since it is mediated by phonons, and it is attractive, so we can write:

$$U_{kk'} = -|U_0|,$$

which is just an hypothesis, since in real materials the potential can have a more complicated form, but it is a good approximation to understand the physics of Cooper pairs and superconductivity. When this interaction is constant and attractive in the narrow shell around the Fermi surface, we can write the equation for $g(\mathbf{k})$ as:

$$(2E_k - E)g(\mathbf{k}) - |U_0| \sum_{k'} g(\mathbf{k}') = 0.$$

Now we have to make use of the symmetry of the spatial part of the wavefunction, which is symmetric for the singlet case and antisymmetric for the triplet case. In the triplet case, we can write:

$$E = 2E_k,$$

since the summation goes to zero due to the antisymmetry of the spatial part, and we have just a sum of their energy: it is not a bound state, since the energy is just the sum of the two electrons, and it is not lower than the energy of two free electrons. In the singlet case, we can write:

$$E = 2E_k - |U_0|N,$$

where K is just a constant to highlight that the energy is lower than the energy of two free electrons, and it is a bound state (since the summation of symmetric functions gives a nonzero finite result).

We can write the equation for $g(\mathbf{k})$ as:

$$\sum_k g(\mathbf{k}) = \sum_k \frac{|U_0| \sum_{k'} g(\mathbf{k}')}{2E_k - E},$$

but since the two summations are equal to N we can write the expression for the energy which solves our schrodinger equation:

$$1 = |U_0| \sum_k \frac{1}{2E_k - E},$$

and it is often reffered to as the **gap equation**, since it gives the energy of the bound state, which is the energy gap of the superconductor. The solution of this equation gives us the energy of the Cooper pair, which is lower than the energy of two free electrons, and it is a bound state. The existence of this bound state is a key ingredient for the formation of Cooper pairs and superconductivity.

We are summing over $\mathbf{k}$ functions of the energy, so we can use the density of states to write the summation as an integral over energy:

$$\sum_k \mathcal{F}(E_k) = \sum_k \int \mathrm{d}\omega\, \delta(\omega - E_k)\mathcal{F}(\omega) = \int \mathrm{d}\omega \left[\sum_k \delta(\omega - E_k)\right] \mathcal{F}(\omega) = \int \mathrm{d}\omega\, D(\omega)\mathcal{F}(\omega),$$

where the formal expression in square brackets is the definition of the density of states $D(\omega)$. Then we can write the gap equation as:

$$1 = |U_0| \int_{E_F}^{E_F + \hbar\omega_D} \mathrm{d}\omega\, \frac{D(\omega)}{2\omega - E},$$

where we restricted the integration to the narrow shell around the Fermi surface(since even in the starting summation over $\mathbf{k}$ we were selecting only the momenta which correspond to energies in that shell).

[*Draw of density of states as a function of energy, with a narrow shell around the Fermi level*]

If we plot for example the density of states of the standard jellium model in 3D, we can see that it is a smooth function of energy, and it is finite at the Fermi level; since the Debye frequency is much smaller than the Fermi energy, we can approximate the density of states as a constant in the narrow shell around the Fermi surface, and we can write:

$$1 = |U_0| D(E_F) \int_{E_F}^{E_F+\hbar\omega_D} \mathrm{d}\omega \, \frac{1}{2\omega - E},$$

which can be easily computed knowing that the primitive is a logarithm, and we get:

$$1 = \frac{1}{2} |U_0| D(E_F) \ln\left(\frac{2E_F + 2\hbar\omega_D - E}{2E_F - E}\right).$$

Now inverting this equation, we can write the energy of the bound state as:

$$\frac{2E_F - E}{2E_F + 2\hbar\omega_D - E} = e^{-\frac{2}{|U_0|D(E_F)}},$$

and it is now clear why we need to be on top of the fermi surface: if we were in the vacuum we couldn't have made the approximation of a constant density of states, since the density of states in the vacuum is zero at zero energy, and it increases as a power law with energy. In the presence of a finite density of states at the Fermi level, we can obtain a finite value for the energy of the bound state, which is lower than the energy of two free electrons, and it is a key ingredient for the formation of Cooper pairs and superconductivity.

The numerator of the left hand side is the bounding energy of the Cooper pair, which is the energy gap between the energy of the bound state and the energy of two free electrons, and we call it Δ_b. Thus we can write

$$\frac{\Delta_b}{\Delta_b + 2\hbar\omega_D} = e^{-\frac{2}{|U_0|D(E_F)}},$$

and we can write the energy of the bound state as:

$$\Delta_b = \frac{\hbar\omega_D e^{-\frac{1}{|U_0|D(E_F)}}}{\sinh(1/|U_0|D(E_F))},$$

and since $\sinh(x) \sim x$ for $x \to 0$, we can write the energy of the bound state as:

$$\Delta_b \simeq 2\hbar\omega_D e^{-\frac{1}{|U_0|D(E_F)}},$$

which is the famous BCS formula for the energy gap of the superconductor, and it shows that the energy gap is exponentially small in the inverse of the product of the attractive interaction and the density of states at the Fermi level. This means that even a weak attractive interaction can lead to a finite energy gap, and it is a key ingredient for the formation of Cooper pairs and superconductivity.

If you in the system there are thermal fluctuations, they can break the Cooper pairs, and they can lead to a finite temperature at which the superconducting state is destroyed. Ideally, starting from an high temperature, we can cool down the system, and at a certain critical temperature T_c the Cooper pairs will start to form, and the system will undergo a phase transition to a superconducting state. The critical temperature can be estimated from the energy gap, and it is of the order of few Kelvins, since we have an exponential dependence on the inverse of the product of the attractive interaction and the density of states at the Fermi level, which are both small quantities. The thermal energy has to be smaller than the energy gap in order to have a stable superconducting state.

To form a good superconductor, we need a large density of states at the Fermi level, and a strong attractive interaction mediated by phonons. In real materials, the situation is more complicated, since there are many other factors that can affect the superconducting properties, such as impurities, disorder, and other types of interactions. However, if we have a term $|U_0|$ which is even only infinitesimally different from zero, we can have a finite energy gap, and thus a superconducting state. This is the key message of the BCS theory of superconductivity, which is one of the most successful theories in condensed matter physics, and it has been confirmed by many experiments.

We cannot get that term taylor expanding the effective interaction mediated by phonons, since it is zero at zero energy and momentum, but we can get it by considering infinite order perturbation theory, but as presented here it is an analytic result, since we have just made some approximations to solve the Schrodinger equation for two electrons interacting via an attractive potential mediated by phonons. The key ingredient is the existence of a finite density of states at the Fermi level, which allows us to obtain a finite energy gap even for a weak attractive interaction.

Electron-phonon coupling is crucial, because the larger it is the larger the attractive interaction mediated by phonons, and thus the larger the energy gap and the critical temperature of the superconductor. Good conductors (not super-) have a very very small electron-phonon coupling, and thus they do not have a superconducting state (if they had high electron-phonon coupling, they would not conduct well, since the electrons would be scattered by phonons). On the other hand, good superconductors have a large electron-phonon coupling, which leads to a strong attractive interaction mediated by phonons, and thus to a large energy gap and a high critical temperature, but they are not good conductors, since the electrons are strongly scattered by phonons. This is a trade-off between conductivity and superconductivity, and it is one of the reasons why we do not have room temperature superconductors yet.

3.7.1 | Do not know

From a many body perspective, imagine to have

[*two electron lines entering in Lambda box, opposite spins and opposite momenta and fermionic frequencies* $= \tilde{k} = (\mathbf{k}, ik_\nu)$ *and* $\tilde{P} =$ effective interaction + two effective interactions + 3...]

The Λ box is called **scattering vertex** It is a sort of selection of diagrams, in which we are not including, for example, the cross diagrams or others, we are making a subselection (only *ladder diagrams*) to try and find the form for the scattering vertex, which is the effective interaction between two electrons in the Cooper channel, which is the channel in which the two electrons have opposite momenta and opposite spins.

In order to find an expression for the scattering vertex, we want to cut off the contributions of the external legs...

At a critical temperature, the scattering vertex diverges, and this is a signature of the instability of the normal state towards the formation of Cooper pairs and superconductivity. The divergence of the scattering vertex is a key ingredient for understanding the mechanism of superconductivity.

We can try to study the scattering vertex, and since every time we have some instabilities of the ground state (the fermi sphere) we have a divergence of the scattering vertex, we can try to find the critical temperature at which this divergence occurs, and thus we can find the critical temperature of the superconducting transition.

Dyson equation for the 2 particle scattering vertex in the Cooper channel can be written diagrammatically as: [*Lambda box = vertical wiggle + vertical wiggle attached to lambda box*]

where we used the usual trick of (after having cut off the external legs) to write the scattering vertex as a sum of the effective interaction plus the effective interaction attached to the scattering vertex itself, which is a sort of self-consistent equation for the scattering vertex, and it is a sort of geometric series, since we can iterate this equation and we get an infinite series of diagrams, which can be summed up to get a closed form for the scattering vertex.

we are just considering the particle-particle channel, in which the two electrons have opposite momenta and opposite spins, and we are not considering other channels, such as the particle-hole channel, in which the two electrons have opposite momenta but the same spin. in functional renormization group theory, it is possible to consider all the channels at the same time, and to study the flow of the scattering vertex in all the channels, but here we are just considering the particle-particle channel, which is the most relevant for superconductivity.

$$\Lambda(\mathbf{k},\mathbf{p}) = V + \frac{1}{\beta\Omega} \sum_{|i\omega_n|<\omega_D} \sum_{\mathbf{k}_1:\epsilon_{\mathbf{k}_1}<\omega_D} \mathcal{G}_{0\uparrow}(\mathbf{k}_1, i\omega_n)\mathcal{G}_{0\downarrow}(-\mathbf{k}_1, -i\omega_n)\Lambda(\mathbf{k}_1,\mathbf{p}),$$

where we can assume the interaction potential to be a constant in the narrow shell around the Fermi surface, and also that everything is different from zero; since there is no $\mathbf{k}$ dependence under this assumption (it would have been given by the interaction potential, but we just assumed it constant), we can write the scattering vertex as a constant, and we can write the equation for the scattering vertex as:

$$\Lambda = \frac{V}{1 - \frac{V}{\beta}\sum_{i\omega_n}\frac{1}{\Omega}\sum_{\mathbf{k}_1}\mathcal{G}_{0\uparrow}(\mathbf{k}_1, i\omega_n)\mathcal{G}_{0\downarrow}(-\mathbf{k}_1, -i\omega_n)},$$

which is the RPA expression for the scattering vertex in the Cooper channel, and we can see that the scattering vertex diverges when the denominator goes to zero, which is a signature of the instability of the normal state towards the formation of Cooper pairs and superconductivity.

In the denominator we have substantially a bubble diagram, which is the product of two Green's functions, and it is called the particle-particle bubble, since it describes the propagation of two particles (the two electrons) in the Cooper channel. The divergence of the scattering vertex occurs when the particle-particle bubble becomes large enough to compensate the attractive interaction V, and this is a signature of the instability of the normal state towards the formation of Cooper pairs and superconductivity.

[*Particle-Particle Bubble vs particle-hole bubble, with the two Green's functions going in the same direction for the particle-particle bubble, and in opposite direction for the particle-hole bubble*]

when the denominator goes to zero (with β) we can find the critical temperature at which the superconducting transition occurs, which is given by the same critical temperature that we found from the gap equation, since they are both describing the same physics of the formation of Cooper pairs and superconductivity. It was not clear at the beginning, but now we can see that the gap equation and the divergence of the scattering vertex are two different ways to describe the same physics of the formation of Cooper pairs and superconductivity, and they give us the same critical temperature for the superconducting transition.

We need to compute the bubble and see when it is equal to 1, in order to find the critical value of the interaction V at which the scattering vertex diverges, and thus we can find the critical temperature for the superconducting transition. The bubble can be computed by performing the summation over Matsubara frequencies and the integration over momenta, and it gives us a function of temperature, which we can then use to find the critical temperature at which the scattering vertex diverges.

[*Particle-Particle Bubble, with* $\mathbf{k}_1, i\omega_n$ *and* $-\mathbf{k}_1, -i\omega_n$ *pointing in the same direction* =]

$$diag = \frac{1}{\beta}\sum_{i\omega_n}\frac{1}{\Omega}\sum_{\mathbf{k}_1}\mathcal{G}_{0\uparrow}(\mathbf{k}_1, i\omega_n)\mathcal{G}_{0\downarrow}(-\mathbf{k}_1, -i\omega_n) = \frac{1}{\beta}\sum_{i\omega_n}\frac{1}{\Omega}\sum_{\mathbf{k}_1}\frac{1}{i\omega_n - \xi_{\mathbf{k}_1}}\frac{1}{-i\omega_n - \xi_{\mathbf{k}_1}},$$

we consider the following integral in the complex plane:

$$I = \int \frac{\mathrm{d}z}{2\pi i}\frac{1}{z - \xi_{\mathbf{k}_1}}\frac{1}{-z - \xi_{\mathbf{k}_1}}n_F(z),$$

which is the integral of the product of the same two Green's functions and the Fermi distribution function, and it has poles at $z = \xi_{\mathbf{k}_1}$ and $z = -\xi_{\mathbf{k}_1}$, which are the poles of the Green's functions, and it has also poles at the Matsubara frequencies $z = i\omega_n$, which are the poles of the Fermi distribution function. We can compute this integral by closing the contour in the upper half plane, as we already did when we computed the summation over Matsubara frequencies, and we get:

$$bubble = \frac{1}{\Omega}\sum_{\mathbf{k}_1}\frac{1 - 2n_F(\xi_{\mathbf{k}_1})}{2\xi_{\mathbf{k}_1}},$$

where the temperature dependence is given by the Fermi distribution function, and it is crucial for finding the critical temperature at which the scattering vertex diverges, and thus for finding the critical temperature for the superconducting transition. The bubble is a function of temperature, and it increases as the temperature decreases, and it diverges at zero temperature, which is a signature of the instability of the normal state towards the formation of Cooper pairs and superconductivity at zero temperature.

The critical temperature can be found by solving the equation $1 = V \cdot bubble(T_c)$, which gives us the critical temperature at which the scattering vertex diverges, and thus at which the superconducting transition occurs. If we now transform the summation over momenta into an integral over energy, we can write the bubble as:

$$bubble = \int \mathrm{d}\epsilon\, D(\epsilon)\frac{1 - 2n_F(\epsilon)}{2\epsilon} = \int_{E_F - \omega_D}^{E_F + \omega_D} \mathrm{d}\epsilon\, D(\epsilon)\frac{1 - 2n_F(\epsilon)}{2\epsilon},$$

where we have restricted the integration to the narrow shell around the Fermi surface, and we can approximate the density of states as a constant in this narrow shell, and we can write:

$$bubble = D(E_F)\int_{E_F}^{E_F + \omega_D} \mathrm{d}\epsilon\, \frac{1 - 2n_F(\epsilon)}{\epsilon} = D(E_F)\int_0^{\omega_D} \mathrm{d}\epsilon\, \frac{1 - 2n_F(\epsilon)}{\epsilon},$$

and since $\omega_D \sim 100K$, we can split the integral into two parts, one from 0 to k_BT, and one from k_BT to ω_D (assuming $k_BT \ll \omega_D$ and $k_BT = 0$ on the Fermi surface), and we can write:

$$bubble = D(E_F)\left[\int_0^{k_BT} \mathrm{d}\epsilon\, \frac{1 - 2n_F(\epsilon + E_F)}{\epsilon} + \int_{k_BT}^{\omega_D} \mathrm{d}\epsilon\, \frac{1 - 2n_F(\epsilon + E_F)}{\epsilon}\right],$$

[*some drawing of integral plot*]

since we are considering integrals of a sort of hyperbolic tangent (looking at the numerator), which it is approximately equal to 1 for $\epsilon \gg k_B T$, and it is approximately equal to -1 for $\epsilon \ll k_B T$ (furthermore linear in ϵ for $\epsilon \sim k_B T$) we can understand that

- the first integral gives a contribution of the order of $D(E_F)(k_B T)$, since in the regime of the first integral the hyperbolic tangent is approximately linear with ϵ, which after the integration gives a contribution of the order of $k_B T$;
- the second integral gives a contribution of the order of $D(E_F)\ln\left(\frac{\omega_D}{k_B T}\right)$, which is positive and finite, since the integral is dominated by the ending part of the hyperbolic tangent, which is approximately equal to 1 for $\epsilon \gg k_B T$: we are left with a logarithmic divergence.

thus we can write the bubble as:

$$\textit{bubble} = D(E_F)\left[k_B T + \ln\left(\frac{\omega_D}{k_B T}\right)\right] \sim D(E_F)\ln\left(\frac{\hbar\omega_D}{k_B T}\right),$$

and since we are looking for the critical temperature at which the scattering vertex diverges, we can write the equation for the critical temperature as:

$$VD(E_F)\ln\left(\frac{\hbar\omega_D}{k_B T_c}\right) = 1,$$

which gives us the critical temperature as:

$$T_c = \frac{\hbar\omega_D}{k_B} e^{-\frac{1}{VD(E_F)}},$$

which is the same critical temperature that we found from the gap equation, and it shows that the critical temperature is exponentially small in the inverse of the product of the attractive interaction and the density of states at the Fermi level, which is a key ingredient for the formation of Cooper pairs and superconductivity. This is a remarkable result, since it shows that even a weak attractive interaction can lead to a finite critical temperature, and thus to a superconducting state.

If one had computed the integral of the fermi function precisely, one would have found a prefactor of 1.14 in front of the Debye frequency, but since we are just interested in the order of magnitude of the critical temperature, we can neglect this prefactor.

3.8 | BCS theory

[...]

We can define the coopers pair operator as

$$\hat{C}^\dagger_{\mathbf{k}\uparrow}\hat{C}^\dagger_{-\mathbf{k}\downarrow}$$

[...]

We can write the BCS Hamiltonian as

$$H_{BCS} = \sum_{\mathbf{k},\sigma} \xi_{\mathbf{k}\sigma} \ldots\ldots\ldots$$

where the interacting term has theh creation and annihilation of cooper pairs multiplied by the potential. The first term is the kinetic energy of the electrons, while the second term is the interaction between the electrons that leads to the formation of cooper pairs.

Diagrammatically, we can represent the BCS Hamiltonian as

[*the hamiltonian tales a cooper pair and scatter it back (through the potential part) in another cooper pair*]

We have an interaction term that scatters a cooper pair from one state to another. We cannot solve this hamiltonian exactly, but we can use some techniques to find an approximate solution some of which we already know.

We will use the **mean field approximation** to solve this hamiltonian. If we have two operators $\hat{A}$ and $\hat{B}$, we can write their product as

$$\hat{A}\hat{B} \simeq \langle\hat{A}\rangle\hat{B} + \hat{A}\langle\hat{B}\rangle - \langle\hat{A}\rangle\langle\hat{B}\rangle,$$

in order to have only mean field terms of single operators.

We already encountered an approximation of this kind: recalling the coulmb potential

$$V(q)C^\dagger_{\mathbf{k}_1+\mathbf{q}\uparrow}C^\dagger_{\mathbf{k}_2-\mathbf{q}\downarrow}C_{\mathbf{k}_1\downarrow}C_{\mathbf{k}_2\uparrow},$$

and we wrote it as the mean field approximation, finding some averages which enforced $q = 0$ for the hartree term plus an exchange term for Fock.

If we apply the same approximation to the BCS hamiltonian, we can write

$$H_{BCS} \simeq \ldots$$

which in the end gives a constant term plus a term that is quadratic in the creation and annihilation operators (the constant can be reabsorbed in the chemical potential). We have no more interaction term, it is not anymore an interacting hamiltonian. But there are strange terms as strange combinations of creation and annihilation operators.

$\langle C^\dagger_{\mathbf{k}\uparrow}C_{-\mathbf{k}\downarrow}\rangle$ is strange since we have the same state at the edges of the average, but it is not zero: before, if we encountered a term like this, since one operator destroys a particle and the other creates a particle, in different states, we would say that the average is zero. We will see later the consequences of these terms being non zero.

Let us define

$$\begin{cases} \Delta_{\mathbf{k}} = \dfrac{1}{N} \displaystyle\sum_{\mathbf{k}'} V_{\mathbf{k}\mathbf{k}'} \langle C_{-\mathbf{k}'\downarrow} C_{\mathbf{k}\uparrow} \rangle, \\ \Delta^*_{\mathbf{k}} = \dfrac{1}{N} \displaystyle\sum_{\mathbf{k}'} V_{\mathbf{k}\mathbf{k}'} \langle C^\dagger_{\mathbf{k}'\uparrow} C^\dagger_{-\mathbf{k}'\downarrow} \rangle, \end{cases}$$

then we can write the hamiltonian as

$$H_{BCS} \simeq \sum_{\mathbf{k},\sigma} \xi_{\mathbf{k}\sigma} C^\dagger_{\mathbf{k}\sigma} C_{\mathbf{k}\sigma} + \sum_{\mathbf{k}} (\Delta_{\mathbf{k}} C^\dagger_{\mathbf{k}\uparrow} C^\dagger_{-\mathbf{k}\downarrow} + \Delta^*_{\mathbf{k}} C_{-\mathbf{k}\downarrow} C_{\mathbf{k}\uparrow}),$$

which is a quadratic hamiltonian that we can solve exactly: it is a hamiltonian of free particles with an anomalous term that creates and destroys cooper pairs. We can diagonalize this hamiltonian in order to find the energy spectrum of the system and the ground state wavefunction.

If we look at the newly defined $\Delta_{\mathbf{k}}$, we see that it is a sum of the potential multiplied by the average of the cooper pair operator. We can interpret $\Delta_{\mathbf{k}}$ as the **order parameter** of the superconducting phase, since it is non zero only in the superconducting phase and it is zero in the normal phase. The fact that $\Delta_{\mathbf{k}}$ is non zero means that there is a finite average of the cooper pair operator, which means that there is a finite density of cooper pairs in the system.

Imagine to take some normal state, like the fermi sphere

$$\Psi_{FS} = \prod_{\mathbf{k}\in FS} C^\dagger_{\mathbf{k}\uparrow} C^\dagger_{-\mathbf{k}\downarrow} \left|0\right\rangle,$$

but if we try to compute the average of the cooper pair operator in this state, we find that it is zero. This means that the normal state does not have a finite density of cooper pairs, and therefore it is not a superconducting state. It is intuitive, since we did not include any mechanism to form cooper pairs in the normal state. If we now use the following ansatz for the wavefunction

$$\Psi_{BCS} = \prod_{\mathbf{k}\in FS} \left[u_{\mathbf{k}} + v_{\mathbf{k}} C^\dagger_{\mathbf{k}\uparrow} C^\dagger_{-\mathbf{k}\downarrow} \right] \left|0\right\rangle,$$

with the condition of $|u_{\mathbf{k}}|^2 + |v_{\mathbf{k}}|^2 = 1$ for each $\mathbf{k}$. This wavefunction is a superposition of the normal state and the state with cooper pairs, and it is the ground state of the BCS hamiltonian. We do not justify this ansatz, but we will see later that it is the correct one. Even historically, this ansatz was proposed since it explained the experimental results, and then it was justified a posteriori by the diagonalization of the hamiltonian.

If we now try to rewrite it in a vectorial form, we define the **Nambu spinor**

$$\Psi_{\mathbf{k}} = \begin{pmatrix} C_{\mathbf{k}\uparrow} \\ C^\dagger_{-\mathbf{k}\downarrow} \end{pmatrix}, \quad \Psi^\dagger_{\mathbf{k}} = \begin{pmatrix} C^\dagger_{\mathbf{k}\uparrow} & C_{-\mathbf{k}\downarrow} \end{pmatrix},$$

which are very convenient to use in this context since they allow us to write the hamiltonian in a compact form. In order to do so, let us try to rezrite the following term with these spinors

$$\sum_{\mathbf{k}\sigma} \xi_{\mathbf{k}\sigma} C^\dagger_{\mathbf{k}\sigma} C_{\mathbf{k}\sigma} = \sum_{\mathbf{k}} \xi_{\mathbf{k}} (C^\dagger_{\mathbf{k}\uparrow} C_{\mathbf{k}\uparrow} + C^\dagger_{\mathbf{k}\downarrow} C_{\mathbf{k}\downarrow}) = \sum_{\mathbf{k}} \xi_{\mathbf{k}} (C^\dagger_{\mathbf{k}\uparrow} C_{\mathbf{k}\uparrow} + C^\dagger_{-\mathbf{k}\downarrow} C_{-\mathbf{k}\downarrow})$$

where the last step is justified by the fact that we are summing over all $\mathbf{k}$ and that the dispersion is symmetric $\xi_{\mathbf{k}} = \xi_{-\mathbf{k}}$. we can now use the anticommutation relations of the fermionic operators to rewrite

$$C^\dagger_{-\mathbf{k}\downarrow} C_{-\mathbf{k}\downarrow} = 1 - C_{-\mathbf{k}\downarrow} C^\dagger_{-\mathbf{k}\downarrow},$$

so that we can write

$$\sum_{\mathbf{k}\sigma} \xi_{\mathbf{k}\sigma} C^\dagger_{\mathbf{k}\sigma} C_{\mathbf{k}\sigma} = \sum_{\mathbf{k}} \xi_{\mathbf{k}} (C^\dagger_{\mathbf{k}\uparrow} C_{\mathbf{k}\uparrow} - C_{-\mathbf{k}\downarrow} C^\dagger_{-\mathbf{k}\downarrow}) + \sum_{\mathbf{k}} \xi_{\mathbf{k}}.$$

This expression can be written in a compact form using the Nambu spinors as

$$\sum_{\mathbf{k}} \Psi^\dagger_{\mathbf{k}} \begin{bmatrix} \xi_{\mathbf{k}} & 0 \\ 0 & -\xi_{\mathbf{k}} \end{bmatrix} \Psi_{\mathbf{k}} + \sum_{\mathbf{k}} \xi_{\mathbf{k}},$$

or, explicitly writing the components of the spinor

$$\sum_{\mathbf{k}} \begin{pmatrix} C^\dagger_{\mathbf{k}\uparrow} & C_{-\mathbf{k}\downarrow} \end{pmatrix} \begin{bmatrix} \xi_{\mathbf{k}} & 0 \\ 0 & -\xi_{\mathbf{k}} \end{bmatrix} \begin{pmatrix} C_{\mathbf{k}\uparrow} \\ C^\dagger_{-\mathbf{k}\downarrow} \end{pmatrix} + \sum_{\mathbf{k}} \xi_{\mathbf{k}}.$$

For the anomalous term, we can write

$$\begin{aligned} &\sum_{\mathbf{k}} (\Delta_{\mathbf{k}} C^\dagger_{\mathbf{k}\uparrow} C^\dagger_{-\mathbf{k}\downarrow} + \Delta^*_{\mathbf{k}} C_{-\mathbf{k}\downarrow} C_{\mathbf{k}\uparrow}) \\ &= \sum_{\mathbf{k}} \begin{pmatrix} C^\dagger_{\mathbf{k}\uparrow} & C_{-\mathbf{k}\downarrow} \end{pmatrix} \begin{bmatrix} 0 & \Delta_{\mathbf{k}} \\ \Delta^*_{\mathbf{k}} & 0 \end{bmatrix} \begin{pmatrix} C_{\mathbf{k}\uparrow} \\ C^\dagger_{-\mathbf{k}\downarrow} \end{pmatrix}. \end{aligned}$$

Thus we can write the BCS mean field hamiltonian in a compact form as

$$H^{mf}_{BCS} = \sum_{\mathbf{k}} \Psi^\dagger_{\mathbf{k}} \begin{bmatrix} \xi_{\mathbf{k}} & \Delta_{\mathbf{k}} \\ \Delta^*_{\mathbf{k}} & -\xi_{\mathbf{k}} \end{bmatrix} \Psi_{\mathbf{k}} + \sum_{\mathbf{k}} \xi_{\mathbf{k}},$$

which is a quadratic hamiltonian that we can diagonalize to find the energy spectrum and the ground state wavefunction. The matrix that appears in the hamiltonian is called the **Bogoliubov-de Gennes** (BdG) matrix, and it is a 2×2 matrix that describes the coupling between the normal and anomalous terms in the hamiltonian. The eigenvalues of this matrix give us the energy spectrum of the system, while the eigenvectors give us the coefficients $u_{\mathbf{k}}$ and $v_{\mathbf{k}}$ that appear in the BCS wavefunction.

Remark. *One could try to decompose any 2×2 hamiltonian in terms of the identity and the Pauli matrices, since they form a complete basis for 2×2 matrices. In this case, we can write*

$$\ldots$$

then by taking the square of the hamiltonian and using the anticommutation relations of the Pauli matrices, we can find the energy spectrum of the system:

$$H^2 = (\alpha^2 + \beta^2 + \gamma^2)\sigma_0 \implies E = \pm\sqrt{\alpha^2 + \beta^2 + \gamma^2}.$$

We can use the Pauli matrices trick to find the hamiltonian in the form

$$\ldots$$

Now we can diagonalize the hamiltonian by finding the eigenvalues and eigenvectors of the BdG matrix. The eigenvalues are given by

$$E_{\mathbf{k}} = \pm\sqrt{\xi^2_{\mathbf{k}} + |\Delta_{\mathbf{k}}|^2},$$

where $|\Delta_{\mathbf{k}}|^2$, since we wrote $\Delta = \Delta_1 - i\Delta_2$ is given by

$$(\Delta_1 - i\Delta_2)(\Delta_1 + i\Delta_2).$$

Thus the eigevectors are two dimensional vectors that can be written as

$$\begin{pmatrix} \sqrt{\frac{1}{2}\left[1+\frac{\xi_{\mathbf{k}}}{E_{\mathbf{k}}}\right]} \\ \sqrt{\frac{1}{2}\left[1-\frac{\xi_{\mathbf{k}}}{E_{\mathbf{k}}}\right]} \end{pmatrix}, \quad \begin{pmatrix} -\sqrt{\frac{1}{2}\left[1-\frac{\xi_{\mathbf{k}}}{E_{\mathbf{k}}}\right]} \\ \sqrt{\frac{1}{2}\left[1+\frac{\xi_{\mathbf{k}}}{E_{\mathbf{k}}}\right]} \end{pmatrix},$$

which are orthogonal and normalized. We can now identify the coefficients $u_{\mathbf{k}}$ and $v_{\mathbf{k}}$ that appear in the BCS wavefunction with the components of the eigenvectors, so that we can write

$$u_{\mathbf{k}} = \sqrt{\frac{1}{2}\left[1+\frac{\xi_{\mathbf{k}}}{E_{\mathbf{k}}}\right]}, \quad v_{\mathbf{k}} = \sqrt{\frac{1}{2}\left[1-\frac{\xi_{\mathbf{k}}}{E_{\mathbf{k}}}\right]}.$$

Remember that we are working in the Nambu spinor space, so that when we apply the eigenvectors to these spinors

$$\Psi^\dagger_{\mathbf{k}} \begin{pmatrix} u_{\mathbf{k}} \\ v_{\mathbf{k}} \end{pmatrix} = u_{\mathbf{k}} C^\dagger_{\mathbf{k}\uparrow} + v_{\mathbf{k}} C_{-\mathbf{k}\downarrow} = a^\dagger_{\mathbf{k}\uparrow},$$

and

$$\Psi^\dagger_{\mathbf{k}} \begin{pmatrix} -v_{\mathbf{k}} \\ u_{\mathbf{k}} \end{pmatrix} = u_{\mathbf{k}} C_{-\mathbf{k}\downarrow} - v_{\mathbf{k}} C^\dagger_{\mathbf{k}\uparrow} = a_{-\mathbf{k}\downarrow},$$

which are the **Bogoliubov operators** that diagonalize the hamiltonian (they are fermionic, thus they satisfy the fermionic anticommutation relations). Thus we have found the coefficients $u_{\mathbf{k}}$ and $v_{\mathbf{k}}$ that appear in the BCS wavefunction by diagonalizing the hamiltonian, which justifies the ansatz that we made for the wavefunction. Thus we can write the BCS (mean field) hamiltonian in the diagonal form as

$$H^{mf}_{BCS} = \sum_{\mathbf{k}} E_{\mathbf{k}} a^\dagger_{\mathbf{k}\sigma} a_{\mathbf{k}\sigma}.$$

Stufying the spectrum, instead, let us assume momentaneously that $\Delta_{\mathbf{k}} = \Delta$ is a constant, then we can write the spectrum as

$$E_{\mathbf{k}} = \pm\sqrt{\xi^2_{\mathbf{k}} + \Delta^2},$$

then the spectrum will be given by two branches, one with positive energy and one with negative energy, and there will be a gap of 2Δ between the two branches.

[*draw of the energy spectrum, two inverted parabulas symmetric in y and with a gap of 2delta between ...*]

This gap is called the **superconducting gap**, and it is a direct consequence of the formation of cooper pairs in the system. The presence of this gap means that there is a finite energy cost to create excitations in the system, which is a hallmark of the superconducting phase: it is an insulator for the presence of the gap, since we need to pay an energy cost to create excitations, but it is a superconductor since we have a finite density of cooper pairs that can condense and flow without resistance. The excitations are called **Bogoliubov quasiparticles**, and they are a superposition of electrons and holes, since they are created by the Bogoliubov operators that are a linear combination of creation and annihilation operators.

If we compute the density of states of the system, we find that since we have the gap, far from it we have only a parabolic dispersion (density is square root) ...

[*draw of the density of states, with ....*]

Keep in mind that the gap is very small compared to the ground state energy, so that the spectrum is almost continuous, but there is a small gap around the Fermi level that is a signature of the superconducting phase. Furthermore, remember that we did not treat the whole energy of the system, but only the superconducting part, so that the total energy of the system is given by the sum of the superconducting part and the normal part, which is given by the kinetic energy of the electrons. The normal part of the spectrum is given by the parabolic dispersion, while the superconducting part is given by the gapped spectrum that we found.

TODO: check, professor was supercazzoling us

Example (STS spectrum of superconductors). The superconducting gap can be measured experimentally using Scanning Tunneling Spectroscopy (STS), which is a technique that allows us to measure the local density of states of a material. In a superconductor, the STS spectrum will show a gap around the Fermi level, with peaks at the edges of the gap corresponding to the coherence peaks of the superconducting state. The size of the gap can be extracted from the STS spectrum, and it provides information about the strength of the superconducting pairing.

[*Image of sts*]

In a metal we have a finite density of states at the Fermi level, while in a superconductor we have a gap, so the density of states is zero at the Fermi level. The peaks at the edges of the gap are called coherence peaks, and they are a signature of the superconducting state. The size of the gap can be extracted from the distance between the coherence peaks, and it provides information about the strength of the superconducting pairing.

Taking again the definition of the order parameter, we want to show that it is non zero in the superconducting phase (below the critical temperature), and that it is zero in the normal phase (above the critical temperature).

In order to do so, we can write the expectation value of the bogoliubov operators we find the density of cooper pairs in the system, which is given by [...]

Now taking thus the definition of the order parameter, we can write

$$\begin{aligned}
\Delta_{\mathbf{k}} &= \frac{1}{N}\sum_{\mathbf{k}'} V_{\mathbf{k}\mathbf{k}'}\langle C_{-\mathbf{k}'\downarrow}C_{\mathbf{k}\uparrow}\rangle \\
&= -\frac{1}{N}\sum_{\mathbf{k}'} V_{\mathbf{k}\mathbf{k}'}\langle (v_{\mathbf{k}'}a^\dagger_{\mathbf{k}'\uparrow} + u_{\mathbf{k}'}a_{-\mathbf{k}'\downarrow})(u_{\mathbf{k}'}a^\dagger_{\mathbf{k}'\uparrow} - v_{\mathbf{k}'}a_{-\mathbf{k}'\downarrow})\rangle \\
&= -\frac{1}{N}\sum_{\mathbf{k}'} V_{\mathbf{k}\mathbf{k}'}\left[v_{\mathbf{k}'}u_{\mathbf{k}'}\dots\right]
\end{aligned}$$

[...]

4 Diagrammatic Monte Carlo

elevtron electron, electron phonon, Landau 1933 polaron H^{pol}_{el-ph}

we will see different hamiltonians we are not able to solve analytically

montecarlo to diagrammatic montecarlo

4.1 | Review of the Polaron Problem

We have a lattice composed by different types of atoms forming a crista: many body system. We dope the system, injectinng excess charges: it can be either an electron or an hole. for example injecting an electron in a crystal, if the cristal is either ionic or polarizable, equally charged particle will be displaced away from the electron, while oppositely charged particles will be attracted towards the electron. This injection breaks the local symmetry of the system, and this local rearrangement generates a potential, which actually creates a trap for the electron itself. The electron is then trapped in this potential, in the quantum well, and it is not free to move around the crystal. This forms a quasiparticle, which is called polaron. The polaron is the electron dressed by the phonon cloud, and it is a composite particle. The phonons can cause vibrations locally, thus an electron can bond with phonons to create this localized distortion, and when the dimension of the polaron is smaller than the lattice constant, we call it small polaron, while if the dimension of the polaron is larger than the lattice constant, we call it large polaron. The nature of the polaron is determined by the strength of the electron-phonon coupling, which is a measure of how strongly the electron interacts with the lattice vibrations. If a material is dominated by short range el-ph interactions is more prone to form small polarons, while if a material is dominated by long range el-ph interactions is more prone to form large polarons.

From the point of view of the numerical and analytical methods, we can divide in two main categories:

- we can contruct a schrodinger equation, where the hamiltonian contains all the dof of the system, huge object, but if we are able to contruct it we have this deterministic first principle method, which is very powerful. We can gain lots of insight, but we have to make a lot of approximations.
- We can contruct a hamiltonian, which is an effective hamiltonian, where we consider just the more important dof of the system, in a restricted hilbert space, and the ingredients are rather clear: electronic dof (one single electronic dispersion band) + phonons (summed over momenta, these are the phonon which interact to form the polaron) + the interaction between the two (mixed term of both types of operators). This is an effective hamiltonian, like the BCS, but in this case we are focusing on just a few particles and tune the adjustable parameters of the hamiltonian to reproduce the physics of the system. Furthermore, many body correlations are included here and were not present in the first approach, and we can solve this hamiltonian with different methods, either analytical or numerical. This is a more phenomenological approach, but it is more flexible and it is more powerful to capture the physics of the system.

From a pictorial point of view, from the first principle approach, we can plot the potential energy landscape of the system, and we can see how the electron is trapped in the potential well, and we can see how the phonons are distributed around the electron. There is no preferential side for the electron to be trapped, its probability distribution to be found in each site of the lattice is a uniform pdf. This comes immediatly from the solution of the schrodinger equation, which is a delocalized wavefunction, and the probability distribution is uniform. In the reciprocal k-space the electron will occupy the first free band of the system. uniform distribution will correspond to a minimal energy which we can calculate, and if we switch on the el-ph coupling, the electron will be trapped in a potential well, and the probability distribution will be localized around the center of the well, and the energy will be lower than the free electron case. The stronger the el-ph coupling, the more localized the electron will be, and the lower the energy will be. Localization in real space corresponds to delocalization in k-space, and the electron will occupy completely the first band (which will result in a flat band below the fermi surface). We can calculate the energy of the configuration with the distorted lattice, and if it is less than the energy of the free electron, then the polaron is stable, and the system will prefer to be in this configuration (the material is prone to form polarons).

The polaron hamiltonian considers:

- incoming electron, which is a free electron, with a dispersion ϵ_k, which is the energy of the electron as a function of its momentum k.
- the vertex in which the electron interacts with the phonon, which is the el-ph coupling g: the phonon propagates with momentum q and the electron with momentum $k - q$.
- the phonon propagator, which is the energy of the phonon as a function of its momentum q, and it is given by ω_q.
- the outgoing electron, which is the same as the incoming electron, with momentum k, after the interaction with the phonon.

This dispersion relation for the electron is just for one band, which is the one in which the polaron forms. We can consider both tight band or a parabolic band, but the physics is the same. For the large polaron (on which we will focus on), the dispersion is parabolic, and the el-ph coupling is long range, and it is given by the Fröhlich hamiltonian. Indeed the coupling constant is proportional to the inverse of the momentum, which means that the coupling is stronger for small momenta, which corresponds to long range interactions in real space. The strebght is also given by the α coefficient, depending on the material and in particular on its dielectric properties. These are the main ingredients defining the hamiltonian.

There is a possibility to solve this hamiltonian analytically, but it is not exact, and it is based on some approximations. We want to compute properties of this quasiparticle: since the polaron formation causes a change inn the band dispersion, we are then interested in computing the polaron dispersion; similarly, if a material want to form a polaron, it means that the energy of the polaron is lower than the energy of the free electron or other type of solutions, so we want to compute the polaron energy; many of this quantities pass by the computation of greens functions and self energy, which are the main quantities we want to compute.

One idea is to employ perturbation theory, which is a powerful method to compute the properties of the system, but it is based on the assumption that the el-ph coupling is weak, and it is not able to capture the physics of the system when the coupling is strong. we have seen that the interaction strenght is given by alpha, then we can use it compared to 1 to determine if the coupling is weak or strong, and in the strong coupling regime we have to use other methods, which are non perturbative. Weak distorsion are associated to large polarons, while strong distorsion are associated to small polarons, and the physics is different in the two cases.

The variational approach, instead, is generally more applicable to the strong coupling regime, and it is based on the idea of constructing a trial wavefunction, which is a guess for the true wavefunction of the system, and then we minimize the energy with respect to the parameters of the trial wavefunction. This method can capture some of the physics of the system, but it is not exact, and it is based on some assumptions about the form of the trial wavefunction.

Results from perturbation theory and variational principle, different dipendences on momenta and alpha.

The phonons causes a renormalization of the electron mass, which is called the effective mass of the polaron, and it is larger than the bare mass of the electron, and it depends on the el-ph coupling strength. The stronger the coupling, the larger the effective mass, and this is a signature of the polaron formation. But let us now focus on the energy.

Perturbation theory result (only in a small region of values for alpha) gives only a small fraction of the polaron energy and it is linear, while the variational approach gives a wider range of estimate of the polaron energy, which becomes more and more accurate as the coupling strength increases, and it is quadratic in alpha. The exact solution of the hamiltonian, which we can obtain with diagrammatic montecarlo, gives a result which is in between the two approximations, and it is able to capture the physics of the system for all values of alpha, from weak to strong coupling regime. Diagrammatic Monte Carlo can provide numerically exact solution for all values of the coupling strength, and it is able to capture the physics of the system in both weak and strong coupling regimes, and it can be used to benchmark the results obtained with perturbation theory and variational principle.

There is also a Feynman Path Integral approach, which is a powerful method to compute the properties of the system, and it is based on the idea of expressing the partition function of the system as a path integral over all possible paths of the system, and then we can use this path integral to compute the properties of the system. This method can capture some of the physics of the system, but it is not exact, and it is based on some assumptions about the form of the paths.

4.2 | Monte Carlo Methods

different types of montecaro based on the same common feature: the use of random numbers to perform numerical calculations. The main idea is to use random sampling to estimate the properties of a system, and it is particularly useful for systems with a large number of degrees of freedom, where traditional methods are not able to provide accurate results.

There are problems simply to complicated to be solved analytically methods,even with approximations, and we need to use numerical methods to solve them. A stochastic approach means we relies on random sampling to obtain numerical results, and the extimations obtained can be approximations, wrong, very good approximations or even numerically exact, depending on the method used and the problem at hand.

We can cut off some portions of the phase space, and we can use random sampling to explore the remaining portion of the phase space, and we can use this sampling to estimate the properties of the system.

Monte carlo started as a method to compute integrals, and it is based on the idea of using random sampling to estimate the value of an integral, and it is particularly useful for high-dimensional integrals, where traditional methods are not able to provide accurate results. If we want to explore a configuration space governed by a complex probability distribution, with a form we do not know analitycally, we can use random sampling to explore this configuration space, and we can use Markov Chain Monte Carlo (MCMC) methods to generate a sequence of samples from this distribution, and we can use these samples to estimate the properties of the system. Doing so we hop from one configuration to another, and we can use this sequence of samples to estimate the properties of the system. This is a powerful method to explore complex configuration spaces, and it is widely used in many fields, including physics, chemistry, biology, and machine learning.

It is also used in optimization problems, where we want to find the minimum of a function, and we can use random sampling to explore the configuration space of the function, and we can use this sampling to estimate the minimum of the function. We will not explore this type of applications, but it is worth mentioning that montecarlo is a powerful method to solve optimization problems, and it is widely used in many fields.

There are several different types of montecarlo methods, ll implied in different applications, and they are based on the same common feature of using random sampling to estimate the properties of a system. Some of the most common types of montecarlo methods include: Path integral Monte Carlo, Variational Monte Carlo, Diffusion Monte Carlo, and Diagrammatic Monte Carlo. Each of these methods has its own advantages and disadvantages, and it is suitable for different types of problems.

4.2.1 | Basic Statistics

Probability density function (pdf) are normalized and positice definite, and they can be used to describe the probability of finding a system in a particular state.

The expectation value of a variable X with respect to a pdf $\rho(x)$ is given by the integral of the product of the function and the pdf over the entire configuration space. The variable x is distributed according to the pdf $\rho(x)$, and the expectation value is a measure of the average value of the function X over the configuration space, weighted by the probability of finding the system in each state.

For continuous or discrete variables, we can use the same formula to compute the expectation value, but we have to replace the integral with a sum in the case of discrete variables. For a function of the variable X, we can compute the expectation value of the function by replacing X with the function in the formula for the expectation value.

in montecarlo, working with discrete variables, we just compute a mean. The montecarlo extimator of the expectation value is given by the average of the function over a set of samples drawn from the pdf. The law of large numbers states that as the number of samples increases, the montecarlo estimator converges to the true expectation value, and it is a powerful result that allows us to use montecarlo methods to estimate the properties of a system with a high degree of accuracy. The same applies to funxtions. Montecarlo does not compute explicitly neither the sum neither the integral, it just sumsover the samples chosen randomly from the pdf, and it is a powerful method to estimate the properties of a system, especially when the configuration space is large and complex. How this is achieved in practice is not trivial

Example (Monte Carlo Extimator: trivial example). we consider the expectation value of a function,

If we knew the pdf we could sample more values from the region in which the weight is higher, and we could obtain a better estimate of the expectation value. But we do not know the pdf, and we have to use random sampling to explore the configuration space.

The key idea is that the expectation value is similar to the sample average as long as the samples are drawn from the correct distribution. If we can generate samples from the correct distribution, then we can use the sample average to estimate the expectation value, and this is the basis of montecarlo methods.

We can imagine to sample uniformly from the configuration space, and we can use these samples to estimate the expectation value, but this is not efficient, because we are sampling from regions of the configuration space that have a low probability of being occupied by the system, and we are not sampling enough from regions that have a high probability of being occupied by the system.

MC extimates drawing many random numbers in this region, and thus as N increases, the estimate becomes more and more accurate, and it converges to the true expectation value. This is a powerful result that allows us to use montecarlo methods to estimate the properties of a system with a high degree of accuracy, even when the configuration space is large and complex. This can be proven by the law of large numbers, which states that as the number of samples increases, the sample average converges to the true expectation value.

Knowing the pdf, then we can obtain the exact expectation value by sampling from the correct distribution, and this is the basis of montecarlo methods.

We will always asses the quality of the montecarlo estimate by comparing it with the exact solution, and this can be done through variance and standard deviation, which are measures of the spread of the distribution of the samples around the true expectation value. The variance is defined as the average of the squared difference between the samples and the true expectation value, and it is a measure of how much the samples deviate from the true expectation value. The standard deviation is the square root of the variance, and it is a measure of how much the samples deviate from the true expectation value in units of the original variable. A smaller variance and standard deviation indicate that the montecarlo estimate is more accurate, while a larger variance and standard deviation indicate that the montecarlo estimate is less accurate.

4.2.2 | Calculating π with Monte Carlo

This will be our first example of a montecarlo method, and it is a simple example to illustrate the basic idea of montecarlo methods. We can use random sampling to estimate the value of π, which is a fundamental constant in mathematics and physics, and it is defined as the ratio of the circumference of a circle to its diameter.

The basic idea is to take the quarter unit circle and unit square in the positive quarter plane, and we can use random sampling to estimate the area of the quarter unit circle, which is given by $\pi/4$. We can generate N random points (x_i, y_i) in the 2D unit square, uniformly distributed (it is not the best approach, but we will come back later to this) and we can count how many of these points fall inside the quarter unit circle. The ratio of the number of points that fall inside the quarter unit circle to the total number of points generated will give us an estimate of $\pi/4$, and we can multiply this ratio by 4 to obtain an estimate of π.

After one estimate of π (thermalization is complete), we can repeat the process many times to obtain a distribution of estimates for π, and we can compute the mean and standard deviation of this distribution to assess the accuracy of our estimate. As we increase the number of random points N, the estimate of π will become more accurate, and the standard deviation will decrease, indicating that our estimate is converging to the true value of π. This is a powerful demonstration of how montecarlo methods can be used to estimate properties of a system with a high degree of accuracy, even when the configuration space is large and complex.

We need to address how to count the points: how do we distinguish between points that fall inside the quarter unit circle and points that fall outside? We can use the equation of the quarter unit circle, which is given by $x^2 + y^2 \leq 1$. If a point (x_i, y_i) satisfies this equation, then it falls inside the quarter unit circle, and we can count it as a point that falls inside. If a point does not satisfy this equation, then it falls outside the quarter unit circle, and we can count it as a point that falls outside. We thus define a function $f(x_i, y_i)$ that takes the value 1 if the point falls inside the quarter unit circle and 0 if it falls outside

$$f(x,y) = \begin{cases} 1, & \text{if } x^2 + y^2 \leq 1, \\ 0, & \text{otherwise,} \end{cases}$$

and we can use this function to count the number of points that fall inside the quarter unit circle.

The area of the quarter unit circle is given by the following 2D integral:

$$A_c = \frac{\pi}{4} = \int_0^1 \int_0^1 f(x,y)\mathrm{d}x\mathrm{d}y$$

In the MC pov we have to insert in some way the pdf, so we can multiply by $1 = \frac{\rho}{\rho}$ and we can write the area as an expectation value with respect to the pdf $\rho(x,y)$:

$$A_c = \left\langle \frac{f(x,y)}{\rho(x,y)} \right\rangle_\rho \sim \frac{1}{N} \sum_{i=1}^{N} \frac{f(x_i,y_i)}{\rho(x_i,y_i)},$$

where the samples (x_i, y_i) are drawn from the pdf $\rho(x,y)$. In the end we approximated again the expectation value with the sample average, and we can use this formula to estimate the area of the quarter unit circle, and thus to estimate the value of π. The choice of the pdf $\rho(x, y)$ can affect the accuracy of our estimate, and it is important to choose a pdf that is well-suited for the problem at hand. In this case, since we are sampling uniformly from the unit square

$$\rho(x, y) = 1 = \mathcal{U}(0, 1) \times \mathcal{U}(0, 1),$$

we can simplify the formula for the area to

$$A_c = \langle f(x, y) \rangle_\rho \sim \frac{1}{N} \sum_{i=1}^{N} f(x_i, y_i) = \frac{N_c}{N},$$

where N_c is the number of points that fall inside the quarter unit circle, and we can use this formula to estimate the area of the quarter unit circle, and thus to estimate the value of π. This is a powerful demonstration of how montecarlo methods can be used to estimate properties of a system with a high degree of accuracy, even when the configuration space is large and complex.

After just one MC run/extimate, we can repeat the process many times to obtain a distribution of estimates for π, and we can compute the mean and standard deviation of this distribution to assess the accuracy of our estimate. As we increase the number of random points N or the number of runs, the estimate of π will become more accurate, and the standard deviation will decrease, indicating that our estimate is converging to the true value of π. The final distribution of estimates for π will be a normal distribution centered around the true value of π.

A single run with an anormous N will be worse than many runs with a smaller N, for statistical reasons, and furthermore, we would not get a variance, which is important to assess the quality of our estimate.

In an alternative approach, we could use a one dimensional function and inytegral to compute π, for example we could use the function $f(x) = \sqrt{1 - x^2}$, which is the equation of the quarter unit circle, and we could compute the area of the quarter unit circle by integrating this function from 0 to 1. Following the same procedure as before, we can use random sampling to estimate the value of π by generating N random points x_i in the interval [0, 1], and we can compute the function $f(x_i)$ for each of these points. Again we will use the sample average to estimate the area of the quarter unit circle, and all of this can be considered as a further test on our method. The final estimate of π will be given by different formula, but the procedure is the same as before, and we can use this alternative approach to estimate the value of π and to assess the accuracy of our estimate.

we will see that problems arises from the arbitrary choice of the pdf, and we will see how to address these, since it is not the best distribution, but it already gives a good estimate of π.

In MC sampling or integration, pdf is an input of the method: the chosen pdf can affect the accuracy of our estimate, and it is important to choose a pdf that is well-suited for the problem at hand. In our case it would be beter to have a large number of N_i inside the quarter unit circle, and a smaller number of N_o outside the quarter unit circle, so we can have a better estimate of the area of the quarter unit circle. Ou problem hase an integrand with high value near 0 and low value near 1, so it would be better to have a pdf that is peaked near 0 and decays towards 1, so we can have more samples in the region where the integrand is large, and fewer samples in the region where the integrand is small. We can also try to use a pdf that is proportional to the integrand, which is called importance sampling, and it can improve the accuracy of our estimate by reducing the variance of the samples. In this case, we can use a pdf that is proportional to the function $\sqrt{1 - x^2}$. We would need to normalize this pdf, but we cannot always derive the normalization constant simply, in this case we have π as normalization, which is the output of our calculation.

If we were to know the exact pdf then the solution would be exact with zero variance.

4.2.3 | Markov Chains

If we do not know the normalization of the pdf, we can emply markov chain with the metropolis algorithm, starting from the first. a markov chain is a decision, proposal of moving from a point in the configuration space to another point, under some scheme we will introduce later, and we can use this markov chain to generate a sequence of samples from the pdf, and we can use these samples to estimate the properties of the system. The new point has to have some memories of the point we came from, and confronting the local probabilities of the two points, we can decide to accept or reject the move. At the end, iterating this process and counting how many times we have accepted a certain move, we can obtain an histogram of the visited points, which will be proportional to the pdf, and we can use this histogram to estimate the properties of the system.

The major difference is that now the samples are not independent, but they are correlated,but this correlatioon is limited to the present state...

We want our chain to spend more time where the pdf is larger, and less time where the pdf is smaller, so we can have a better estimate of the properties of the system. IN order to understand how to construct a markov chain we need to introduce the transition probabilities corresponding to the probability of moving from one point in the configuration space to another point. We are going to generalize the transition probabilities with the transition probability matrix, which is a matrix that contains the transition probabilities for all possible pairs of points in the configuration space. imagine a system with 4 possible states. The transition probability matrix will be a 4x4 matrix, where the element in the i-th row and j-th column will be the transition probability from state i to state j. This is a sort of generalization of the first example with two states. We still do not know the pdf, but we can use the transition probability matrix to generate a sequence of samples from the pdf, and we can use these samples to estimate the properties of the system.

After iterating this process many times, we can obtain a sequence of samples from the pdf, and applying the trasnition probability matrix many times, adjusting the transition probabilities at each step, we can obtain a sequence of samples that converges to the true pdf, and the markov chain ensures that the probabilities converge and are finite in the end. we obtained the pdf to sample from, and we can use this pdf to estimate the properties of the system, and this is the basis of markov chain montecarlo methods. It should be similar to our target function.

We want to build a Markov chain (which is guaranteed to converge in general, there are some peculiar cases, but we are not interested in those) whose equilibrium distribution is exactly the desired target distribution. Once we have a stationary solution from markov chain which has the structure of our target distribution, then we can use this markov chain to generate samples from the target distribution.

But how do we accept or reject each step in order to obtain a markov chain that converges to the desired target distribution? We can use the Metropolis-Hastings algorithm, which is a practical receipt which suggest us a way to build a markov chain whose equilibrium distribution is exactly the desired target distribution. The problem resides in how to decide to choose the transition probabilities in order to obtain a markov chain that converges to the desired target distribution. We need to define a clever acceptance or rejection scheme.

Main steps:

- define target function, which is the function we want to sample from
- start from an initial point in the configuration space, which can be chosen randomly or based on some prior knowledge of the system
- propose a new point in the configuration space, which can be done by adding a random perturbation to the current point, or by using some other proposal distribution
- important step: we compare the probability of the new point with thw previous ones, based on our target probability (function) (if we take the ratio of two unnormalized probabilities, the normalization constant cancels out, and we can use this ratio to decide whether to accept or reject the new point)
- accept or reject the new point based on the knowledge of the target distribution
- if we accept the new point, we move to the new point, and if we reject the new point, we stay at the current point
- repeat the process many times to generate a sequence of samples from the target distribution

Example (Metropolis-Hastings Algorithm: trivial example). We want to sample from a target distribution, which is the mixing of two gaussian random variables...

We have the target distribution and we choose a starting point, now we have to propose a move. we need to define a procedure to do this proposals: proposal distribution, we define a distribution (simple one, a gaussian) centered in the current point, and we draw a random number from this distribution to propose a new point. we will not complicate the tractation with details on proposals distributions, usually they are local but they can be global, we will not go into details. uniform is not th best choice.

until we are in the admittable sector for our target distribution, we will accept the proposal, and we will move to the new point, and if we are not in the admittable sector, we will reject the proposal, and we will stay at the current point.

We compute our function (which we know, we do not know the normalization constant) in the first and second points, and we can define an acceptance probability, as the minimum between 1 and the ratio of the function evaluated at the new point and the function evaluated at the current point: the acceptance probability must fullfill the axioms of probabilities, and it is a clever choice that ensures the convergence of the markov chain to the desired target distribution.

IN order to accept or reject, we draw a random number from a uniform distribution between 0 and 1, and if this random number is less than the acceptance probability, we accept the new point, and if this random number is greater than the acceptance probability, we reject the new point. At each step we have to draw a random number to propose a new point, and we have to draw another random number to decide whether to accept or reject the new point, and this is the basis of the Metropolis-Hastings algorithm.

points - evaluation at points - ratio - acceptance probability - accept or reject

recap

$$f(x) = \frac{1}{2}\mathcal{N}(-2,1) + \frac{1}{2}\mathcal{N}(2,1) = \frac{1}{2\sqrt{2\pi}}\left[e^{-\frac{(x+2)^2}{2}} + e^{-\frac{(x-2)^2}{2}}\right]$$

the initial point is $x_t = 0.6$ and we choose the proposal distribution as

$$\mathcal{N}(x_t, \sigma^2), \quad \sigma^2 = 0.4,$$

and the proposal is

$$x_{t+1} = x_t + \eta, \quad \eta \sim \mathcal{N}(0, \sigma^2),$$

let us suppose we got $x_{t+1} = 1.2$, now we compute $f(x_t)$ and $f(x_{t+1})$, and we can compute

$$\alpha = \min\left(1, \frac{f(x_{t+1})}{f(x_t)}\right) = \min\left(1, \frac{f(1.2)}{f(0.6)}\right),$$

where we have

$$f(x_t) = f(0.6) = 0.08166, \quad f(x_{t+1}) = f(1.2) = 0.014604,$$

thus

$$\alpha = \min\left(1, \frac{0.014604}{0.08166}\right) = \min(1, 1.79) = 1,$$

now we draw a random number from a uniform distribution between 0 and 1, let us say we drew $u = 0.42$, then the α value is grater than the acceptnce, so we acccept the move

$$(0.6,\ 1.2,\ \dots)$$

if we now start again from the new point $x_{t+1} = 1.2$, we can propose a new point, and we can repeat the process (we will do an example of rejection) let us imagine we got $x_{t+2} = 0$ (which would also be drawn from a larger proposal distribution than the previous one, but this is just an example), now we compute $f(x_{t+1})$ and $f(x_{t+2})$

$$f(0) = 0.05399, \quad f(1.2) = 0.014604$$

and then alpha will be

$$\alpha = \min\left(1, \frac{f(x_{t+2})}{f(x_{t+1})}\right) = \min\left(1, \frac{f(0)}{f(1.2)}\right) = \min(1, 0.37) = 0.37,$$

and if we now draw a random number from a uniform distribution between 0 and 1, let us say we drew $u = 0.72$, then the α value is less than the acceptance, so we reject the move and we stay at the current point

$$(0.6,\ 1.2,\ 1.2,\ \cdots)$$

we continuo thousands of times and at the end we plot the histogram of the visited points, and we can see that the histogram is proportional to the target distribution, and we can use this histogram to estimate the properties of the system, and this is the basis of markov chain montecarlo methods.

When we say that we have to run the markov chain for a long time, we mean that we have to run the markov chain until we reach the stationary distribution, which is the distribution that we want to sample from. The time it takes to reach the stationary distribution depends on the choice of the proposal distribution and the acceptance probability, and it can be quite long for some choices. We will then use this new distribution to sample from for our monte carlo method.

More general forms of the acceotance probability invludes also the conditional probability of the step and its inverse. When we reach the stationary distribution, this has to fullfill the detailed balance condition, which means that the probability of moving from one point to another point is equal to the probability of moving from the second point to the first point, and this is a necessary condition for the markov chain to converge to the desired target distribution. The Metropolis-Hastings algorithm is a clever choice of acceptance probability that ensures that the markov chain satisfies the detailed balance condition, and thus it converges to the desired target distribution.

skeleton of pseudocode for the Metropolis-Hastings algorithm...

We would now like to connect this MCMC to DiagMC approximating a distribution with MCMC basically we are working with a simple free propagator (exponential) ... basic idea... implementation details... we have to discretize the space to feed the computer... the discretization choices affect the result... connecting with the pi example, this approxiamtion of the integral for the propagator is similar to the approximation of the integral for the area of the quarter unit circle, and we can use the same idea to estimate the properties of the system, and this is the basis of diagrammatic montecarlo methods.

A first glimpse of diagmc

the exponential was a feynman diagram of the free propagator, and then the updates we propose are the shape and characteristics of the feynman diagram: we are exploring the space of all possible feynman diagrams, and we are using MCMC to sample from this space, and we are using the samples to estimate the properties of the system. We are generating and sampling feynman diagrams.

As we move on to more complicated diagrams, as diagrammatic series: the order of the diagram gives the number of vertices, and the configuration space becomes much more richer: we can update the momentum and other internal variables, number of vertices, the topology of the diagram, and we can use MCMC to sample from this space, and we can use the samples to estimate the properties of the system. We are generating and sampling feynman diagrams of different orders and topologies, and we are using these samples to estimate the properties of the system, and this is the basis of diagrammatic montecarlo methods. We have increased tremendously the complexity of our space, so we need finer update sttrategies, which usually are divided in two major classes in diagmc: class 1: uodates that do not change the order of the diagram, but they change the internal variables of the diagram, such as the momentum and other internal variables or the topology, and we can use these updates to explore the space of diagrams of a given order. class 2: updates that change the order of the diagram, such as adding or removing a vertex, and we can use these updates to explore the space of diagrams of different orders. We need to define a clever acceptance or rejection scheme for these updates, in order to ensure that the markov chain converges to the desired target distribution, which is the distribution of feynman diagrams that we want to sample from.

Appendices